

ELECTRONICS



ΠΑΝΕΠΙΣΤΗΜΙΟ
ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF COMPUTER AND INFORMATION
TECHNOLOGY AND COMPUTER ENGINEERING

PROJECT 4

R-C CIRCUITS SMOOTHING FILTERS - SCISSORS

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DELIVERY DATE : 30/12/2022

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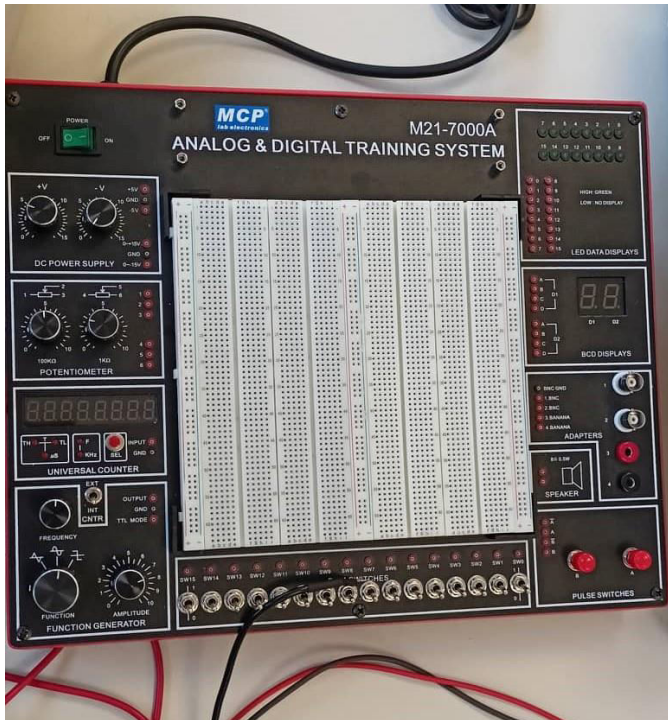
STUDENT PHOTO :



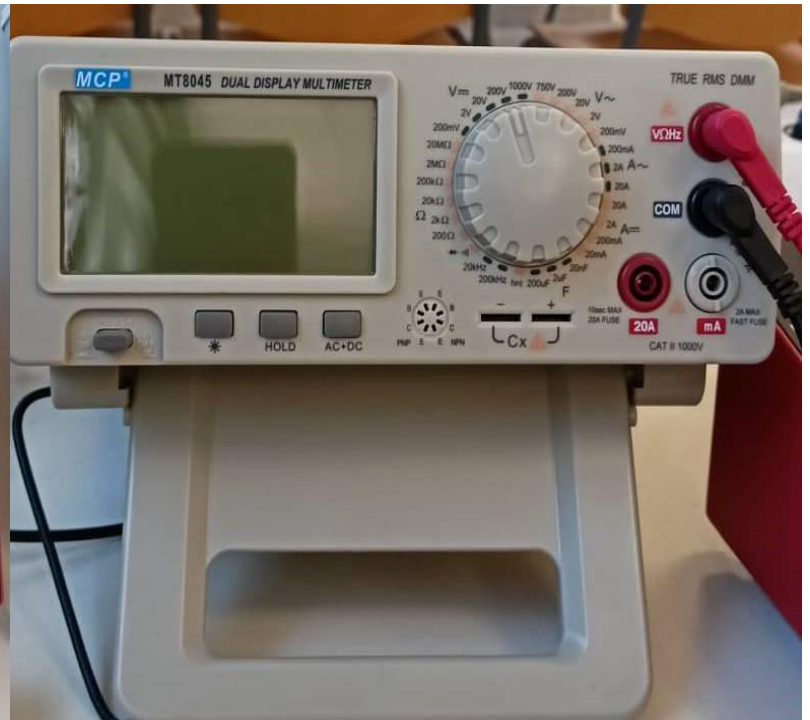
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PHOTOGRAPHS OF EQUIPMENT USED IN THE LABORATORY

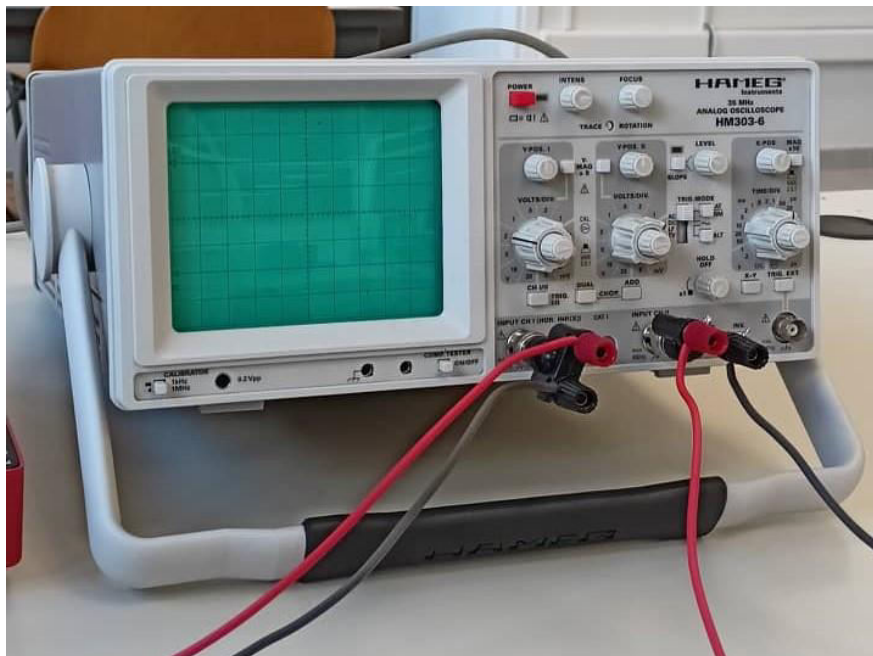
Breadboard M21-7000A



Breadboard M21-7000A



Digital Multimeter Bench MT8045



Oscilloscope HAMEG HM203-5



Cable

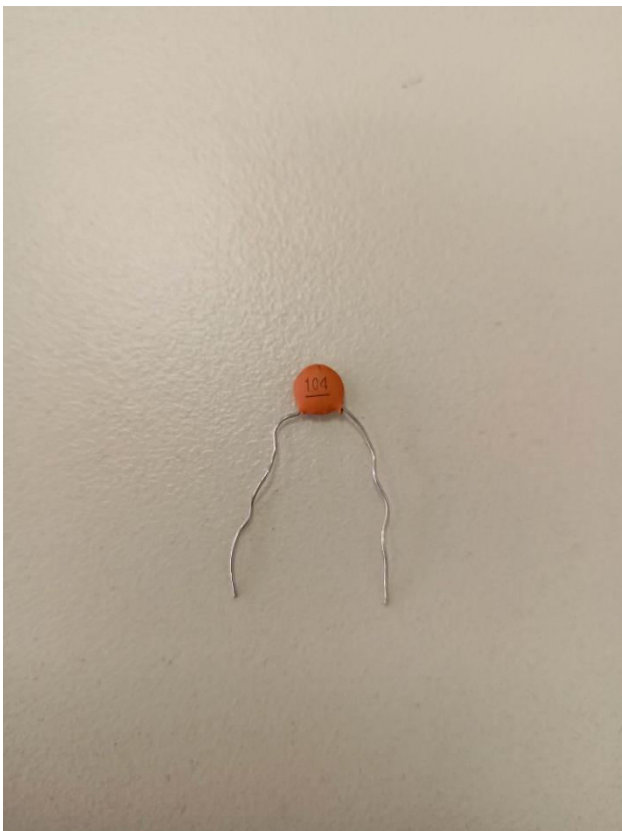
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Resistor of 4.7 KOhm



Silicon diode



Capacitor with a capacitance of 0.1 μF



Resistor of 10 KOhm

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1. R-C CIRCUITS SMOOTHING FILTERS

1.1 SMOOTHING FILTER

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 1.1.1 shows the schematic of the smoothing filter with a 6 V supply_{p-p}, frequency 1 kHz, resistance 4.7 kOhm and capacitance 0.1 μF.

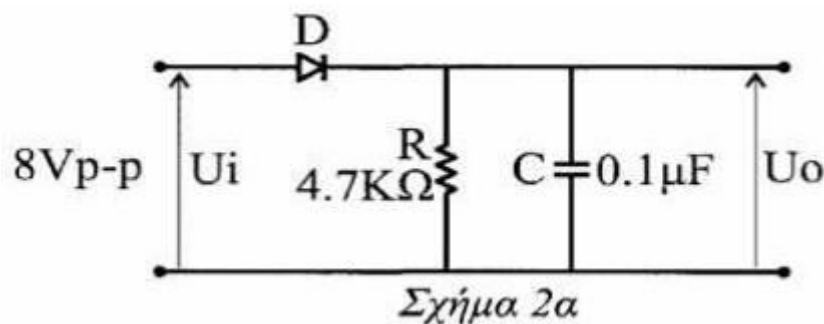


Figure 1.1.1.1 Schematic of the smoothing filter with a frequency of 1 kHz

During the positive half-period the anode of the diode is positively charged and therefore the diode is open. The input signal will be a 3 V_p (6 V_{p-p}) sine wave signal, which is stored by the capacitor through the diode and charged. During the negative half period, the anode of the diode is negatively charged and the capacitor has not had time to discharge and thus the diode does not turn on and with the capacitor discharged into the 4.7 kOhm resistor. Figure 1.1.2 shows the **input** and **output** waveforms.

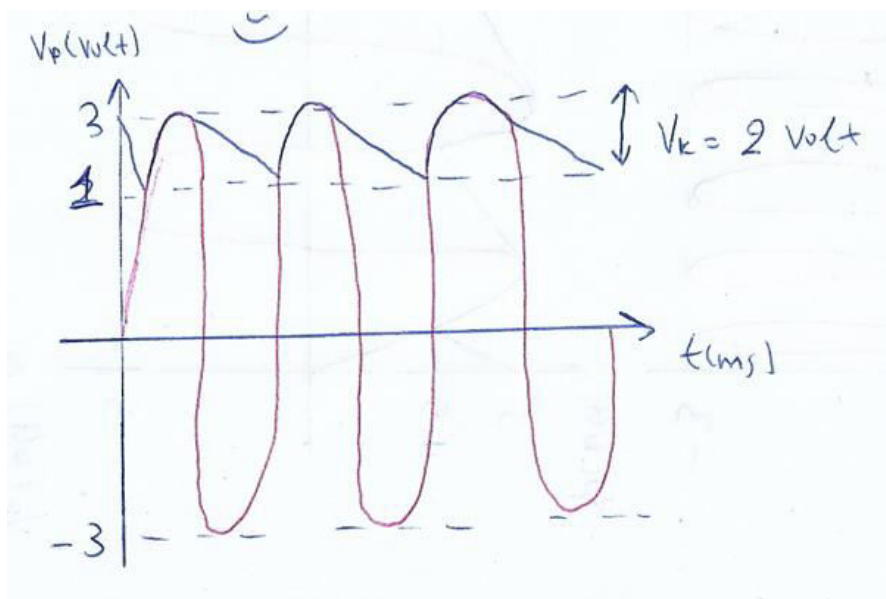


Figure 1.1.2 Smoothing filter waveforms with a frequency of 1 kHz

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The ripple voltage is equal to

$$V_k = \frac{V_p}{fRC} \rightarrow V_k = \frac{1 \text{ V}}{10^3 \text{ Hz} \times 4.7 \times 10^3 \text{ Ohm} \times 0.1 \times 10^{-6} \text{ F}} \rightarrow V_k = 2 \text{ Volt}$$

Simulative Resolution

Figure 1.1.3 shows the simulation of the smoothing filter with 6 V supply_{p-p}, frequency 1 kHz, resistance 4.7 kOhm and capacitance 0.1 μF in the simulation software "Multisim".

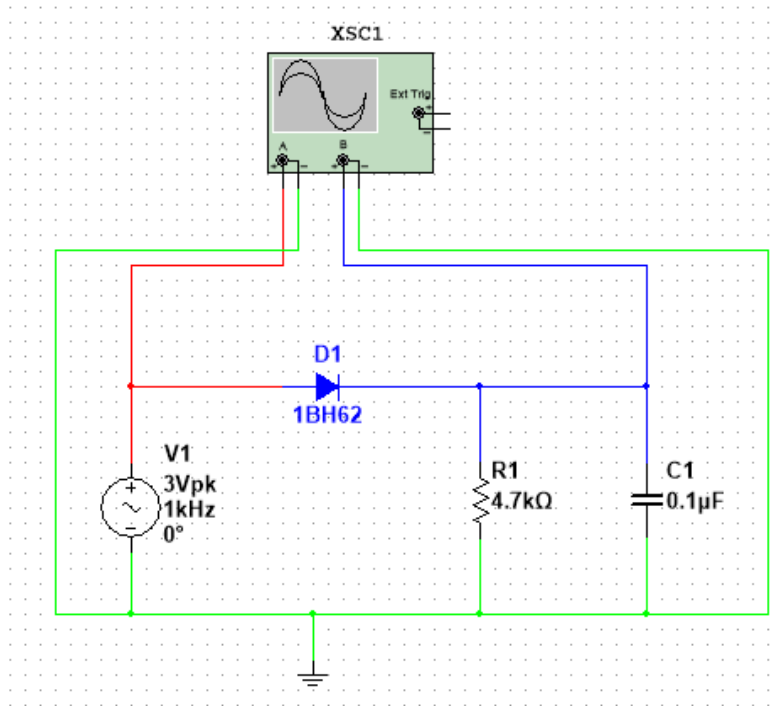


Figure 1.1.3 Simulation of the smoothing filter with a frequency of 1 kHz in "Multisim"

In Figures 1.1.4 and 1.1.5, the ripple voltage V_k calculated in the Theoretical Solution chapter is simulated in the simulation software with the help of the oscilloscope.

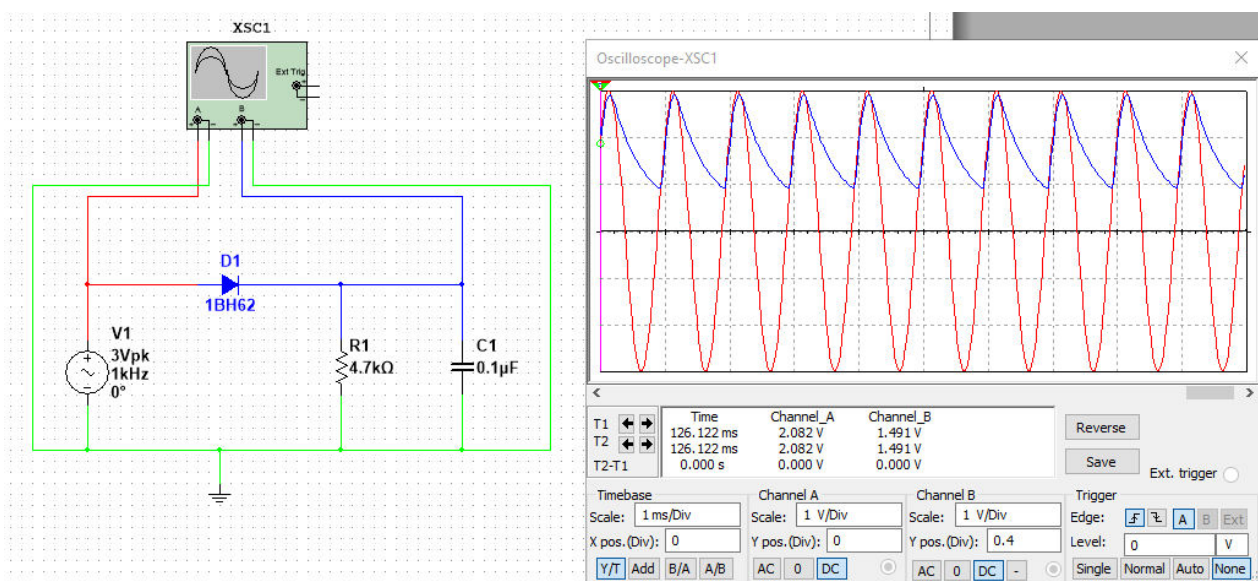


Figure 1.1.4 Waveforms of the smoothing filter with a frequency of 1 kHz in "Multisim"

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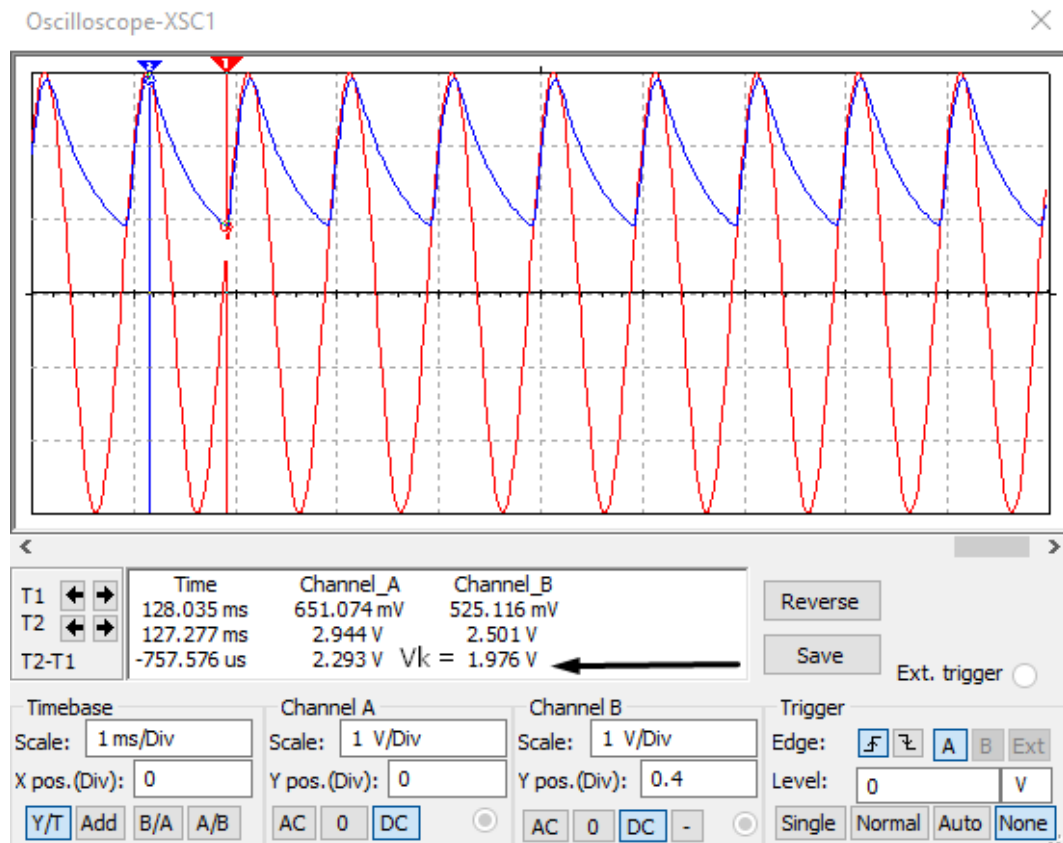


Figure 1.1.1.5 Calculation of V_k of the output signal in the smoothing filter with frequency 1 kHz in "Multisim"

Experimental Solution

Figure 1.1.6 shows the circuit of the smoothing filter with a power supply of 6 V_{p-p}, frequency 1 kHz, resistance 4.7 kOhm and capacitance 0.1 μF on the "breadboard", where it was carried out in the laboratory.

The construction steps of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 1.1.7).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 1.1.8).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 1.1.8).
4. We take channel A of the oscilloscope and connect the terminals to display the input waveform on the screen (Figure 1.1.8).
5. We set the switch to 1 Volts/Div (Figure 1.1.9).
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 6 V_{p-p} (Figure 1.1.9).
7. We align the waveform from the oscilloscope to be correctly positioned on the axes. (Figure 1.1.9).
8. We continue assembling the circuit by taking the silicon diode (Figure 1.1.10) and connecting it in series with the source.
9. We take the 4.7 kOhm resistor (Figure 1.1.11) and connect it in series with the diode.
10. Connect the 0.1 μF capacitor in parallel with the resistor (Figure 1.1.12).
11. We also take channel B of the oscilloscope and connect it in parallel with the terminals of the capacitor to get the output signal (Figure 1.1.13).

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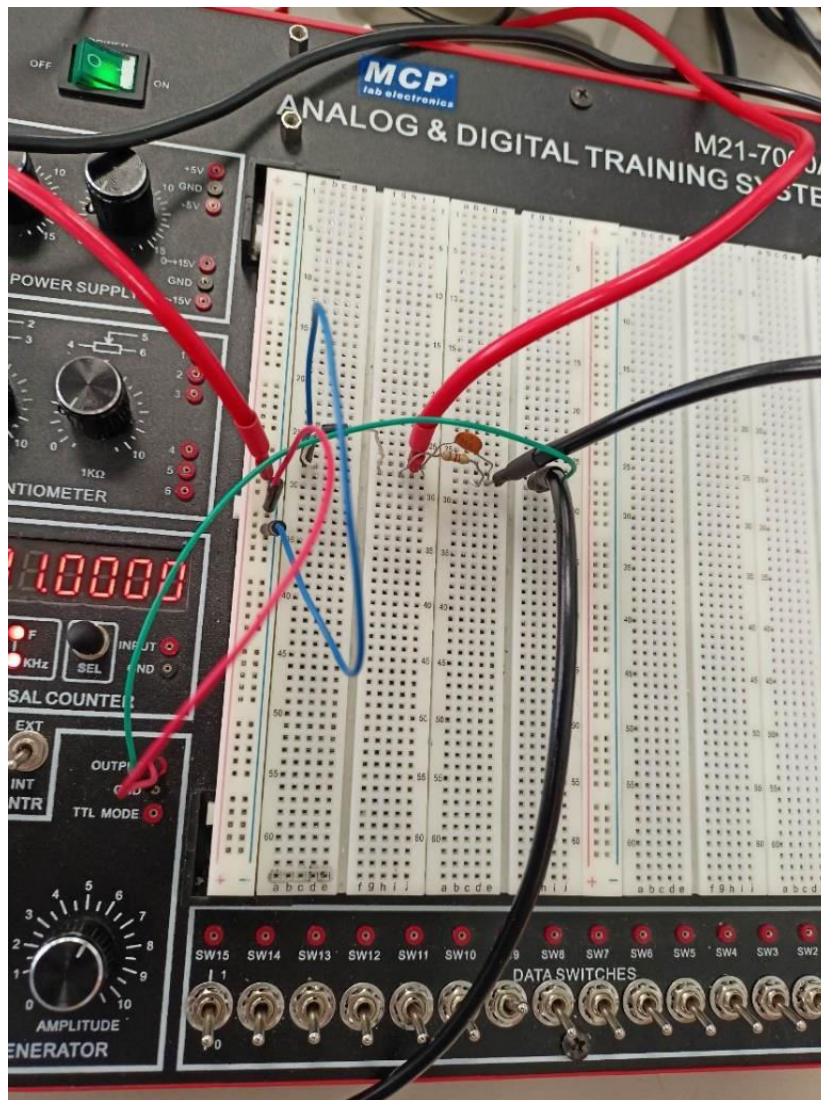


Figure 1.1.1.6 Experimental illustration of the smoothing filter with a frequency of 1 kHz on the "breadboard"



Figure 1.1.1.7 Step 1

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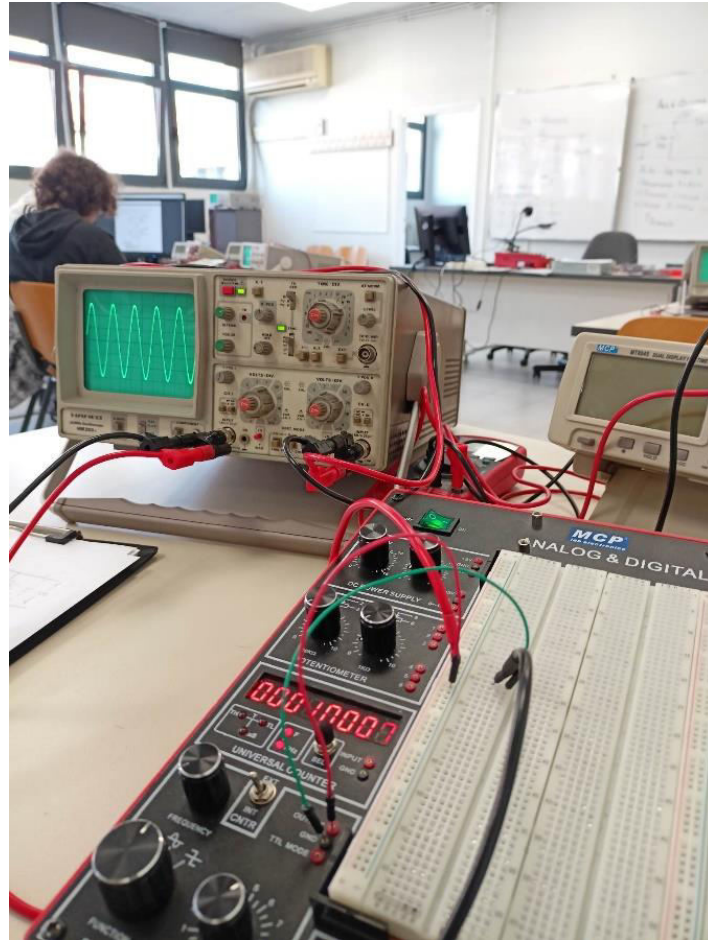


Figure 1.1.8 Step 2, 3, 4

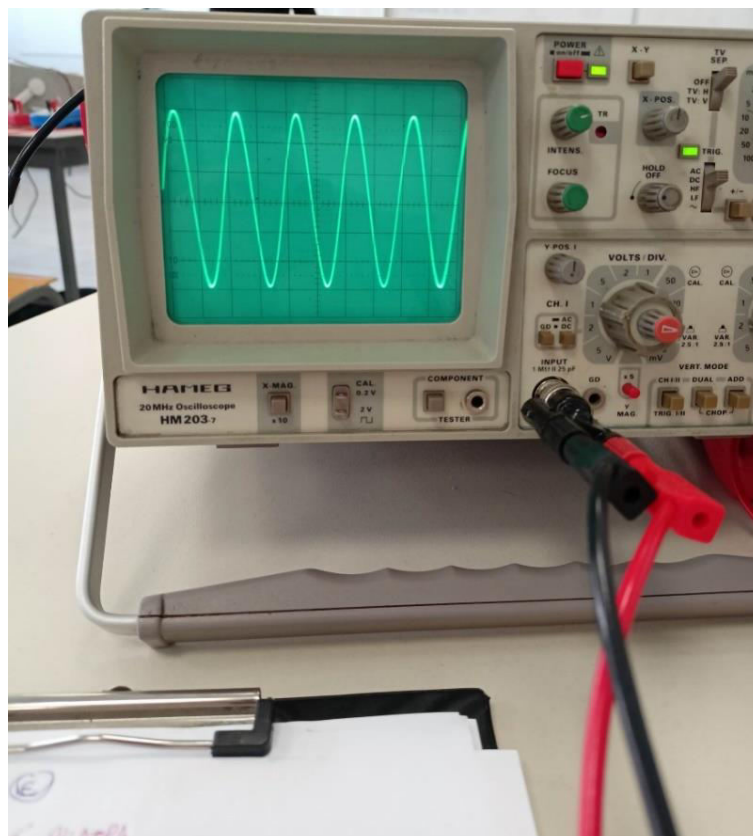


Figure 1.1.9 Step 5, 6, 7

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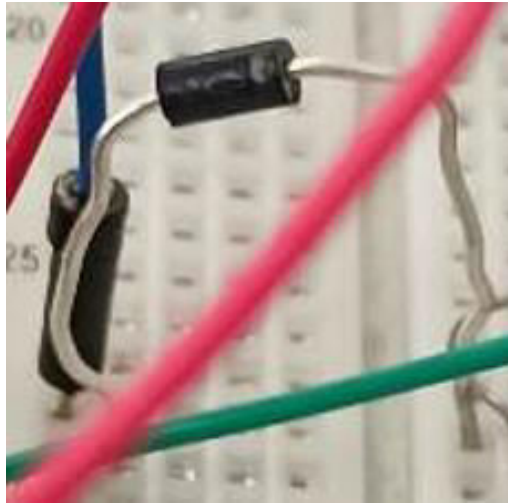


Figure 1.1.10 Step 8



Figure 1.1.11 Step 9



Figure 1.1.12 Step 10

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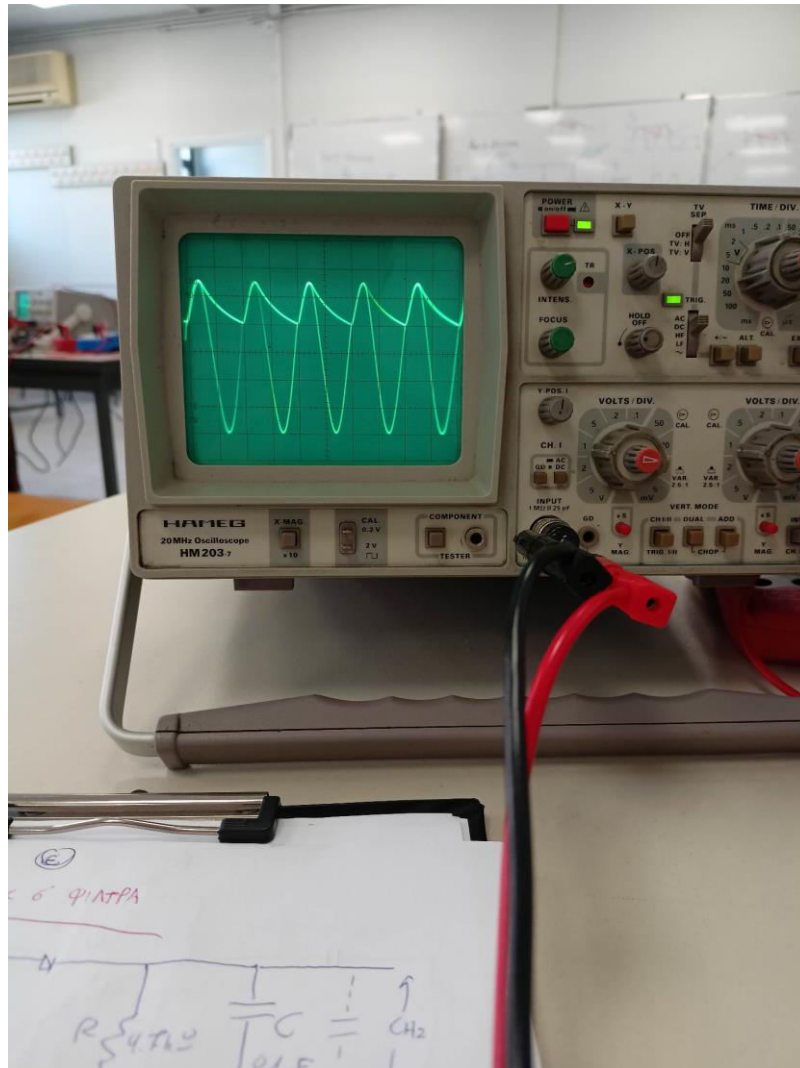


Figure 1.1.13 Step 11

1.2 SMOOTHING FILTER WITH DIFFERENT FREQUENCY

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

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Theoretical Solving

Figure 1.2.1 shows the schematic of the smoothing filter with a 6 V supply_{p-p}, frequency 3 kHz / 500 Hz, resistance 4.7 kOhm and capacitance 0.1 μF.

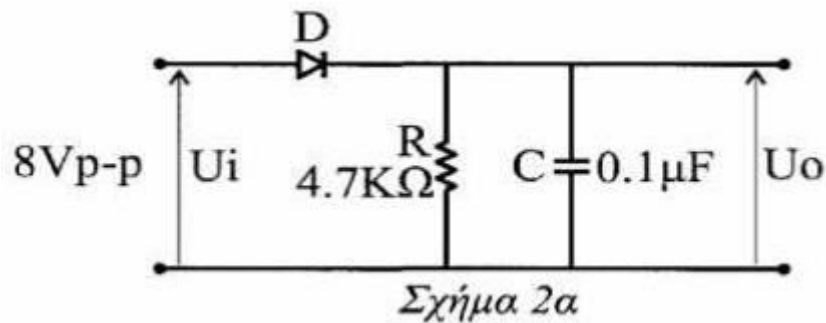


Figure 1.2.1 Schematic of the smoothing filter with a frequency of 3 kHz / 500 Hz

During the positive half-period the anode of the diode is positively charged and therefore the diode is open. The input signal will be a $3 V_p$ ($6 V_{p-p}$) sine wave signal, which is stored by the capacitor through the diode and charged. During the negative half period, the anode of the diode is negatively charged and the capacitor has not had time to discharge and thus the diode does not turn on and with the capacitor discharged into the 4.7 kOhm resistor. Figure 1.2.2 shows the **input** and **output** waveforms for the smoothing filter with a frequency of 3 kHz, while Figure 1.2.3 shows the **input** and **output** waveforms for the smoothing filter with a frequency of 500 kHz.

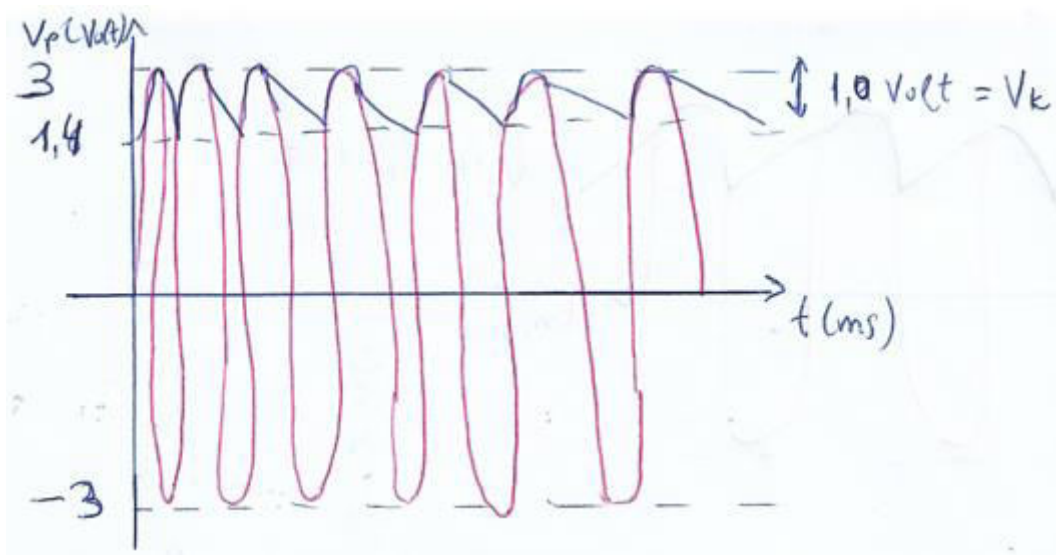


Figure 1.2.2.2 Smoothing filter waveforms with a frequency of 3 kHz

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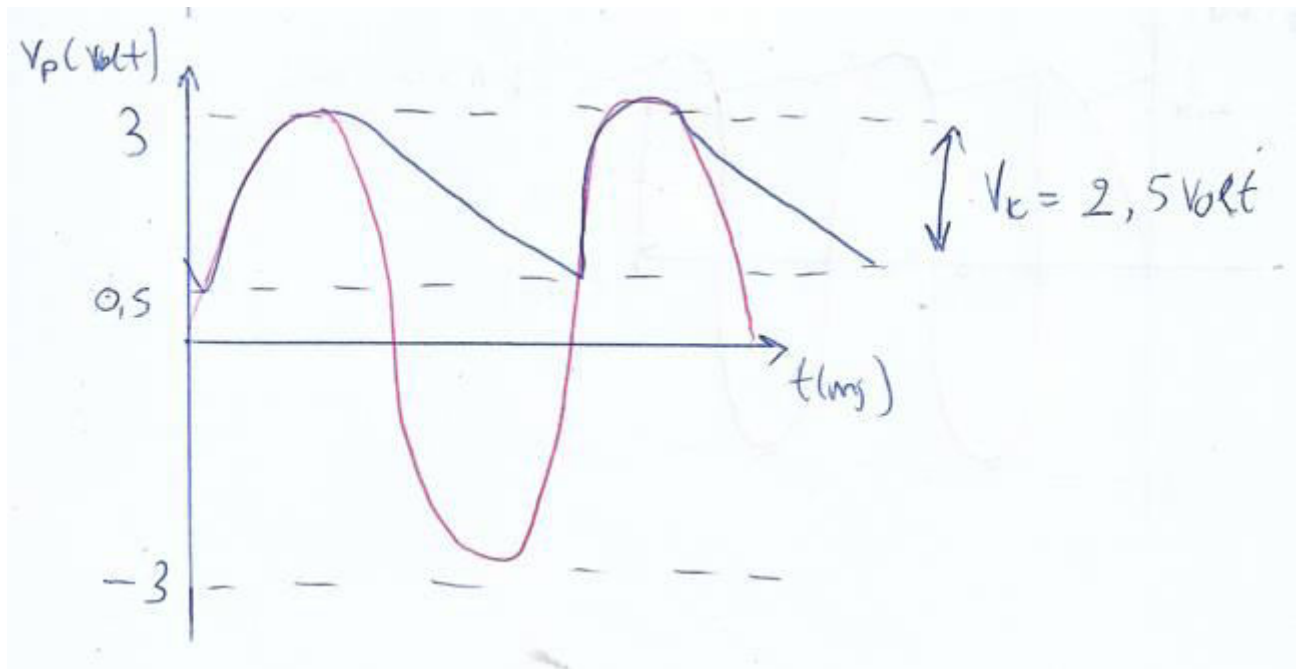


Figure 1.2.3 Smoothing filter waveforms with a frequency of 500 Hz

Increasing the frequency (3 kHz) leads to an increase in the discharge time of the capacitor and, consequently, to a decrease in the ripple voltage which is equal to:

$$V_k = \frac{V_p}{fRC} \rightarrow V_k = \frac{1.4 \text{ V}}{3 \times 10^3 \text{ Hz} \times 4.7 \times 10^3 \text{ Ohm} \times 0.1 \times 10^{-6} \text{ F}} \rightarrow V_k = 1 \text{ Volt}$$

Reducing the frequency (500 Hz) leads to a reduction in the discharge time of the capacitor and, consequently, to an increase in the ripple voltage which is equal to:

$$V_k = \frac{V_p}{fRC} \rightarrow V_k = \frac{0.5 \text{ V}}{5 \times 10^2 \text{ Hz} \times 4.7 \times 10^3 \text{ Ohm} \times 0.1 \times 10^{-6} \text{ F}} \rightarrow V_k = 2.4 \text{ Volt}$$

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Simulative Resolution

Figure 1.2.4 shows the simulation of the smoothing filter with a power supply of 6 V_{p-p}, frequency of 3 kHz, resistance of 4.7 kΩ and capacitance of 0.1 μF in the simulation software "Multisim".

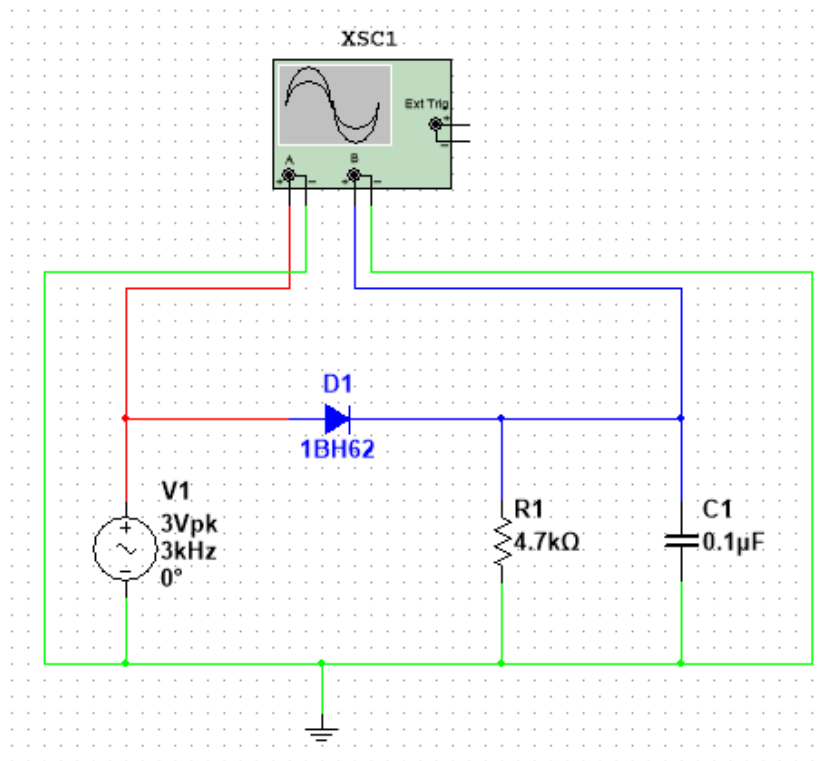


Figure 1.2.4 Simulation of the smoothing filter with a frequency of 3 kHz in "Multisim"

In Figure 1.2.5 and 1.2.6, the ripple voltage V_k calculated in the Theoretical Solution chapter is simulated in the simulation software with the help of the oscilloscope.

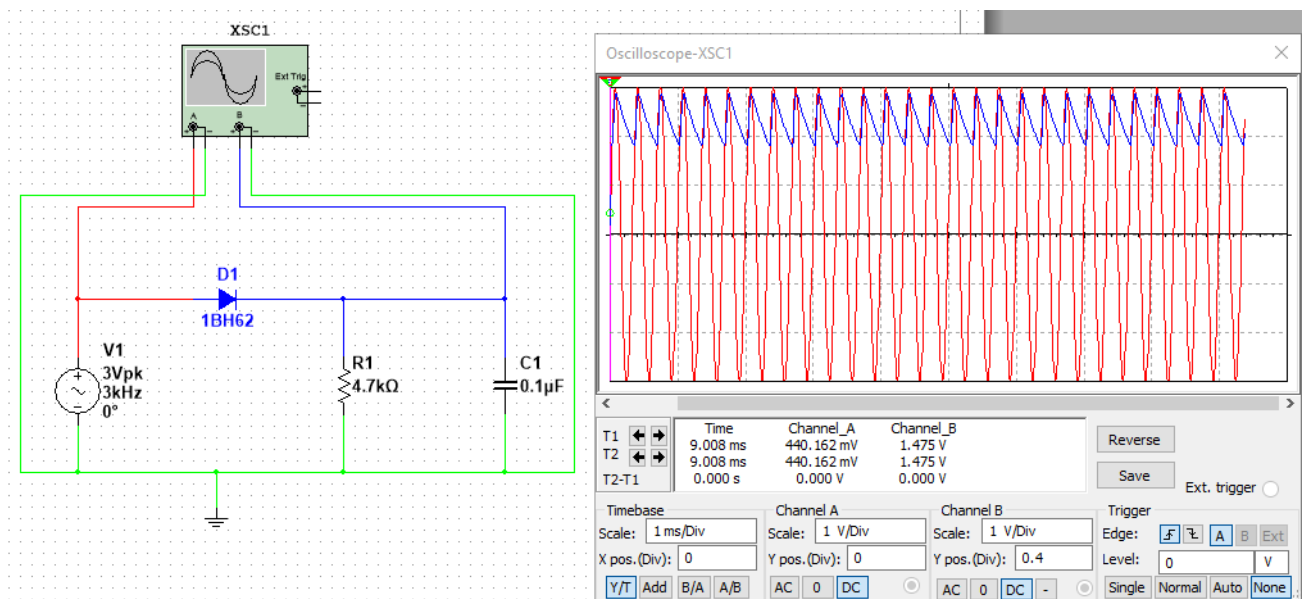


Figure 1.2.5 Waveforms of the smoothing filter with a frequency of 3 kHz in "Multisim"

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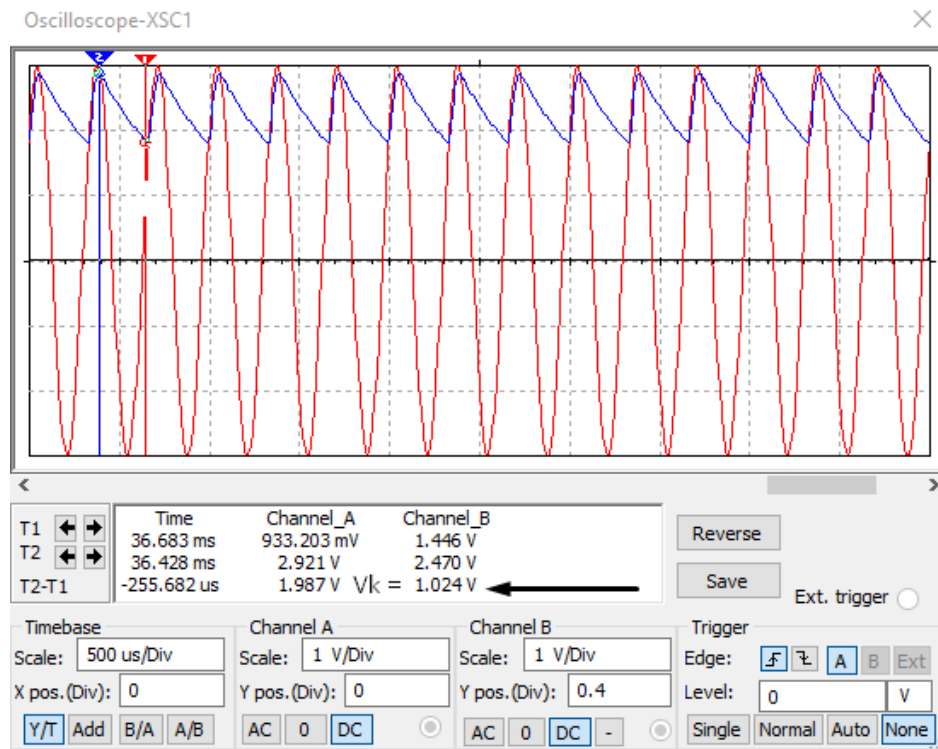


Figure 1.2.6 Calculation of V_k of the output signal in the smoothing filter with frequency 3 kHz in "Multisim"

Figure 1.2.7 shows the simulation of the smoothing filter with a power supply of 6 V_{p-p}, frequency 500 Hz, resistance 4.7 kOhm and capacitance 0.1 μF in the simulation software "Multisim".

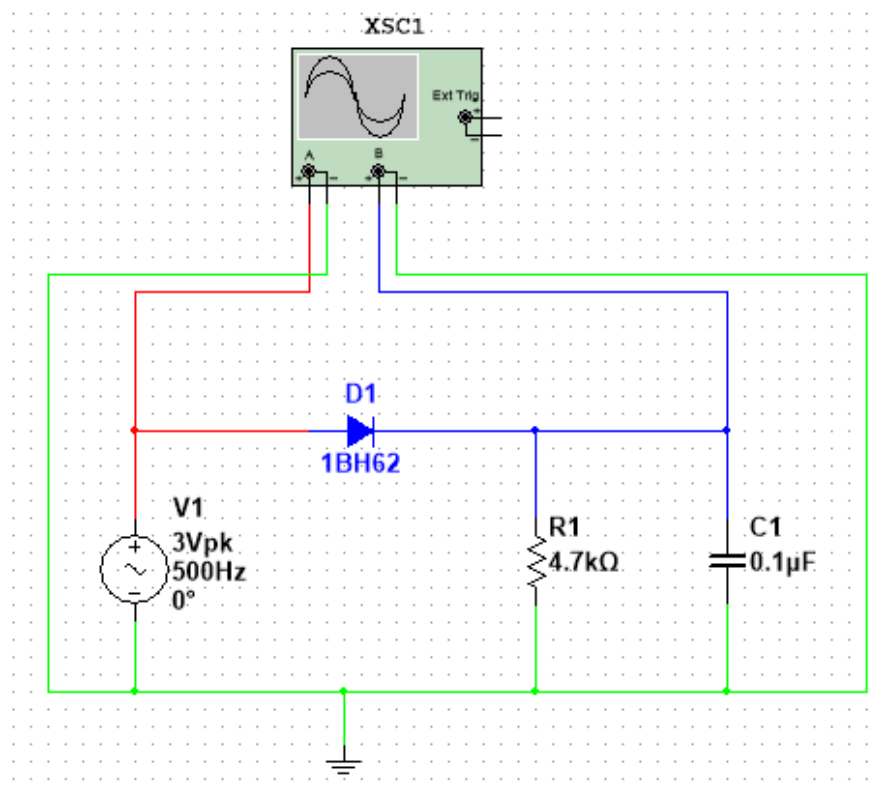


Figure 1.2.7 Simulation of the smoothing filter with a frequency of 3 kHz in "Multisim"

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In Figure 1.2.8 and 1.2.9, the ripple voltage V_k calculated in the Theoretical Solution chapter is simulated in the simulation software with the help of the oscilloscope.

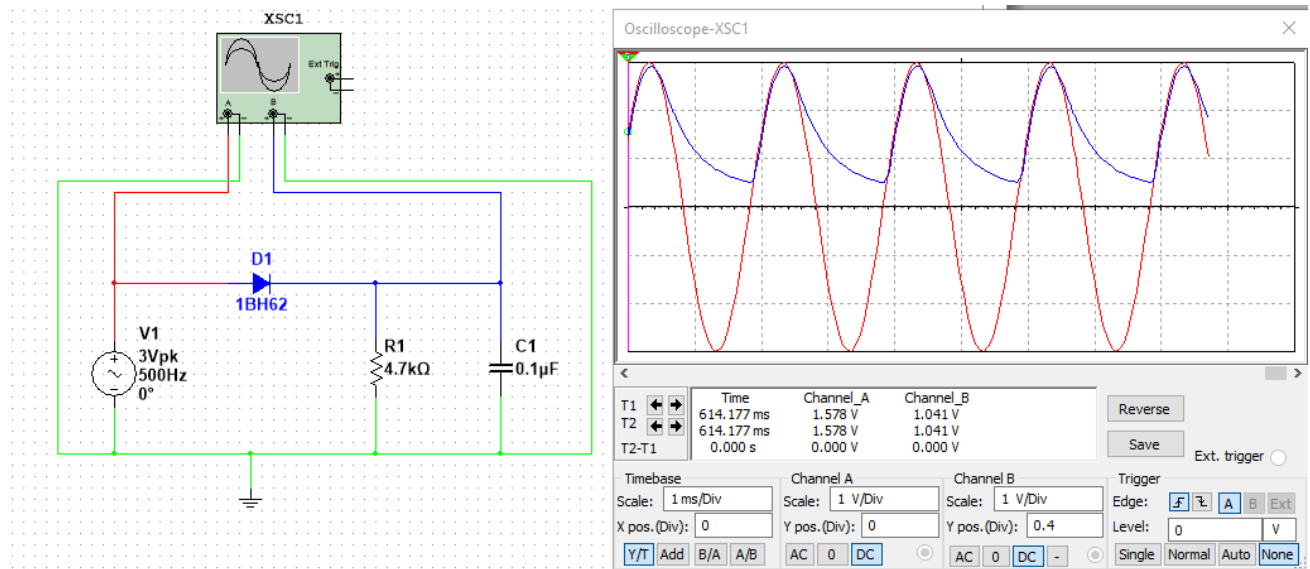


Figure 1.2.8 Waveforms of the smoothing filter with a frequency of 500 Hz in "Multisim"

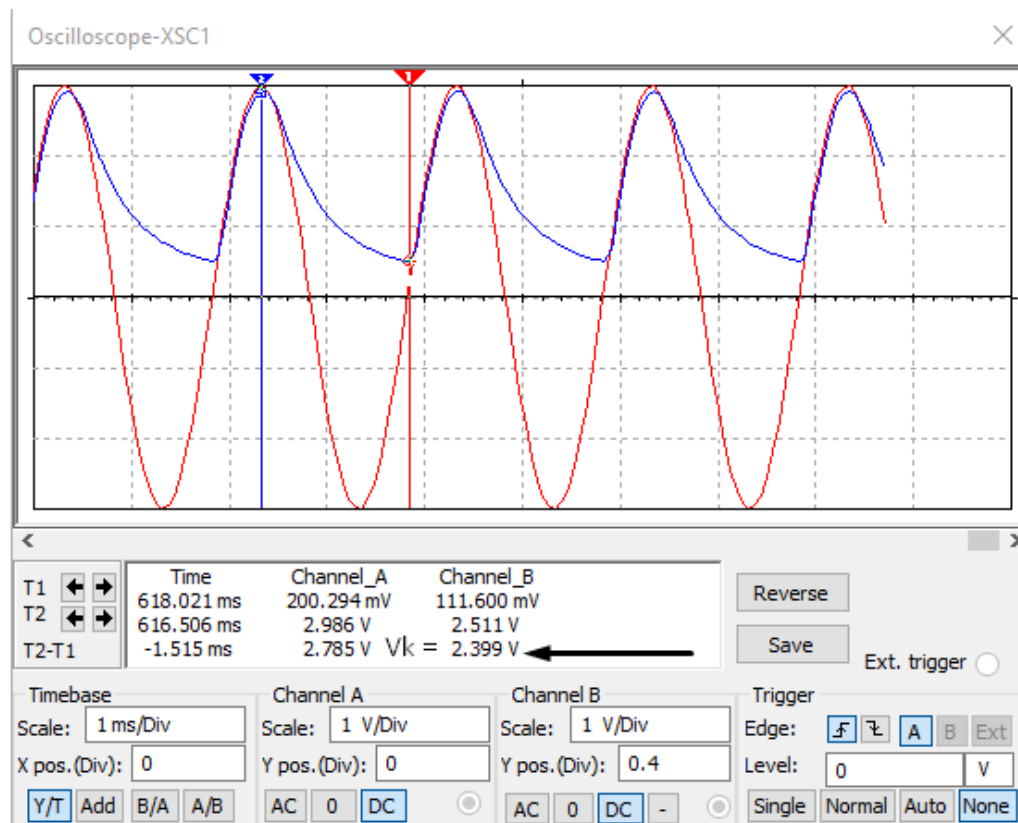


Figure 1.2.9 Calculation of V_k of the output signal in the smoothing filter with a frequency of 500 Hz in "Multisim"

Experimental Solution

Figure 1.2.10 shows the circuit of the smoothing filter with a power supply of 6 V_{p-p}, frequency 3 kHz, resistance 4.7 kΩ and capacitance 0.1 μF on the "breadboard", where it was carried out in the laboratory.

The construction steps of the experiment are as follows:

1. We set the input frequency to 3000 Hz (Figure 1.2.11).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 1.2.12).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 1.2.12).
4. We take channel A of the oscilloscope and connect the terminals to display the input waveform on the screen. (Figure 1.2.12)
5. We set the switch to 1 Volts/Div (Figure 1.2.13).
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 6 V_{p-p} (Figure 1.2.13).
7. We align the oscilloscope so that the waveform is correctly positioned on the axes (Figure 1.2.13).
8. We continue assembling the circuit by taking the silicon diode (Figure 1.2.14) and connecting it in series with the source.
9. We take the 4.7 kΩ resistor (Figure 1.2.15) and connect it in series with the diode.
10. Connect the 0.1 μF capacitor in parallel with the resistor (Figure 1.2.16).
11. We also take channel B of the oscilloscope and connect it in parallel with the terminals of the capacitor to get the output signal (Figure 1.2.17).

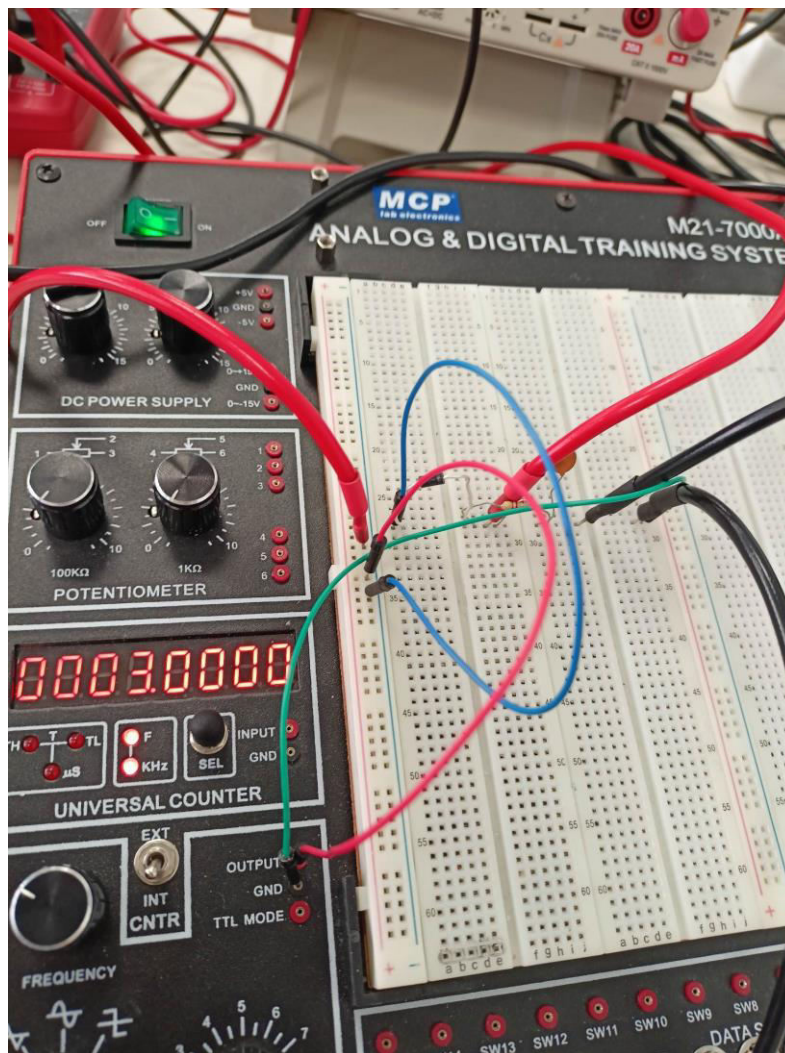


Figure 1.2.10 Experimental illustration of the 3 kHz smoothing filter on the breadboard

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Figure 1.2.11 Step 1

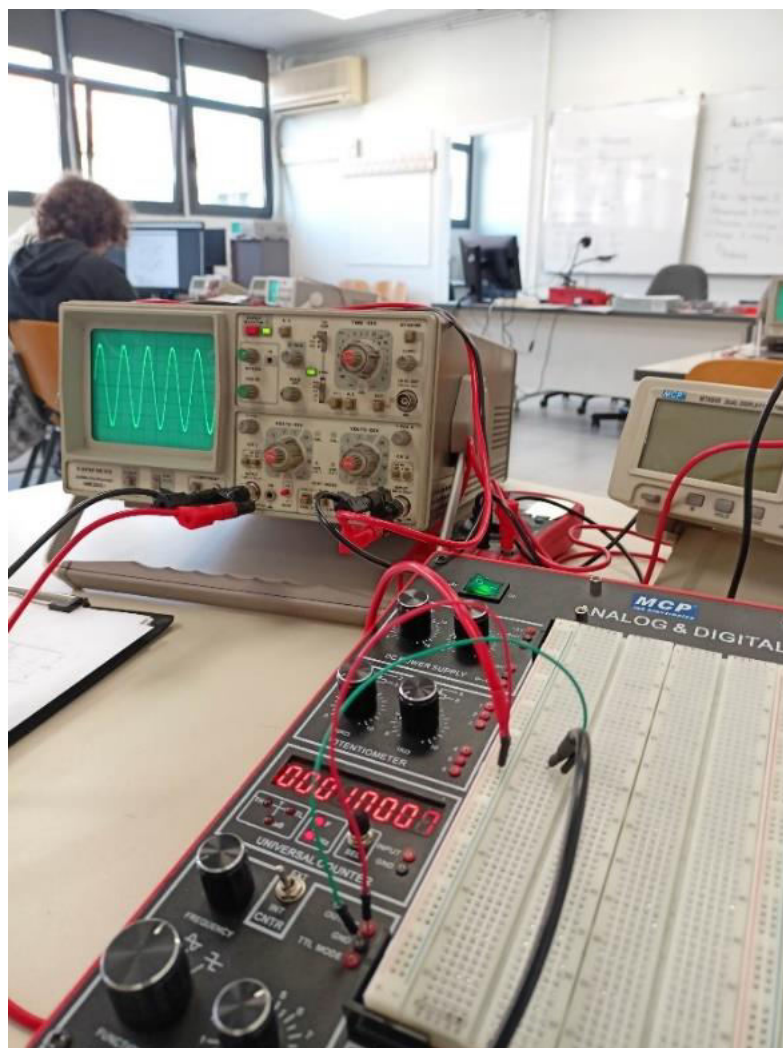


Figure 1.2.12 Step 2, 3, 4

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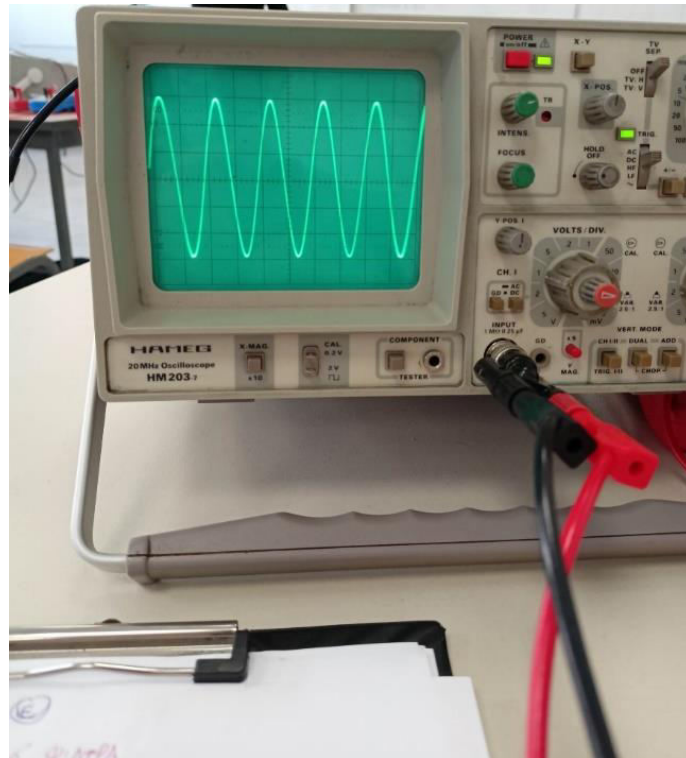


Figure 1.2.13 Step 5, 6, 7

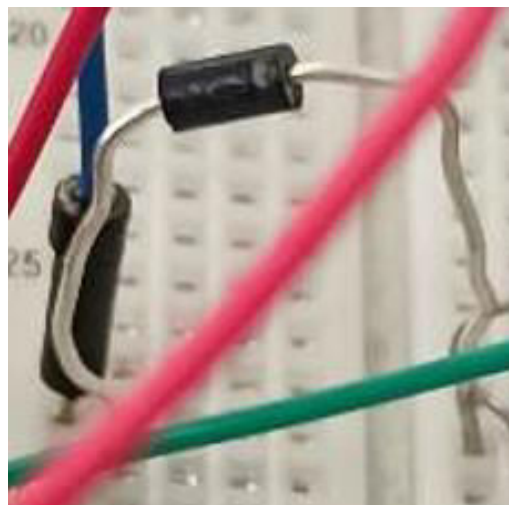


Figure 1.2.14 Step 8



Figure 1.2.15 Step 9

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Figure 1.2.16 Step 10

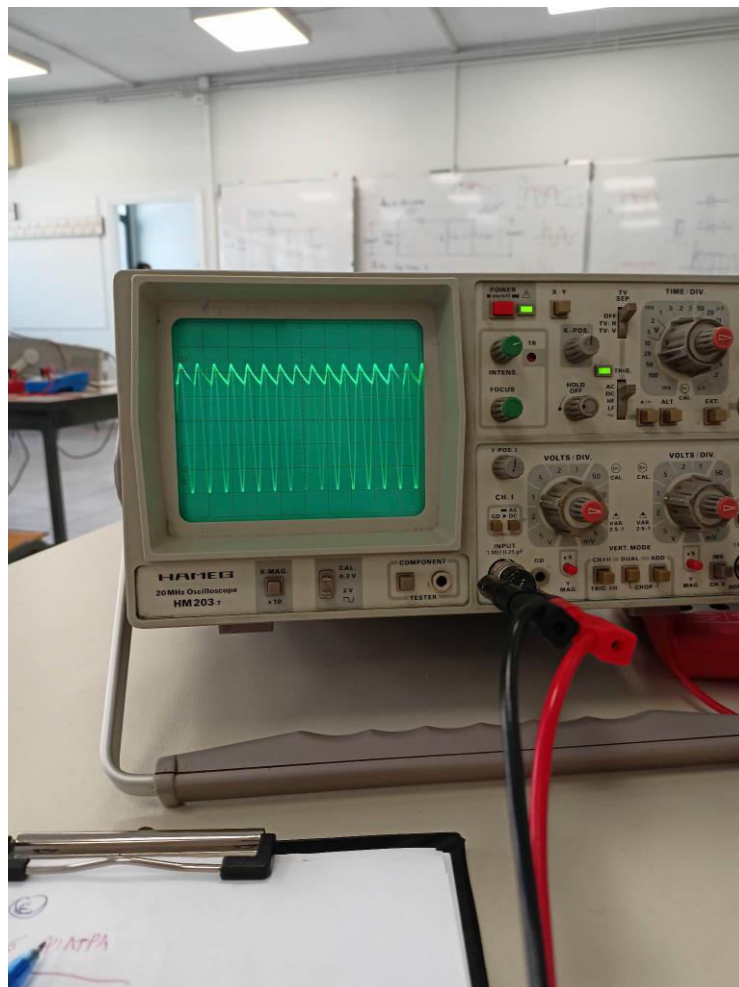


Figure 1.2.17 Step 11

1.3 SMOOTHING FILTER WITH MUTUAL RESISTANCE

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

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Theoretical Solving

Figure 1.3.1 shows the schematic of the smoothing filter with 6 V supply_{p-p}, frequency 1 kHz, resistance 10 kOhm and capacitance 0.1 μF. Figure 1.3.2 shows the schematic of the smoothing filter with 6 V supply_{p-p}, frequency 1 kHz, resistance 100 kOhm and capacitance 0.1 μF

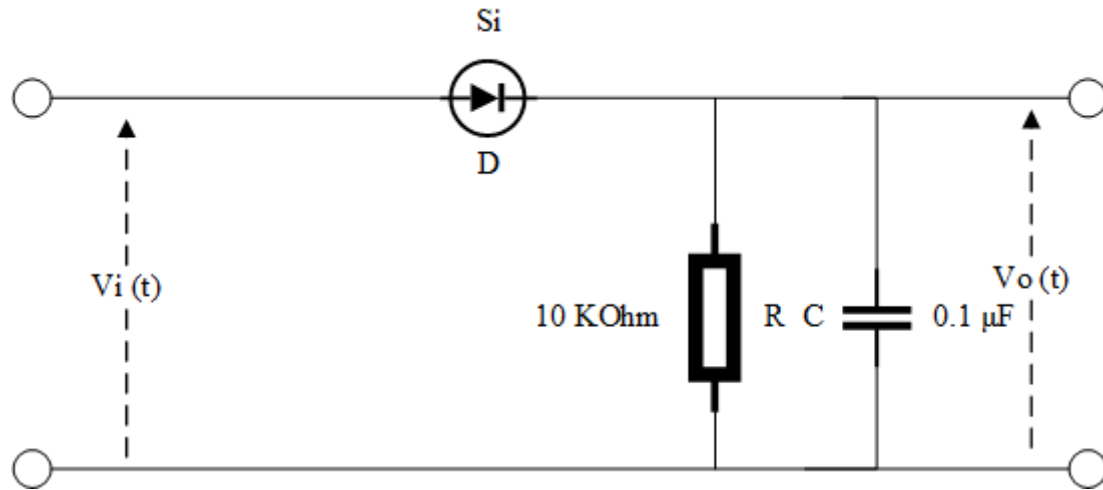


Figure 1.3.1 Schematic of the smoothing filter with a frequency of 1 kHz and a resistance of 10 kOhm

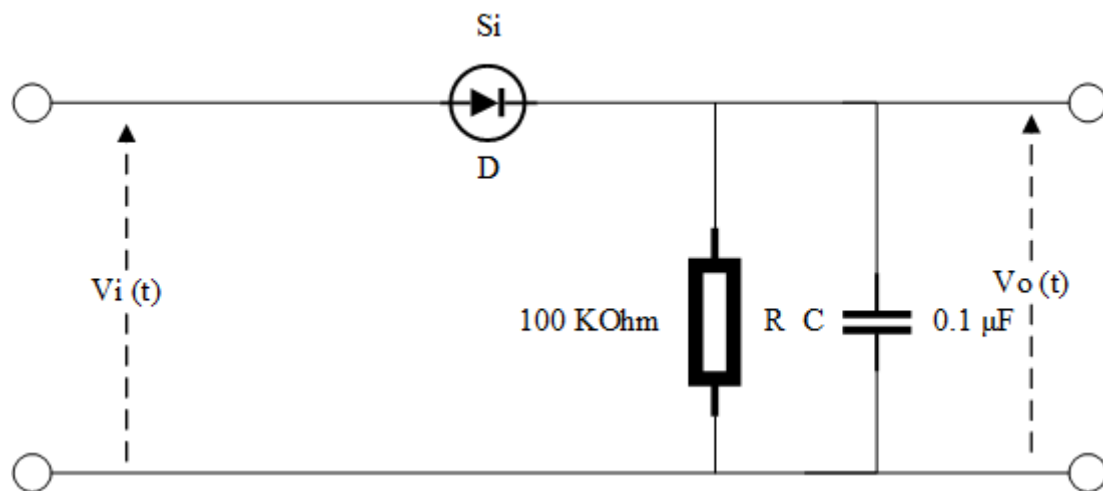


Figure 1.3.2 Schematic of the smoothing filter with a frequency of 1 kHz and a resistance of 100 kOhm

During the positive half-period the anode of the diode is positively charged and therefore the diode is open. The input signal will be a 3 V_p (6 V_{p-p}) sine wave signal, which is stored by the capacitor through the diode and charged. During the negative half period, the anode of the diode is negatively charged and the capacitor has not had time to discharge and thus the diode does not turn on and with the capacitor discharged into the 10 kOhm resistor. Figure 1.3.3 shows the **input** and **output** waveforms for the 10 kOhm resistive smoothing filter, while Figure 1.3.4 shows the **input** and **output** waveforms for the 100 kOhm resistive smoothing filter.

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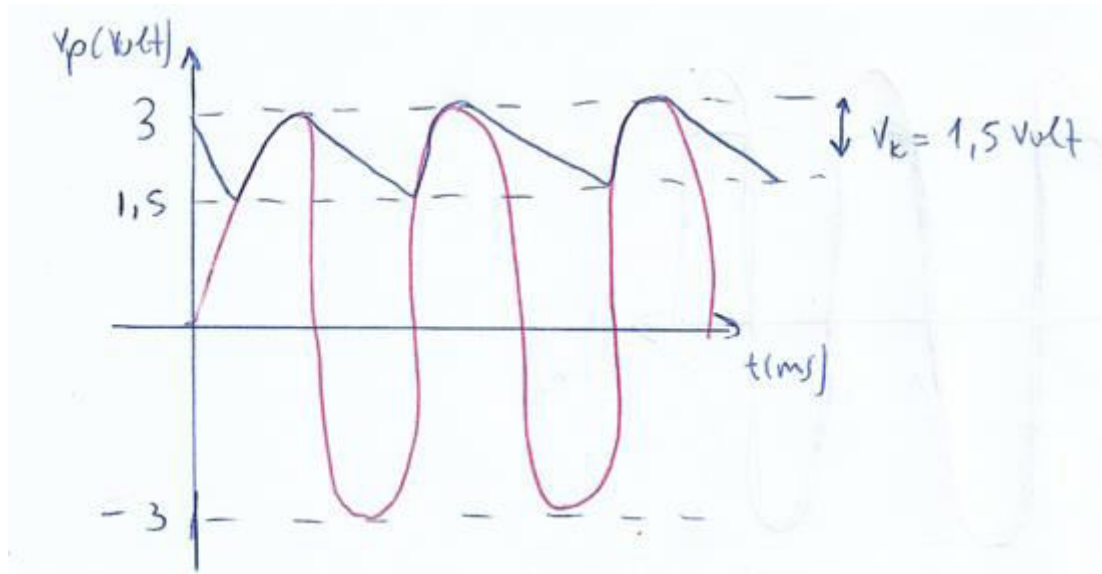


Figure 1.3.3 Waveforms of the smoothing filter with 10 kOhm resistance

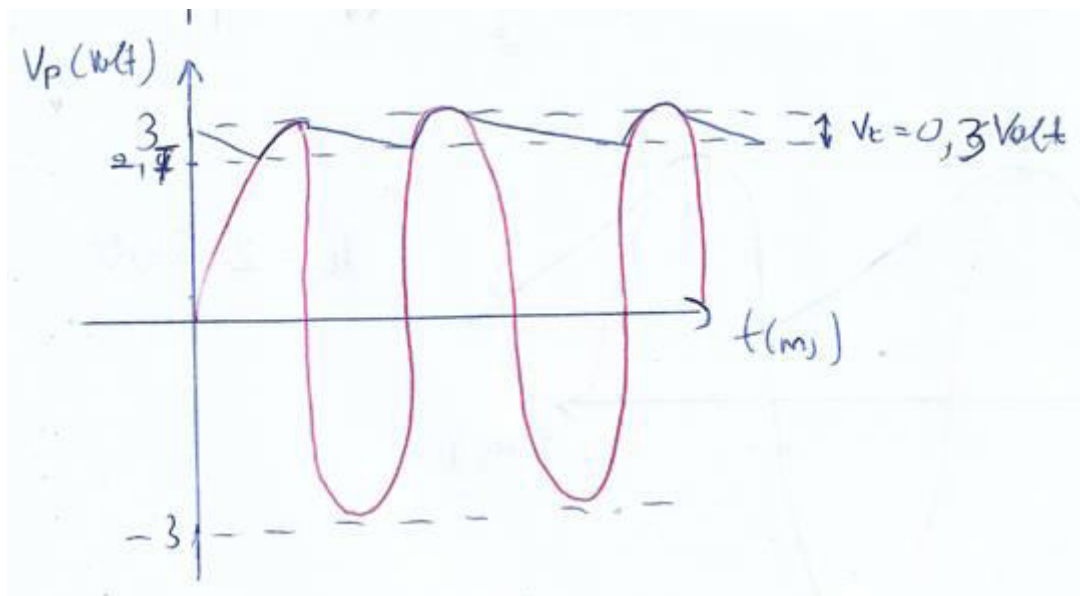


Figure 1.3.4 Waveforms of the smoothing filter with 100 kOhm resistance

The increase in resistance leads to an increase in the discharge time of the capacitor and, consequently, to a decrease in the ripple voltage which is equal to:

$$V_k = \frac{V_p}{fRC} \rightarrow V_k = \frac{1.5 \text{ V}}{10^3 \text{ Hz} \times 10 \times 10^3 \text{ Ohm} \times 0.1 \times 10^{-6} \text{ F}} \rightarrow V_k = 1.5 \text{ Volt}$$

The ripple voltage for 100 kOhm resistor is equal to :

$$V_k = \frac{V_p}{fRC} \rightarrow V_k = \frac{2.7 \text{ V}}{10^3 \text{ Hz} \times 100 \times 10^3 \text{ Ohm} \times 0.1 \times 10^{-6} \text{ F}} \rightarrow V_k = 0.3 \text{ Volt}$$

Simulative Resolution

Figure 1.3.5 shows the simulation of the smoothing filter with a power supply of 6 V_{p-p}, frequency 1 kHz, resistance 10 kOhm and capacitance 0.1 μF in the simulation software "Multisim".

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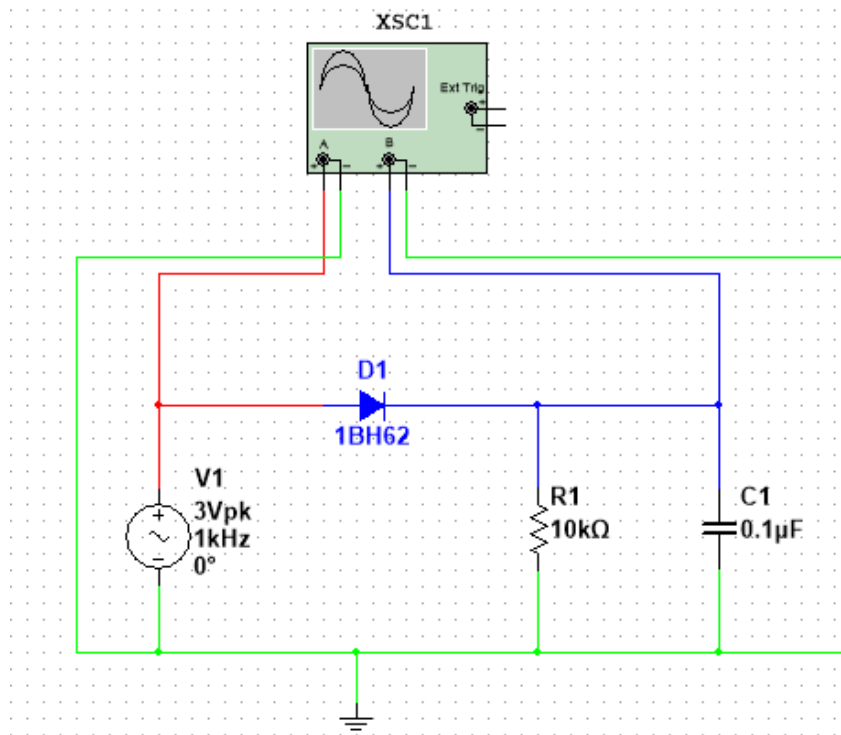


Figure 1.3.5 Simulation of the smoothing filter with 10 kOhm impedance in "Multisim"

In Figure 1.3.6 and 1.3.7, the ripple voltage V_k calculated in the Theoretical Solution chapter is simulated in the simulation software with the help of the oscilloscope.

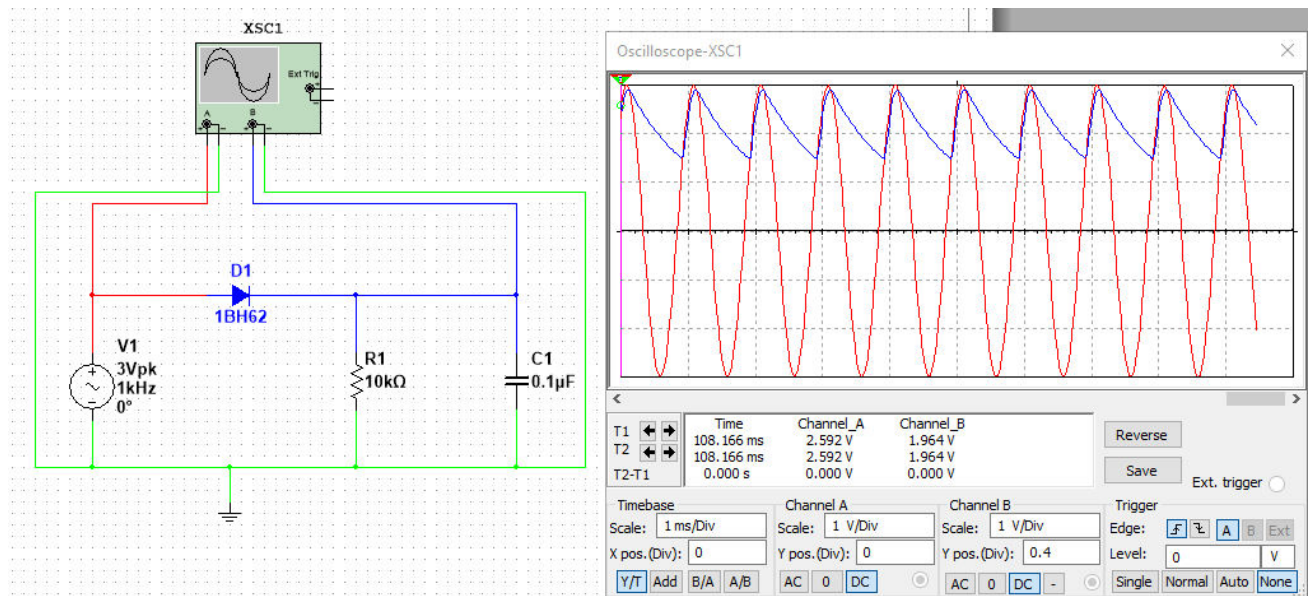


Figure 1.3.6 Waveforms of the smoothing filter with 10 kOhm resistance, in "Multisim"

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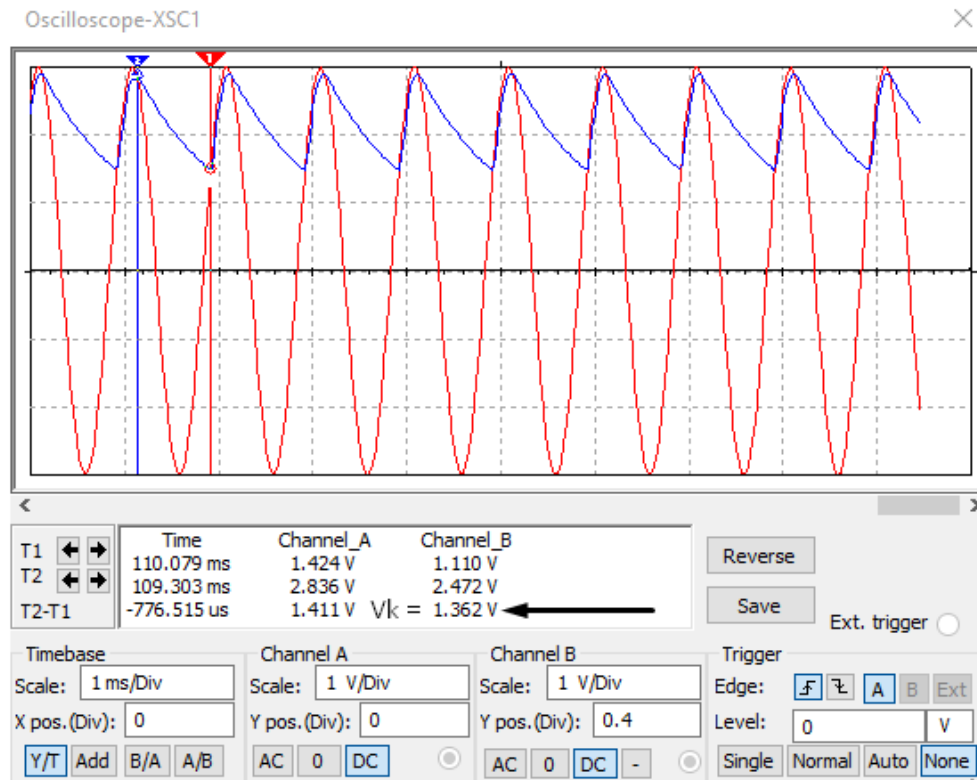


Figure 1.3.7 Calculation of V_k of the output signal in the smoothing filter with 10 kOhm resistor in "Multisim"

Figure 1.3.8 shows the simulation of the smoothing filter with 6 V supply_{p-p}, frequency 1 kHz, resistance 100 kOhm and capacitance 0.1 μF in the simulation software "Multisim".

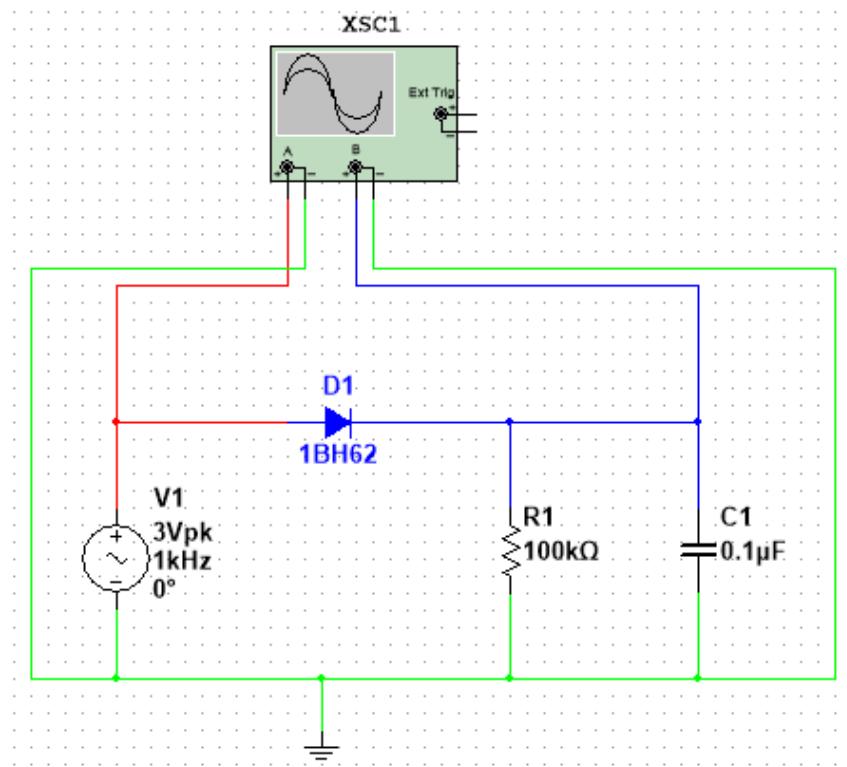


Figure 1.3.8 Simulation of the smoothing filter with 100 kOhm impedance in "Multisim"

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In Figure 1.3.9 and 1.3.10, the ripple voltage V_k calculated in the Theoretical Solution chapter is simulated in the simulation software with the help of the oscilloscope.

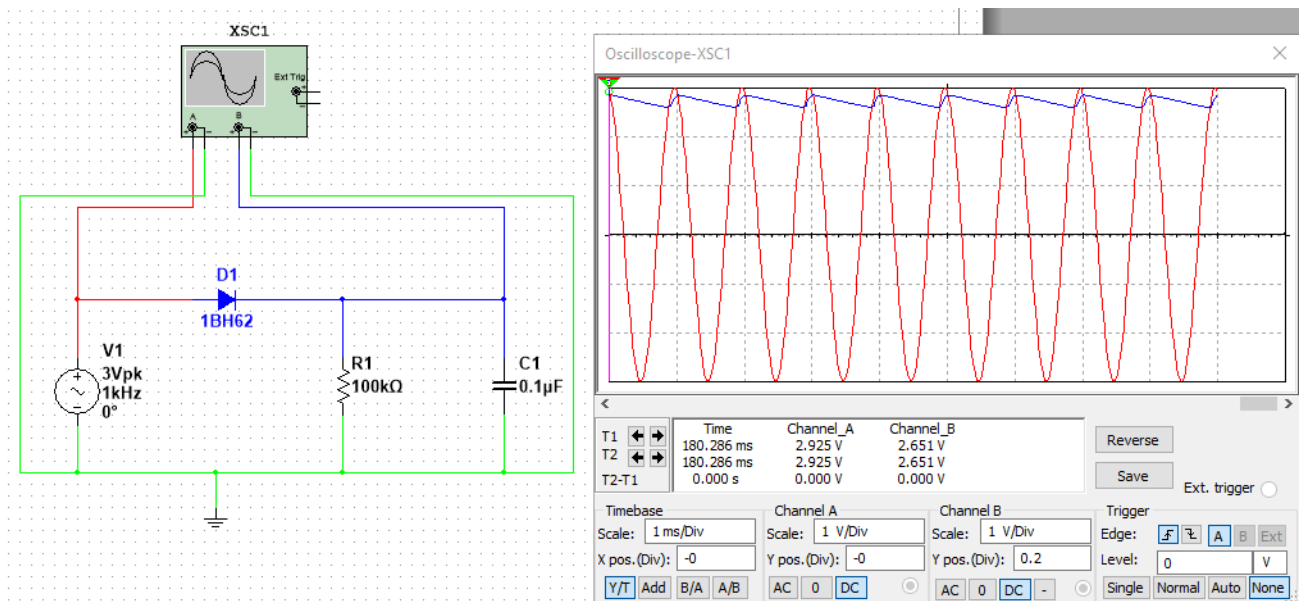


Figure 1.3.8 Waveforms of the smoothing filter with 100 kOhm resistance, in "Multisim"

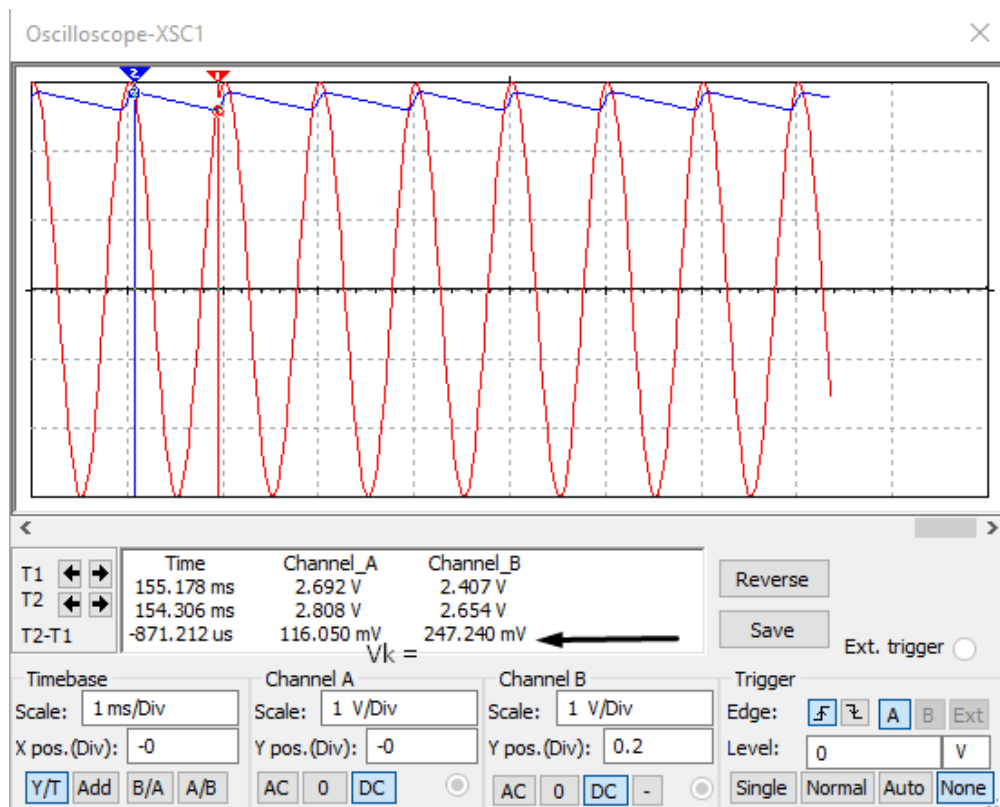


Figure 1.3.10 Calculation of V_k of the output signal in the smoothing filter with 100 kOhm resistance in "Multisim"

Experimental Solution

Figure 1.3.11 shows the circuit of the smoothing filter with a power supply of 6 V_{p-p}, frequency 1 kHz, resistance 10 kOhm and capacitance 0.1 μF on the "breadboard", where it was carried out in the laboratory.

The construction steps of the experiment are as follows:

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1. We set the input frequency to 1000 Hz (Figure 1.3.12).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 1.3.13).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 1.3.13).
4. We take channel A of the oscilloscope and connect the terminals to display the input waveform on the screen (Figure 1.3.13).
5. We set the switch to 1 Volts/Div (Figure 1.3.14).
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 6 V_{p-p} (Figure 1.3.14).
7. We align the oscilloscope so that the waveform is correctly positioned on the axes (Figure 1.3.14).
8. We continue assembling the circuit by taking the silicon diode (Figure 1.3.15) and connecting it in series with the source.
9. We take the 10 kOhm resistor (Figure 1.3.16) and connect it in series with the diode.
10. Connect the 0.1 μ F capacitor in parallel with the resistor (Figure 1.3.17).
11. We also take channel B of the oscilloscope and connect it in parallel with the terminals of the capacitor to get the output signal (Figure 1.3.18).

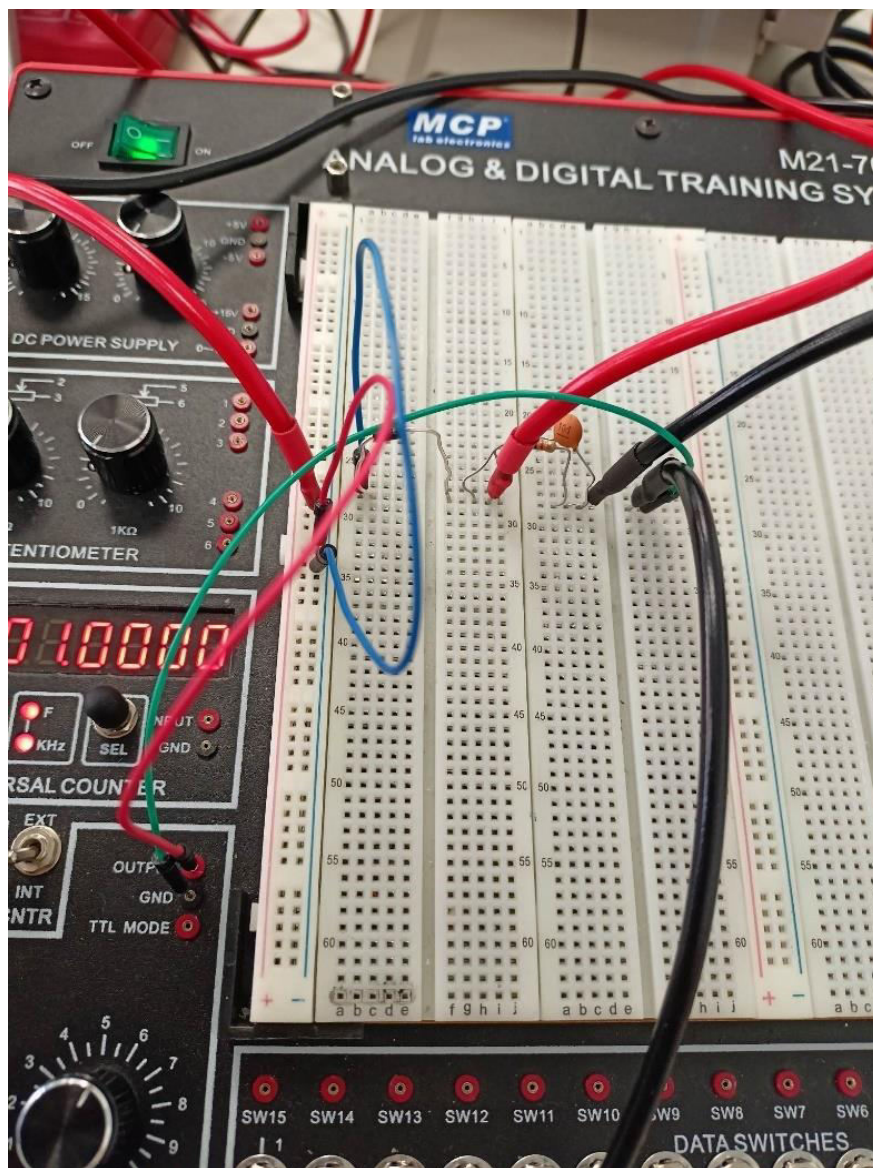


Figure 1.3.11 Experimental illustration of the smoothing filter with 10 kOhm resistance on the "breadboard"

ELECTRONICS



Figure 1.3.12 Step 1

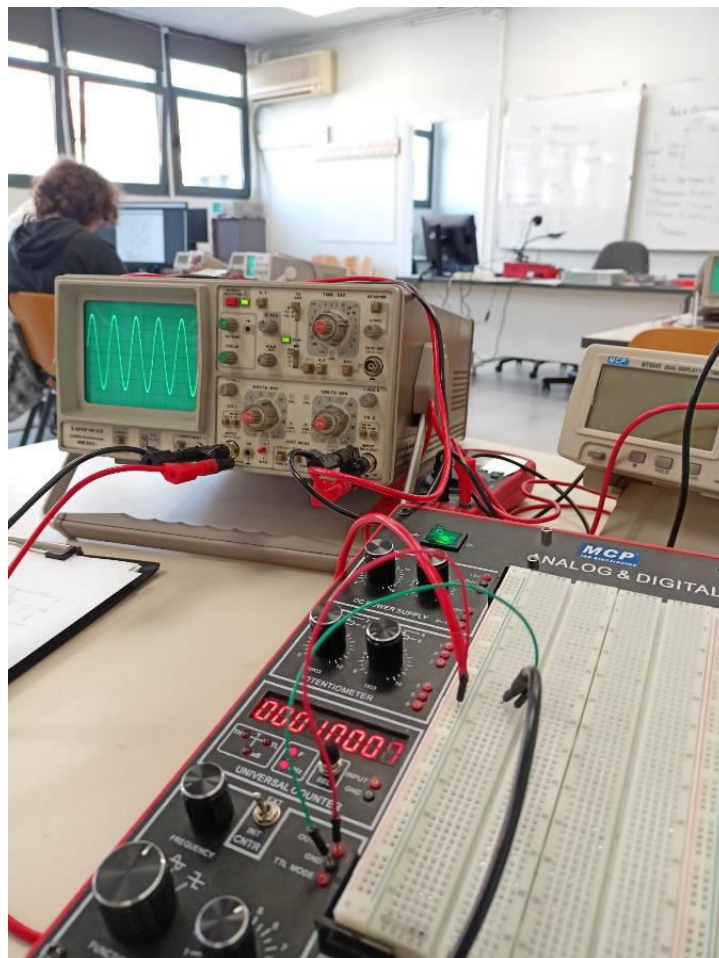


Figure 1.3.13 Step 2, 3, 4

ELECTRONICS

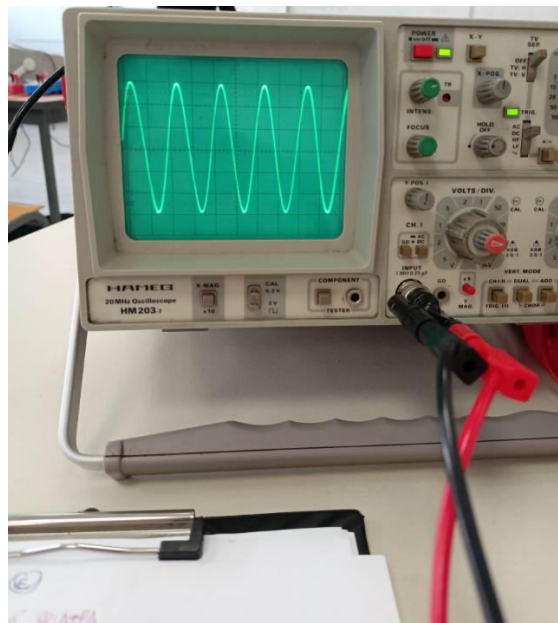


Figure 1.3.14 Step 5, 6, 7

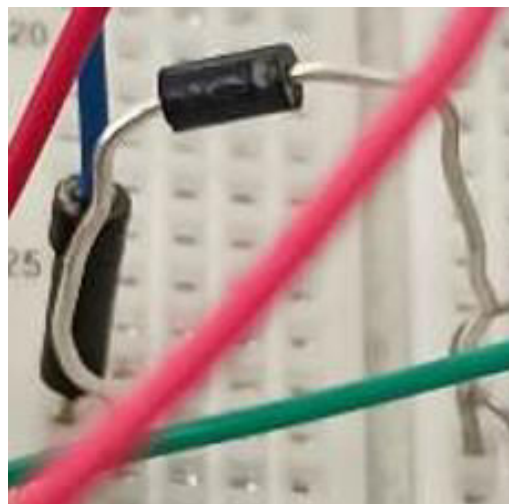


Figure 1.3.15 Step 8



Figure 1.3.16 Step 9

ELECTRONICS

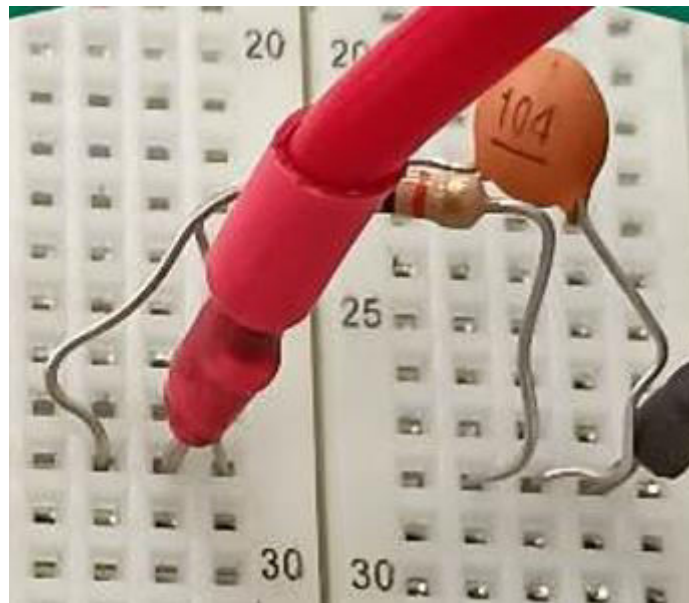


Figure 1.3.17 Step 10

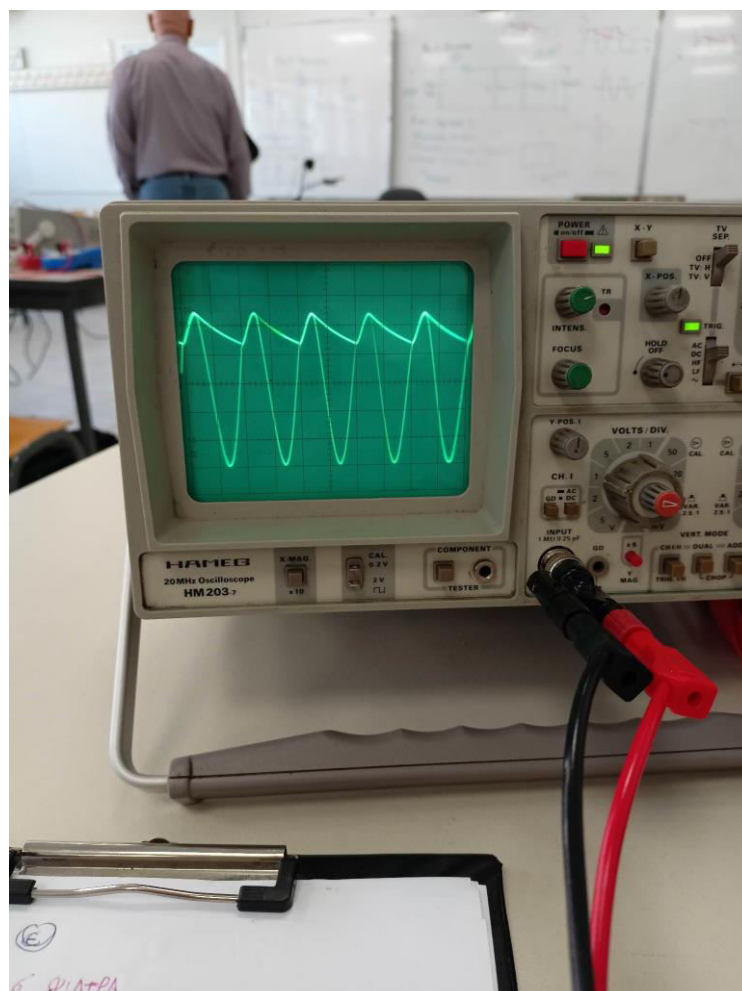


Figure 1.3.18 Step 11

ELECTRONICS

1.4 SMOOTHING FILTER WITH DIFFERENT CAPACITIES

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 1.4.1 shows the schematic of the smoothing filter with a 6 V supply_{p-p}, frequency 1 kHz, resistance 4.7 kOhm and capacitance 0.2 μF .

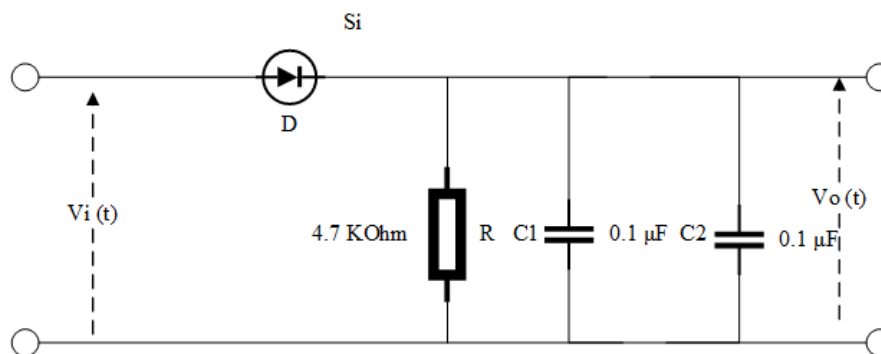


Figure 1.4.1 Schematic of the smoothing filter with a frequency of 1 kHz and a capacitance of 0.2 μF

The equivalent capacitance of two capacitors connected in parallel is equal to the sum of their capacitances, i.e., $C = C_1 + C_2 \rightarrow C = 0.1 + 0.1 \rightarrow C = 0.2 \mu\text{F}$. During the positive half-period the anode of the diode is positively charged and therefore the diode is turned off. The input signal will be a 3 V_p (6 V_{p-p}) sine wave signal, which is stored by the capacitor through the diode and charged. During the negative half period, the anode of the diode is negatively charged and the capacitor has not had time to discharge and thus the diode does not turn on and with the capacitor discharged into the 10 kOhm resistor. Figure 1.4.2 shows the **input** and **output** waveforms.

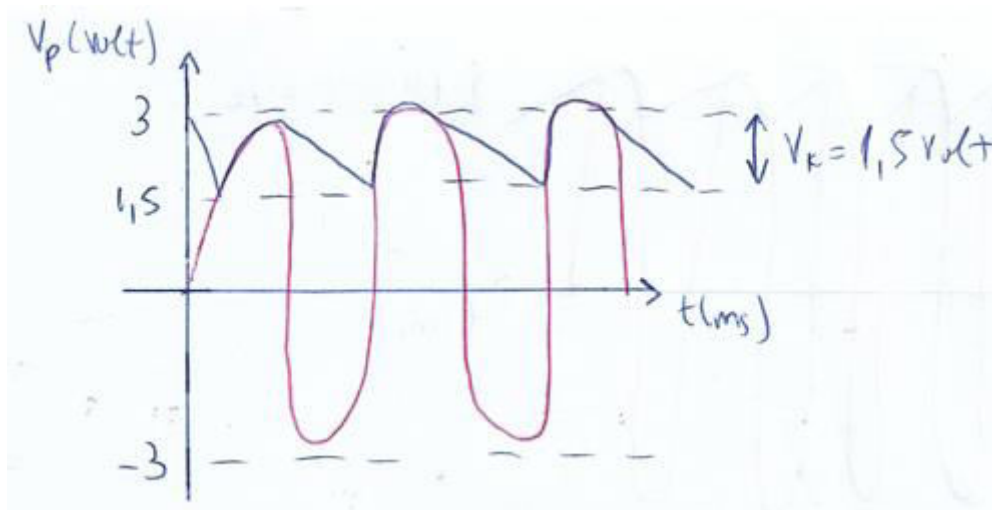


Figure 1.4.2 Waveforms of the smoothing filter with a capacitance of 0.2 μF

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The increase of the capacitance leads to the increase of the discharge time of the capacitor and consequently, to the decrease of the ripple voltage which is equal to:

$$V_k = \frac{V_p}{fRC} \rightarrow V_k = \frac{1.5 \text{ V}}{10^3 \text{ Hz} \times 4.7 \times 10^3 \text{ Ohm} \times 0.2 \times 10^{-6} \text{ F}} \rightarrow V_k = 1.5 \text{ Volt}$$

Simulative Resolution

Figure 1.4.3 shows the simulation of the smoothing filter with 6 V supply_{p-p}, frequency 1 kHz, resistance 4.7 kOhm and capacitance 0.2 μF in the simulation software "Multisim".

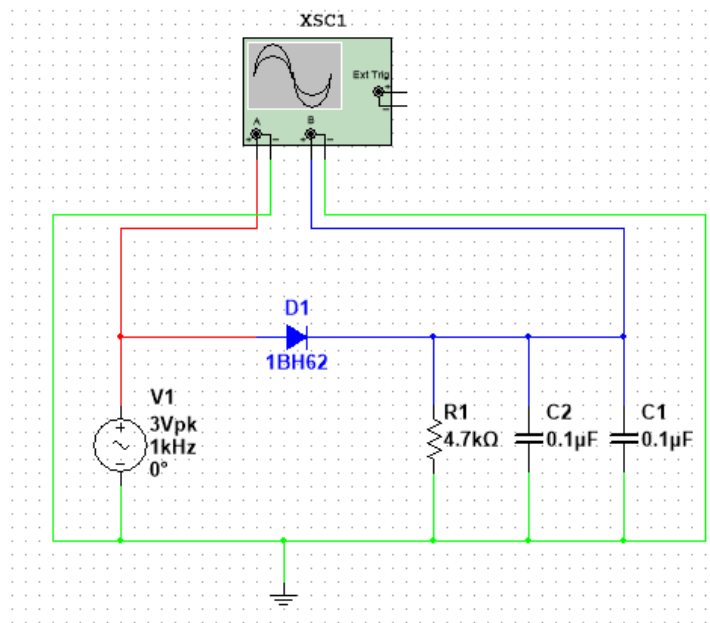


Figure 1.4.3 Simulation of the smoothing filter with a capacitance of 0.2 μF in "Multisim"

In Figure 1.4.4 and 1.4.5, the ripple voltage V_k calculated in the Theoretical Solution chapter is simulated in the simulation software with the help of the oscilloscope.

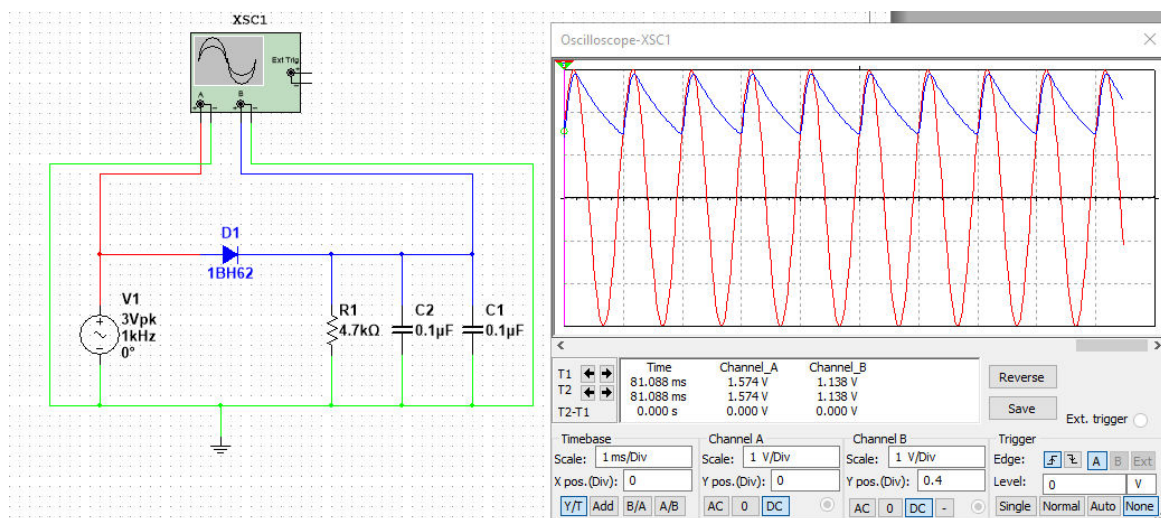


Figure 1.4.4 Waveforms of the smoothing filter with a capacitance of 0.2 μF, in "Multisim"

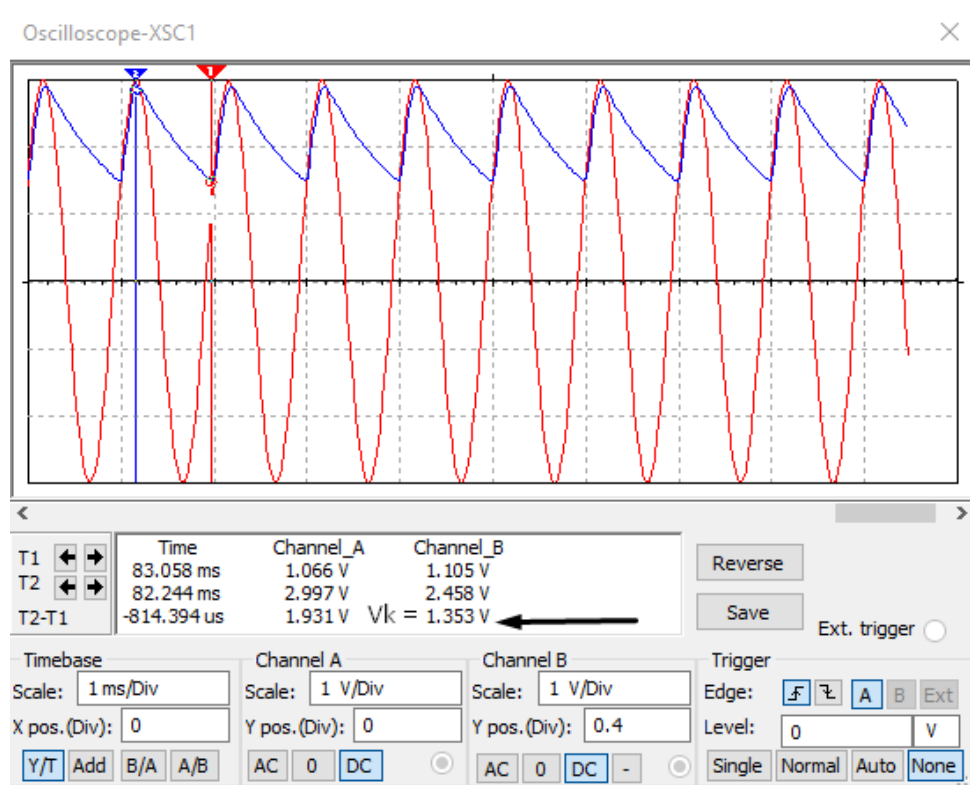


Figure 1.4.5 Calculation of V_k of the output signal in the smoothing filter with 10 kOhm resistor in "Multisim"

Experimental Solution

Figure 1.4.6 shows the circuit of the smoothing filter with a power supply of 6 V_{p-p}, frequency 1 kHz, resistance 4.7 kOhm and capacitance 0.2 μF on the "breadboard", where it was carried out in the laboratory.

The construction steps of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 1.4.7).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 1.4.8).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 1.4.8).
4. We take channel A of the oscilloscope and connect the terminals to display the input waveform on the screen (Figure 1.4.8).
5. We set the switch to 1 Volts/Div (Figure 1.4.9).
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 6 V_{p-p} (Figure 1.4.9).
7. We align the oscilloscope so that the waveform is correctly positioned on the axes (Figure 1.4.9).
8. We continue assembling the circuit by taking the silicon diode (Figure 1.4.10) and connecting it in series with the source.
9. We take the 4.7 kOhm resistor (Figure 1.4.11) and connect it in series with the diode.
10. We connect the 0.1 μF capacitor in parallel with the resistor, as well as another 0.1 μF capacitor in parallel, so that the isodyno capacitance is 0.2 μF, i.e., the sum of the capacitances of the two capacitors (Figure 1.4.12).
11. We also take channel B of the oscilloscope and connect it in parallel with the capacitor terminals to get the output signal (Figure 1.4.13).

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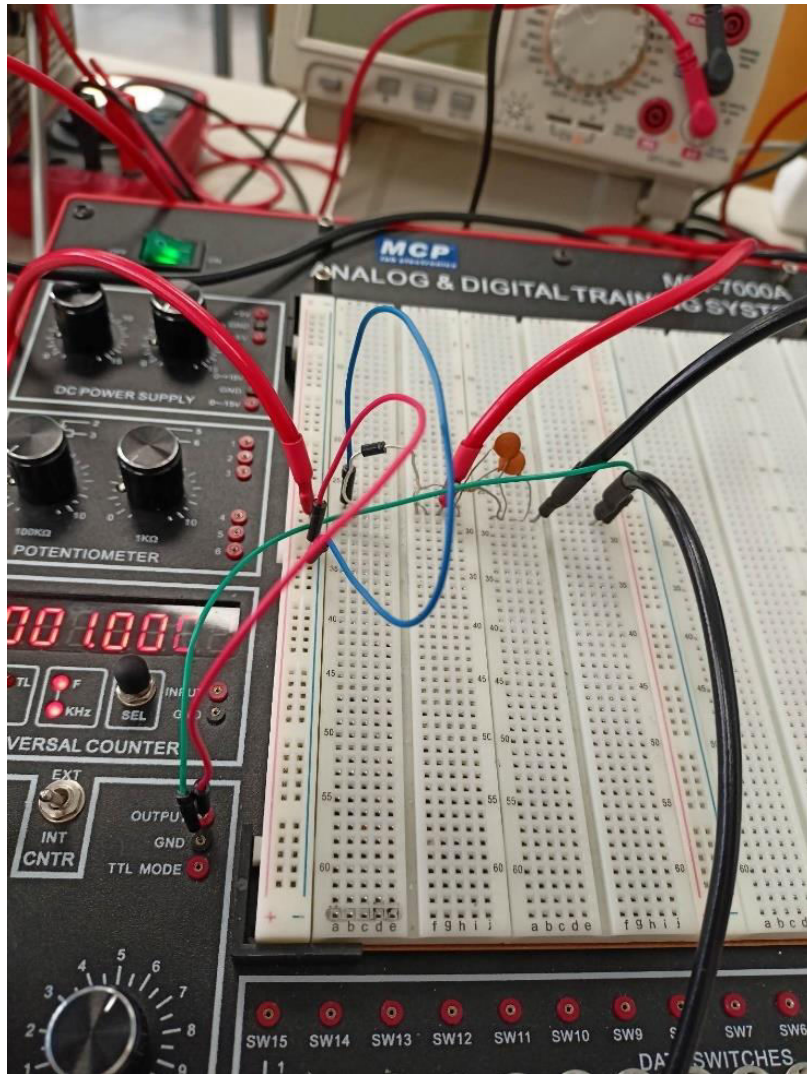


Figure 1.4.6 Experimental illustration of the smoothing filter with a capacitance of $0.2 \mu\text{F}$ on the "breadboard"



Figure 1.4.7 Step 1

ELECTRONICS

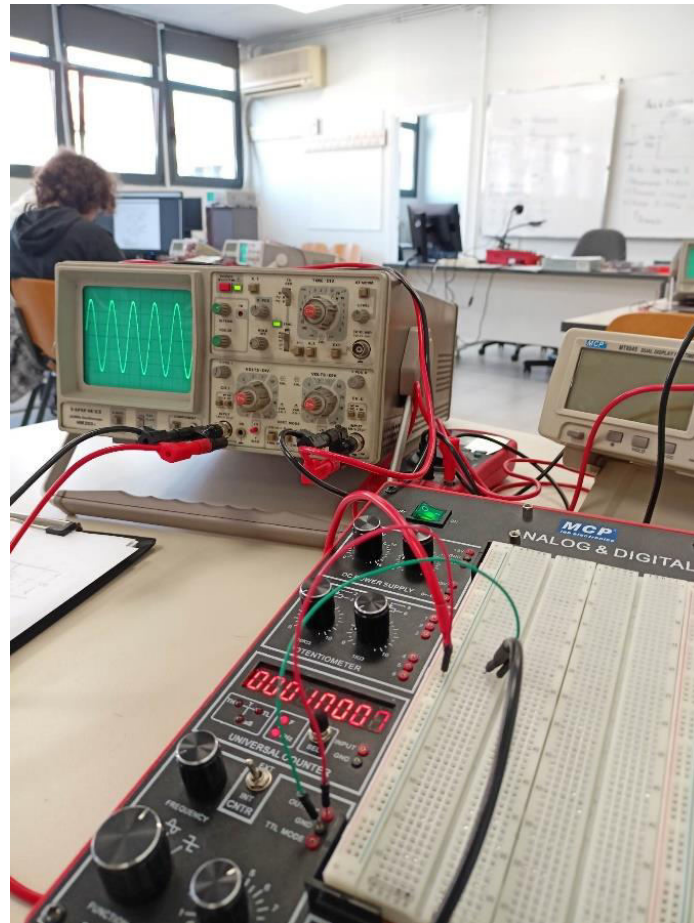


Figure 1.4.8 Step 2, 3, 4

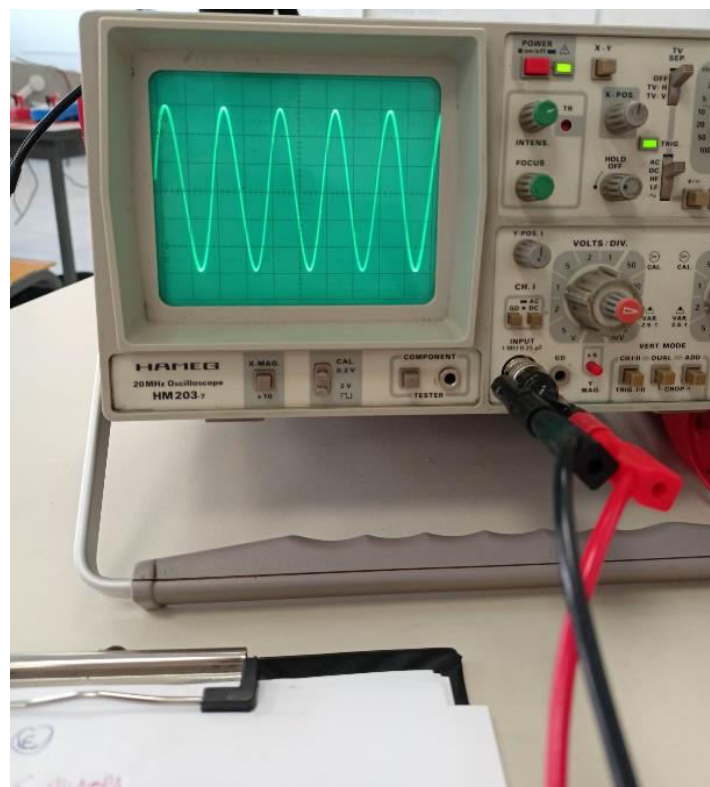


Figure 1.4.9 Step 5, 6, 7

ELECTRONICS



Figure 1.4.10 Step 8



Figure 1.4.11 Step 9

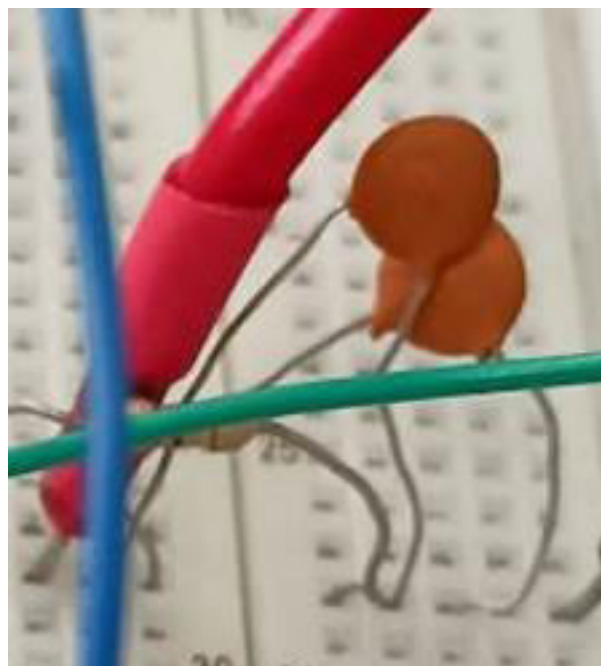


Figure 1.4.12 Step 10

ELECTRONICS

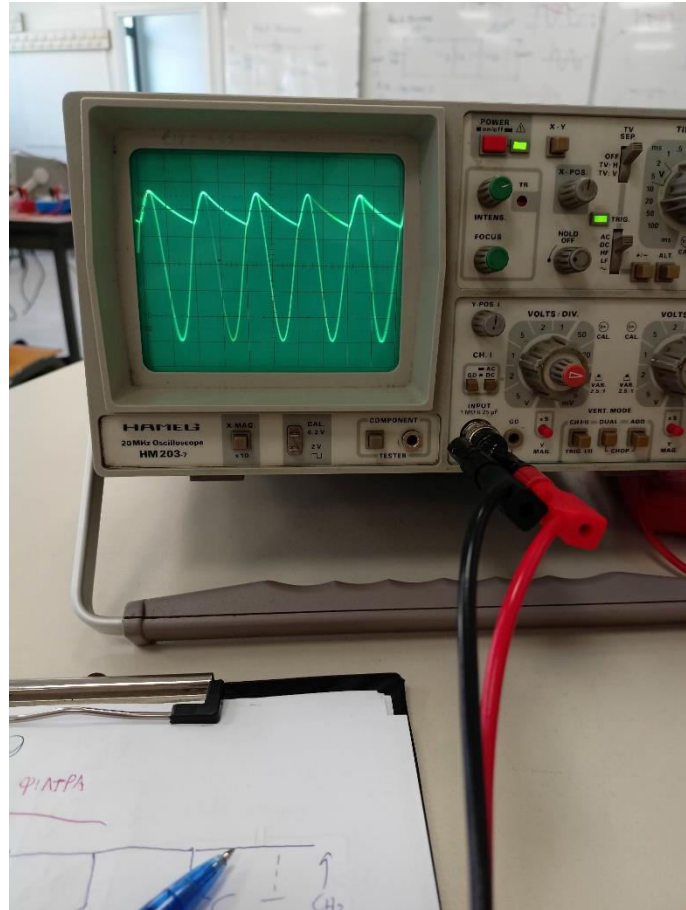


Figure 1.4.13 Step 11

1.5 QUESTIONS

Question 1.5.1

In the circuit in Figure 2a (Figure 1.1.1), what changes do the variations in input frequency bring about and why?

Variations in the input frequency cause changes in the charging and discharging time of the capacitor and, consequently, in the ripple voltage. More specifically, with increasing input frequency the capacitor charging time and hence the ripple voltage decreases, while with decreasing frequency the charging time and the ripple voltage increases.

Question 1.5.2

What does any fluctuation of the capacitor cause and why?

As the capacitance of the capacitor increases, the discharge time of the capacitor increases and therefore the ripple voltage decreases $V_k = \frac{V_p}{fRC}$. Conversely, as the capacitance of the capacitor decreases, the discharge time of the capacitor decreases and therefore the ripple voltage increases.

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Question 1.5.4

How do $R1$ and $R2$ in the circuit in Figure 3a (Figure 1.5.4.1) affect the output and why?

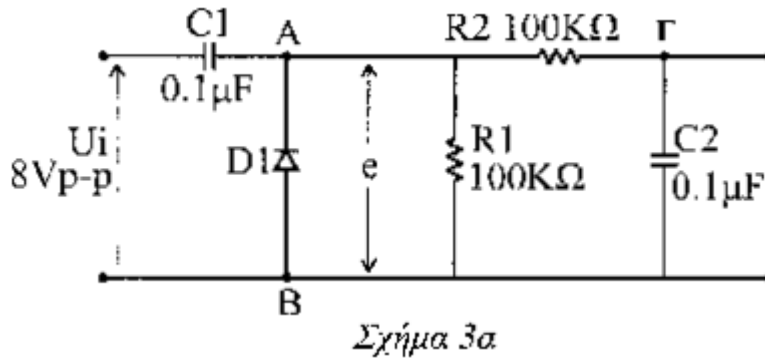


Figure 1.5.4.1 Figure 3a

$R1$ and $R2$ affect how the current is divided at the nodes and the current passing through $R2$ determines the charging time of capacitor $C2$. Therefore, as the resistance becomes larger the capacitor charging time decreases.

Question 1.5.5

Consider a simple rectifier with capacitor filter (see Figure 6, Figure 1.5.5.1) which has an input sinusoidal voltage of 100 Hz and a peak voltage of $V_p = 4$ V. The load $R = 10$ kΩ. Find the value of the capacitance C that will generate a peak-to-peak ripple of 0.2 V.

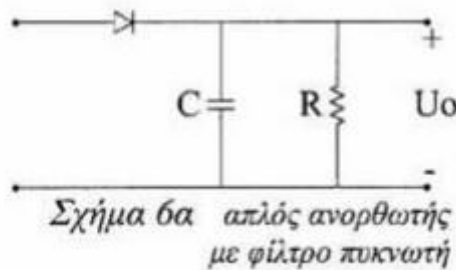


Figure 1.5.5.1 Figure 6a

$$V_k = \frac{V_p}{fRC} \rightarrow 0.2 \text{ V} = \frac{4 \text{ V}}{100 \text{ Hz} \times 10 \times 10^3 \text{ Ohm} \times C} \rightarrow C = \frac{4 \text{ V}}{100 \text{ Hz} \times 10 \times 10^3 \text{ Ohm} \times 0.2 \text{ V}} \rightarrow$$

$$C = 20 \mu\text{F}$$

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2. SCISSORS

2.1 SCISSOR CIRCUIT

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 2.1.1 shows the schematic of the scissor circuit with 8 V_{p-p} power supply, 1 kHz frequency, 4.7 kOhm resistor and 2 Volt DC voltage.

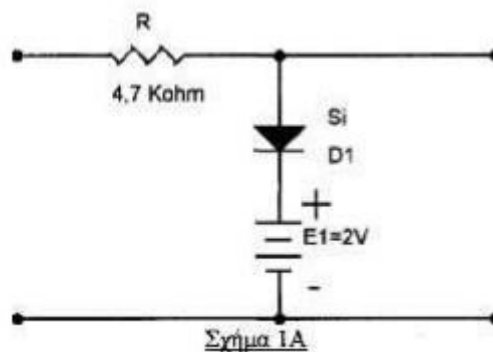


Figure 2.1.1 Schematic illustration of the clipper circuit with 2 Volt DC voltage

Because it is a silicon diode, where the blocking voltage is at 0.6 Volt, the diode does not turn on (reverse bias) until the input voltage reaches 2.6 Volt. This is because the cathode voltage is higher than the anode voltage due to the DC source voltage of 2 Volts, where it is higher than the input signal up to 2.6 Volts (the voltage drop caused by the 4.7 kOhm resistor R is also taken into account). The output will be equal to the input.

Therefore, when the input signal reaches 2.6 Volts, the anode voltage will become higher than the cathode voltage and therefore, the diode will turn off (right polarity) and the output will become equal to the DC source voltage of 2 Volts. In addition, there will be a potential difference of 0.6 volts between the anode and cathode. This occurs until the input voltage drops to 2.5 Volts, at which point the process repeats. Figure 2.1.2 shows the **input** and **output waveforms**.

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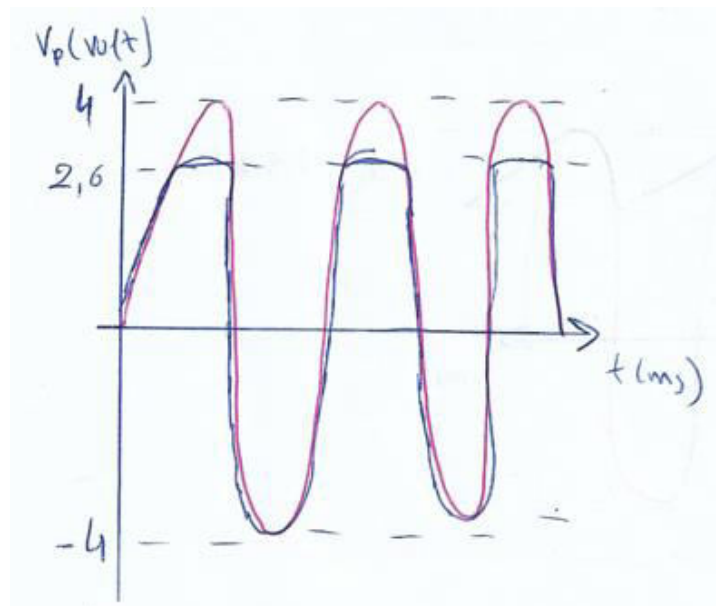


Figure 2.1.2 Waveforms of the scissor circuit with 2 Volt DC voltage

Simulative Resolution

Figure 2.1.3 shows the simulation of the scissor circuit with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kOhm and 2 Volt DC voltage in the simulation software "Multisim".

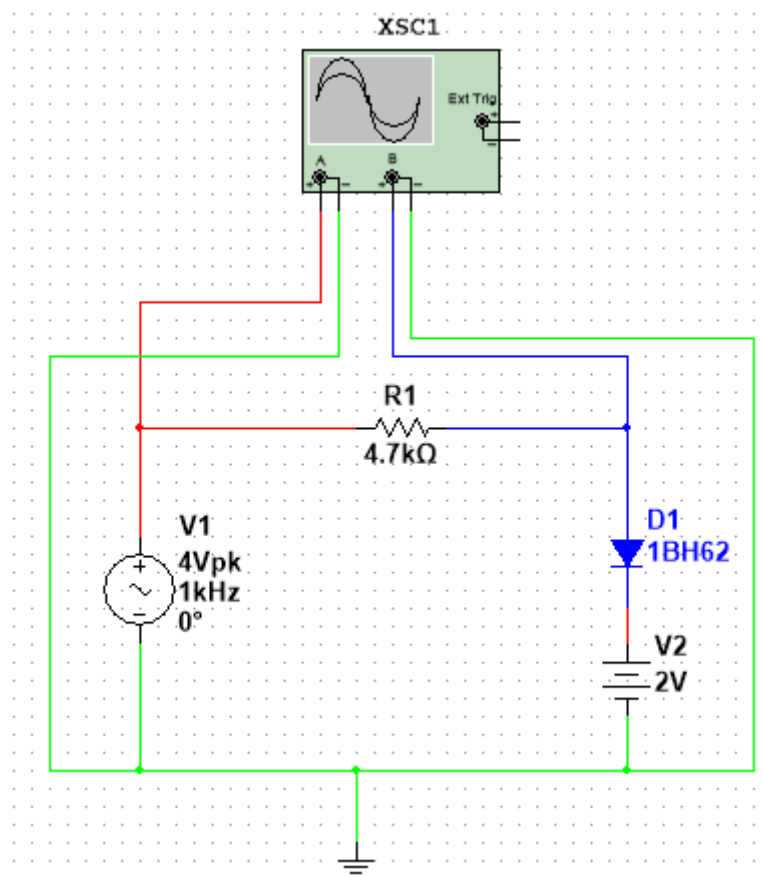


Figure 2.1.3 Simulation of the scissor circuit with 2 Volt DC voltage in "Multisim"

In Figure 2.1.4, the observations made in the Theoretical Solution chapter are verified simulatively in the simulation software with the help of the oscilloscope.

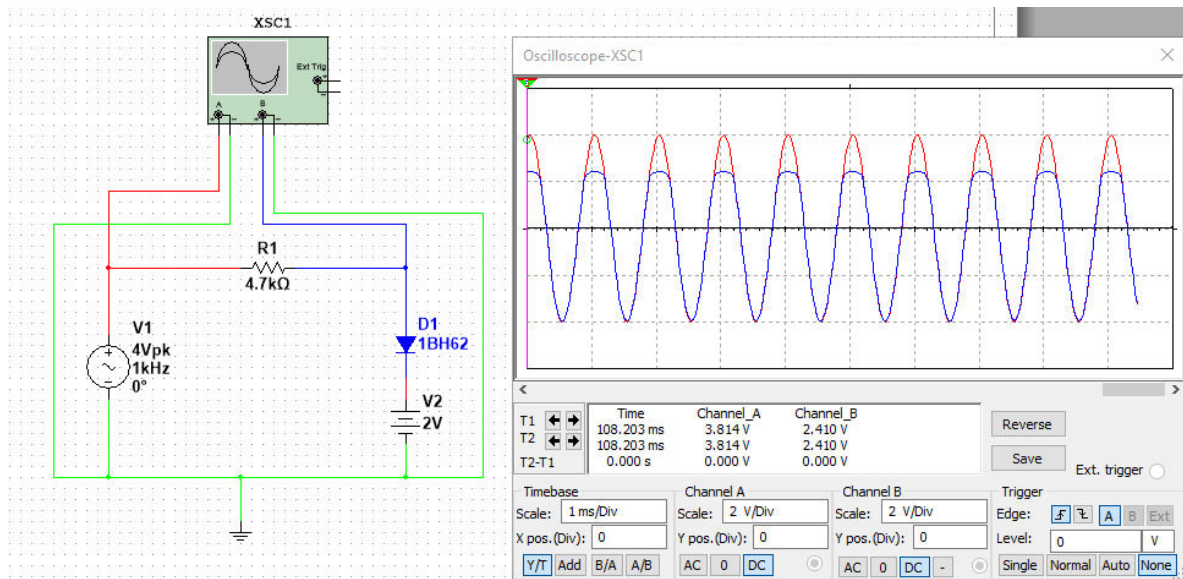


Figure 2.1.4 Waveforms of the scissor circuit with 2 Volt DC voltage, in "Multisim"

Experimental Solution

Figure 2.1.5 shows the scissor circuit with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kΩ and 2 Volt DC voltage on the "breadboard", where it was carried out in the laboratory.

The construction steps of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 2.1.6).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 2.1.7).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 2.1.7).
4. We take channel A of the oscilloscope and connect the terminals to display the input waveform on the screen (Figure 2.1.7).
5. We set the switch to 2 Volts/Div (Figure 2.1.8).
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 8 V_{p-p} (Figure 2.1.8).
7. Align from the oscilloscope the waveform to be correctly positioned on the axes (Figure 2.1.8).
8. We take the 4.7 kΩ resistor (Figure 2.1.9) and connect it in series with the AC source.
9. We continue assembling the circuit by taking the silicon diode (Figure 2.1.10) and connecting it in series with the resistor.
10. We take a cable and connect the DC voltage source 0+15 Volt to the circuit (Figure 2.1.11).
11. We connect to the ends of the DC source a digital bench multimeter configured to calculate DC voltage, so that we can set the DC source to 2 Volts (Figure 2.1.11).
12. We also take channel B of the oscilloscope and connect it in parallel with the anode of the diode and the negative pole of the DC voltage source to get the output signal (Figure 2.1.12).

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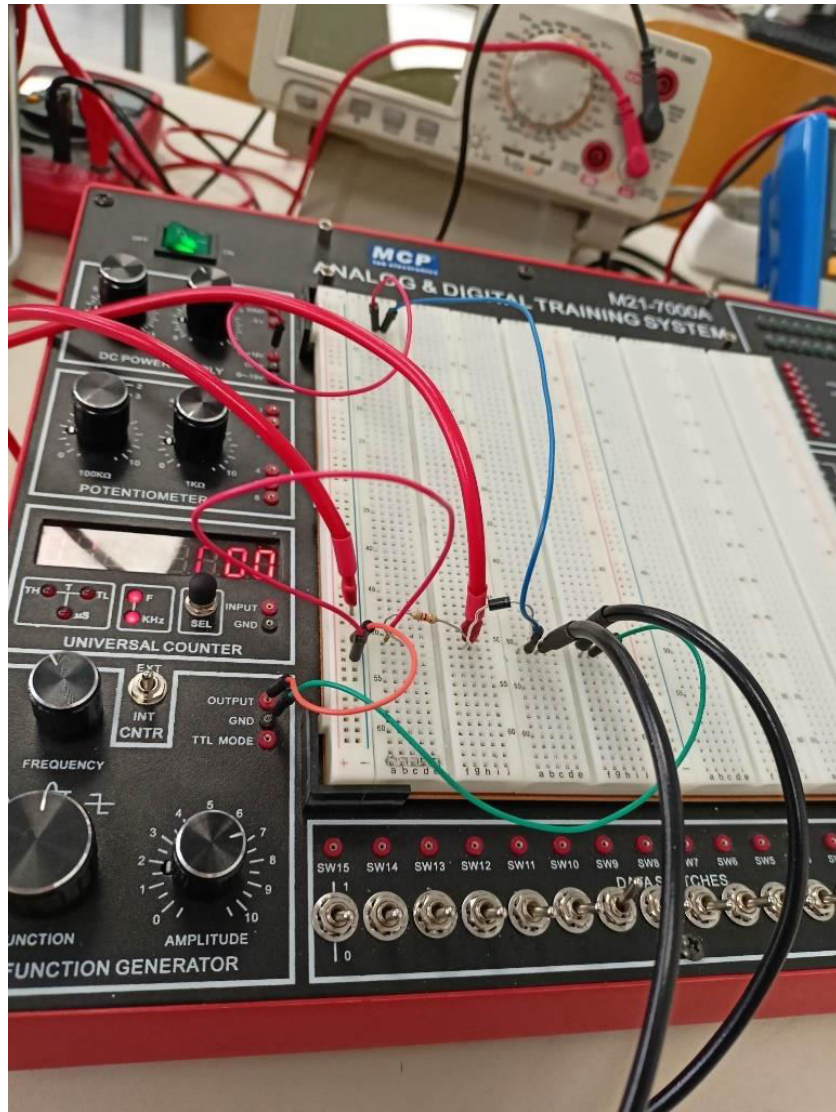


Figure 2.1.5 Experimental illustration of the scissor circuit on the breadboard



Figure 2.1.6 Step 1

ELECTRONICS

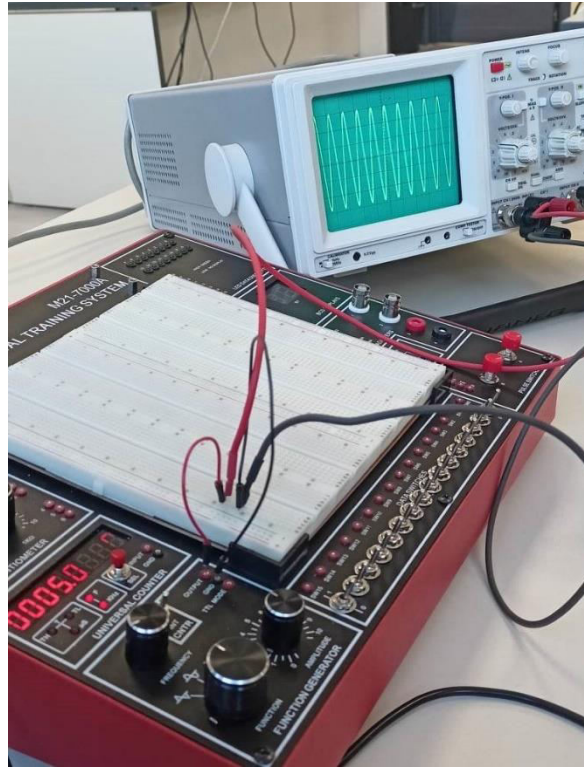


Figure 2.1.7 Step 2, 3, 4

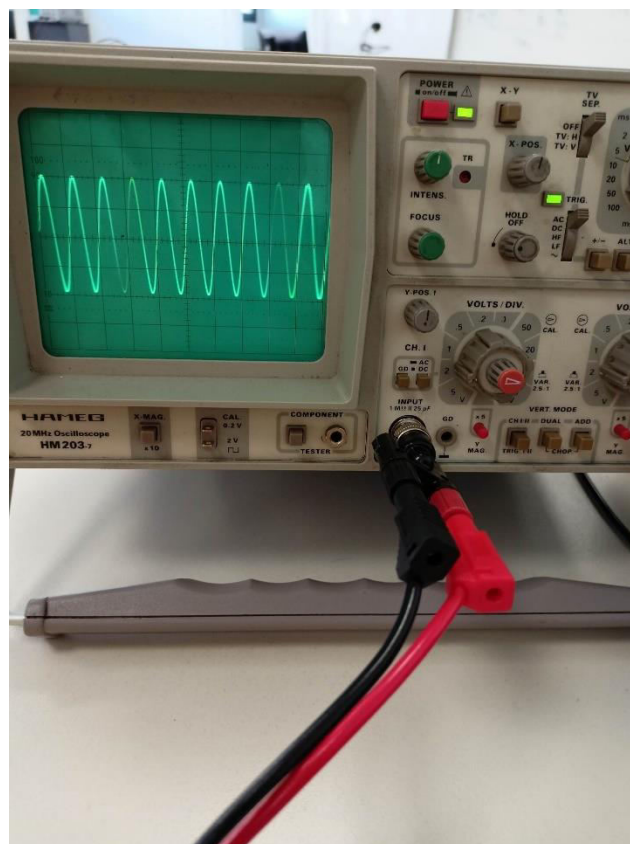


Figure 2.1.8 Step 5, 6, 7

ELECTRONICS

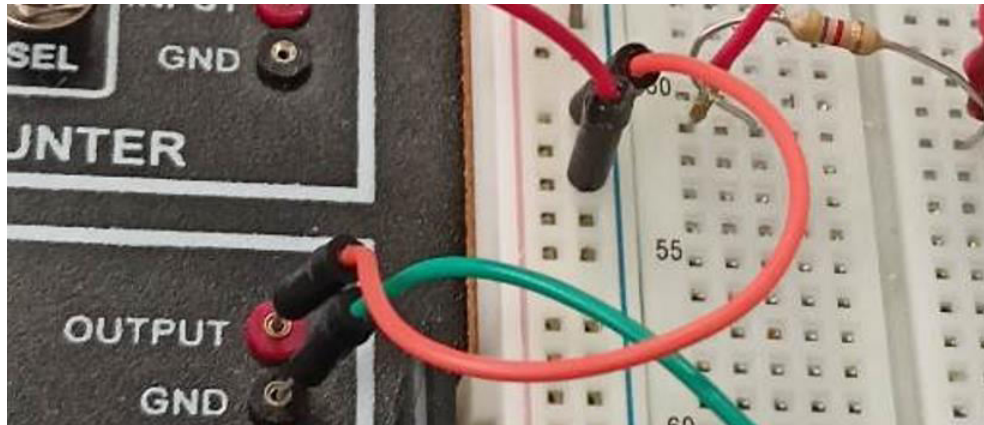


Figure 2.1.9 Step 8



Figure 2.1.10 Step 9

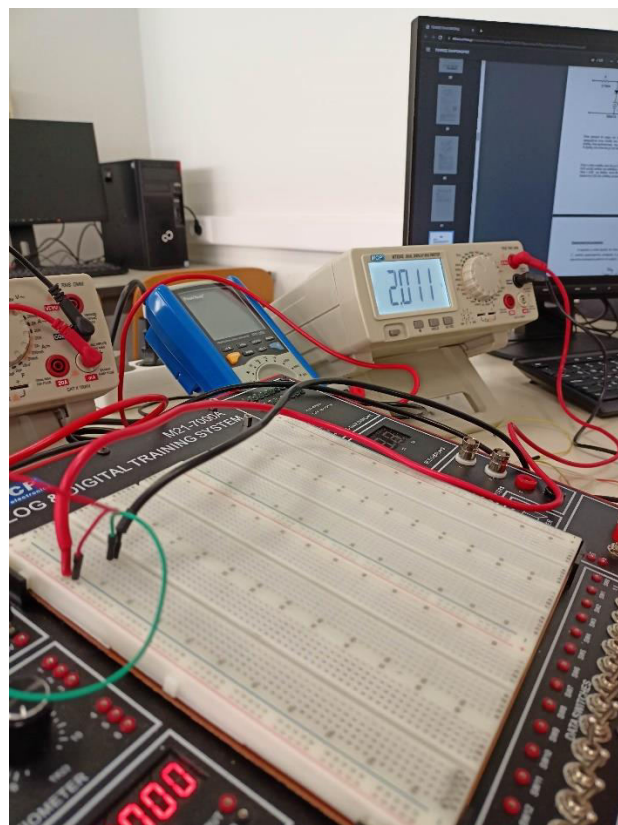


Figure 2.1.11 Step 10, 11

ELECTRONICS

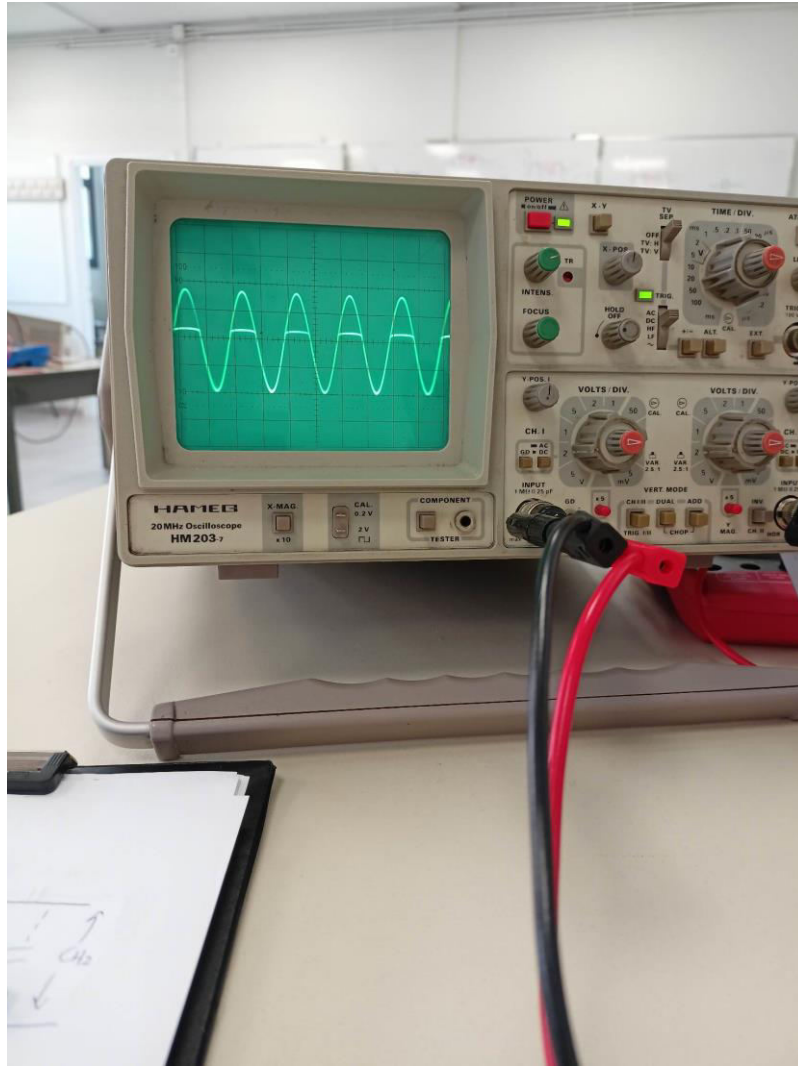


Figure 2.1.12 Step 12

2.2 SCISSORS WITH FLUCTUATIONS IN DC VOLTAGE

General

The chapter "Simulation Solution" presents in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim", while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 2.2.1 shows the schematic of the scissor circuit with 8 V_{p-p} power supply, 1 kHz frequency, 4.7 kOhm resistor and 2 Volt DC voltage.

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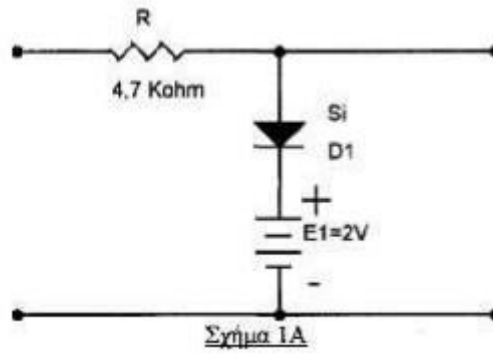


Figure 2.2.1 Schematic illustration of the scissor circuit

Fluctuations in the DC voltage of the source cause changes in the output voltage that are a direct consequence of the diode's conductivity. In other words, the scissor diode turns off for different input voltages and therefore the output voltage that the diode will produce is adjusted.

For example, with a DC voltage of -2 volts, the diode does not turn on (reverse bias) until the input voltage becomes equal to -1.4 volts so that the anode voltage becomes greater than the cathode voltage. Until this voltage is reached, the output follows the input. When the voltage reaches -1.4 Volts, there will be a potential difference of 0.6 Volts between the anode and cathode and the diode will turn off (right polarity). This occurs until the input voltage drops to -1.3 Volts, at which point the process repeats.

For example, with a DC voltage of 2.5 Volts, the diode does not turn on (reverse polarization) until the input voltage becomes equal to 2.8 Volts, so that the anode voltage becomes greater than the cathode voltage. Until this voltage is reached, the output follows the input. When the voltage reaches 2.8 Volts, there will be a potential difference of 0.6 Volts between the anode and cathode and the diode will turn off (right polarity). This occurs until the input voltage drops to 2.7 Volts, at which point the process repeats.

Figure 2.2.2 shows the **input** and **output**

waveforms for the scissor circuit with -2 volts DC voltage, while Figure 2.2.3 shows the **input** and **output** waveforms for the scissor circuit with 2.5 volts DC voltage.

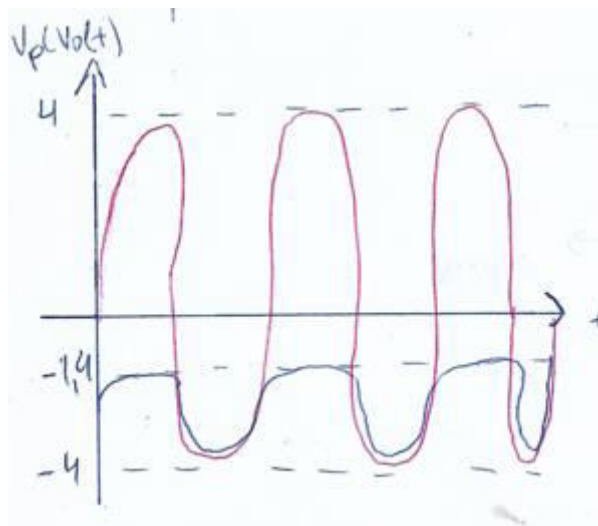


Figure 2.2.2.2 Waveforms of the scissor circuit with -2 Volt DC voltage

ELECTRONICS

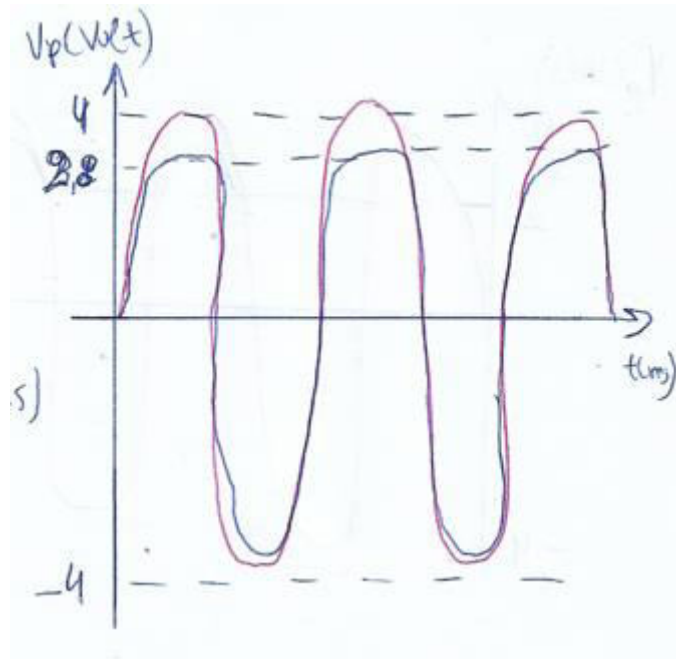


Figure 2.2.3 Waveforms of the scissor circuit with 2.5 Volt DC voltage

Simulative Resolution

Figure 2.2.4 shows the simulation of the scissor circuit with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kOhm and -2 Volt DC voltage in the simulation software "Multisim".

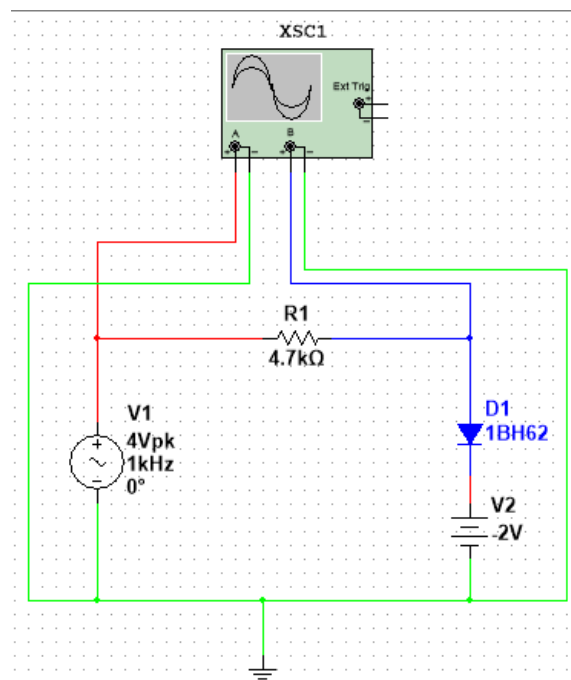


Figure 2.2.4 Simulation of the scissor circuit with DC voltage reduction, in "Multisim"

In Figure 2.2.5, the observations made in the Theoretical Solution chapter are verified simulatively in the simulation software with the help of the oscilloscope.

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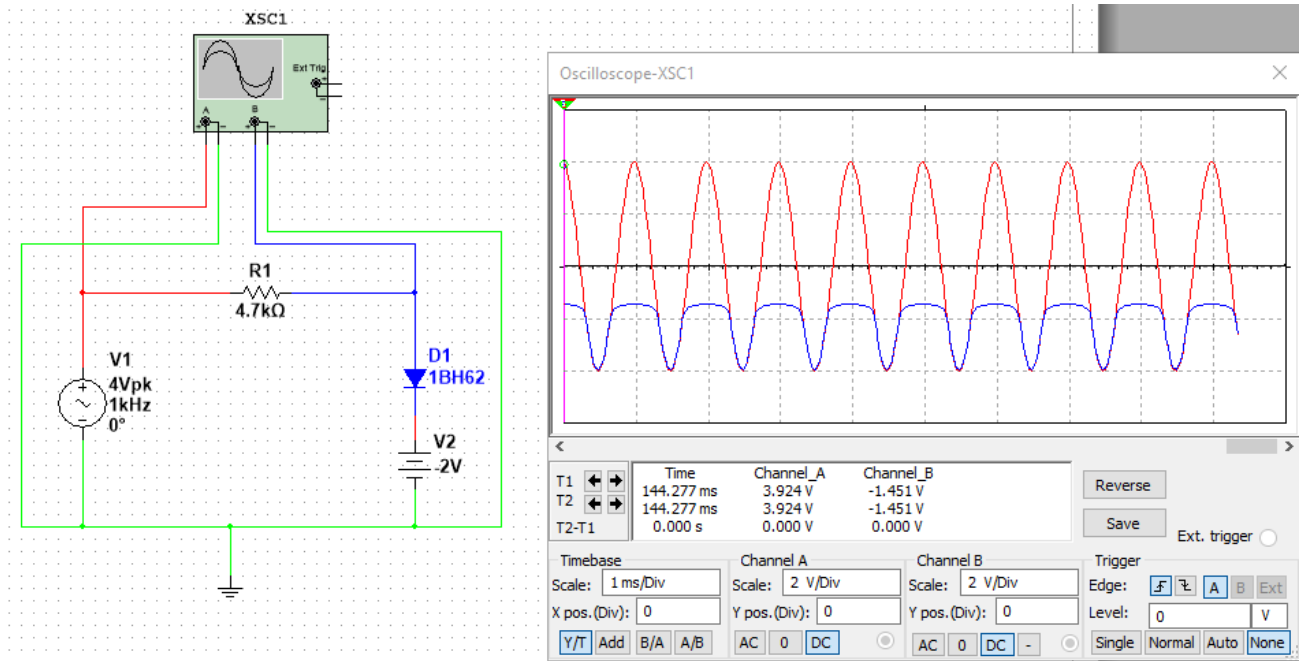


Figure 2.2.5 Waveforms of the scissor circuit with -2 Volt DC voltage, in "Multisim"

Figure 2.2.6 shows the simulation of the scissor circuit with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kΩ and 2.5 Volt DC voltage in the simulation software "Multisim".

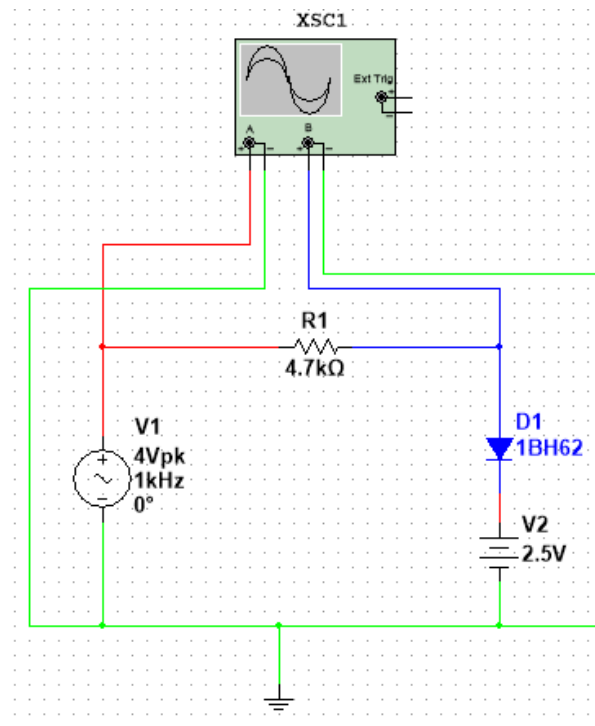


Figure 2.2.2.6 Simulation of the scissor circuit with increasing DC voltage, in "Multisim"

In Figure 2.2.7, the observations made in the Theoretical Solution chapter are verified simulatively in the simulation software with the help of the oscilloscope.

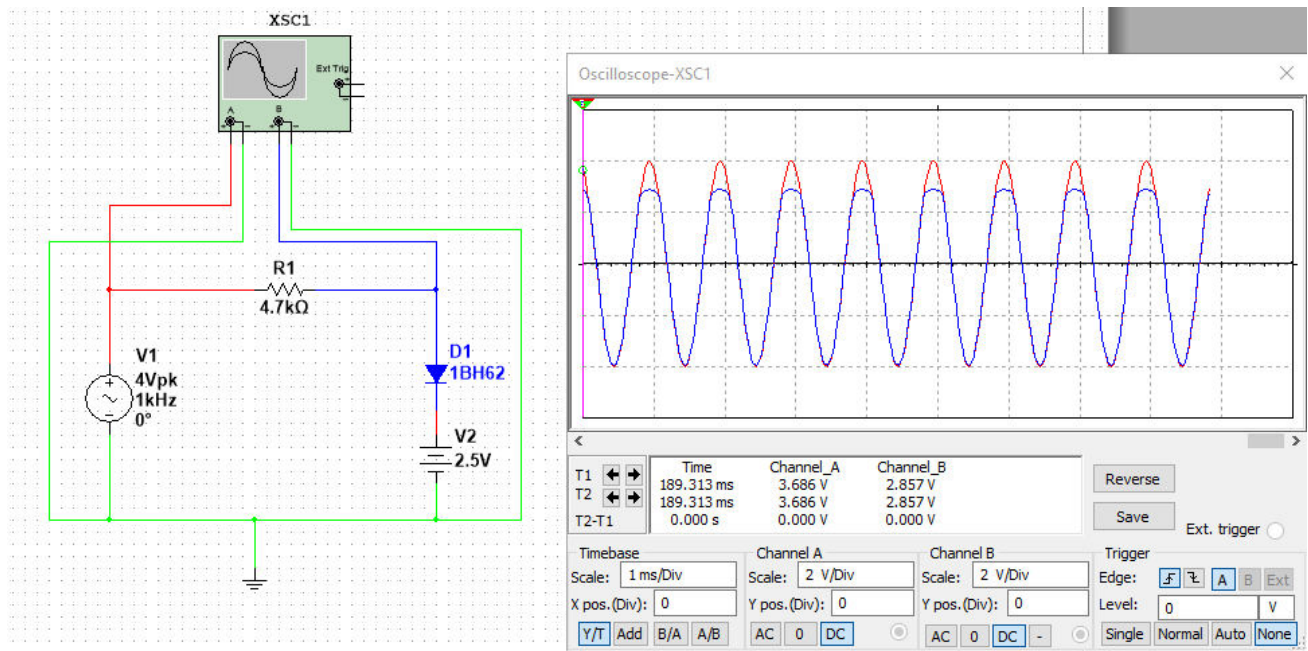


Figure 2.2.2.7 Waveforms of the scissor circuit with 2.5 Volt DC voltage, in "Multisim"

2.3 SCISSOR WITH INVERTED DIODE

General

The chapter "Simulation Solution" presents in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim", while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 2.3.1 shows the schematic of the scissor circuit with 8 V_{p-p} power supply, frequency 1 kHz, 4.7 kΩ resistor, 2 Volt DC voltage and the diode inverted (the cathode faces the AC source, while the anode faces the DC source).

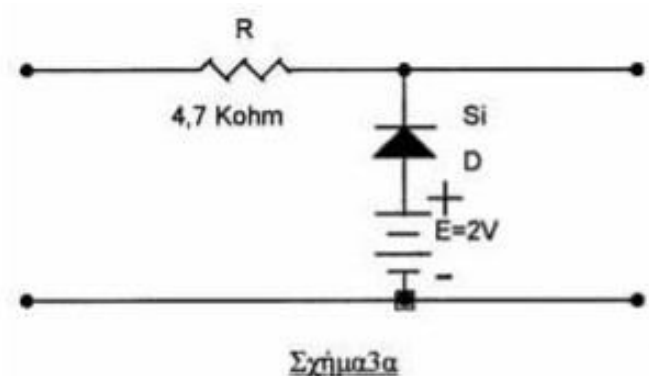


Figure 2.3.1 Schematic of the scissor circuit with the diode inverted

The scenario described in the chapter "Theoretical Solution" of section "2.1 Scissor Circuit" is reversed. Because, it is a silicon diode, where the blocking voltage is at 0.6 Volts, the diode turns on (right polarity) until the input voltage reaches 1.4 Volts. This is due to the fact that the cathode voltage is lower than the anode voltage due to the 2 Volt DC source, where it is lower than the input signal until 1.4 Volt (the voltage drop

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caused by the 4.7 kOhm resistor R is also taken into account). The output will become equal to the DC source voltage of 2 Volts. In addition, there will be a potential difference of 0.6 Volt between the anode and the cathode. This occurs until the input voltage drops to 1.3 Volts

Therefore, when the input signal reaches 1.4 Volts, the cathode voltage will become greater than the anode voltage and therefore, the diode will not turn off (reverse bias) and the output will become equal to the input voltage. Figure 2.3.2 shows the **input** and **output** waveforms.

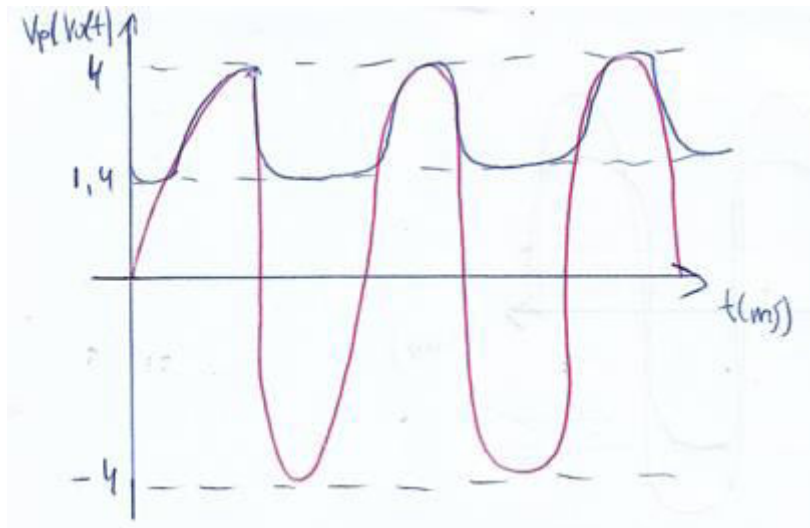


Figure 2.3.2 Waveforms of the scissor circuit with the diode inverted

Simulative Resolution

Figure 2.3.3 shows the simulation of the scissor circuit with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kOhm, 2 Volt DC voltage and the diode inverted, in the simulation software "Multisim".

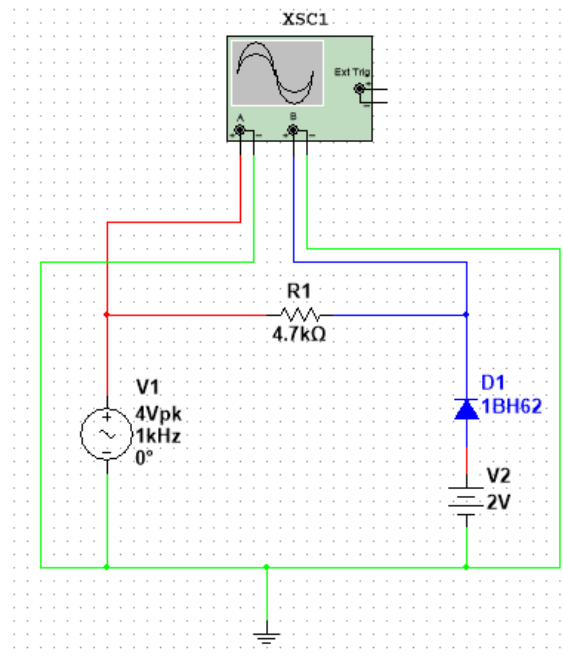


Figure 2.3.3 Simulation of the scissor circuit with the diode inverted, in "Multisim"

In Figure 2.3.4, the observations made in the Theoretical Solution chapter are verified simulatively in the simulation software with the help of the oscilloscope.

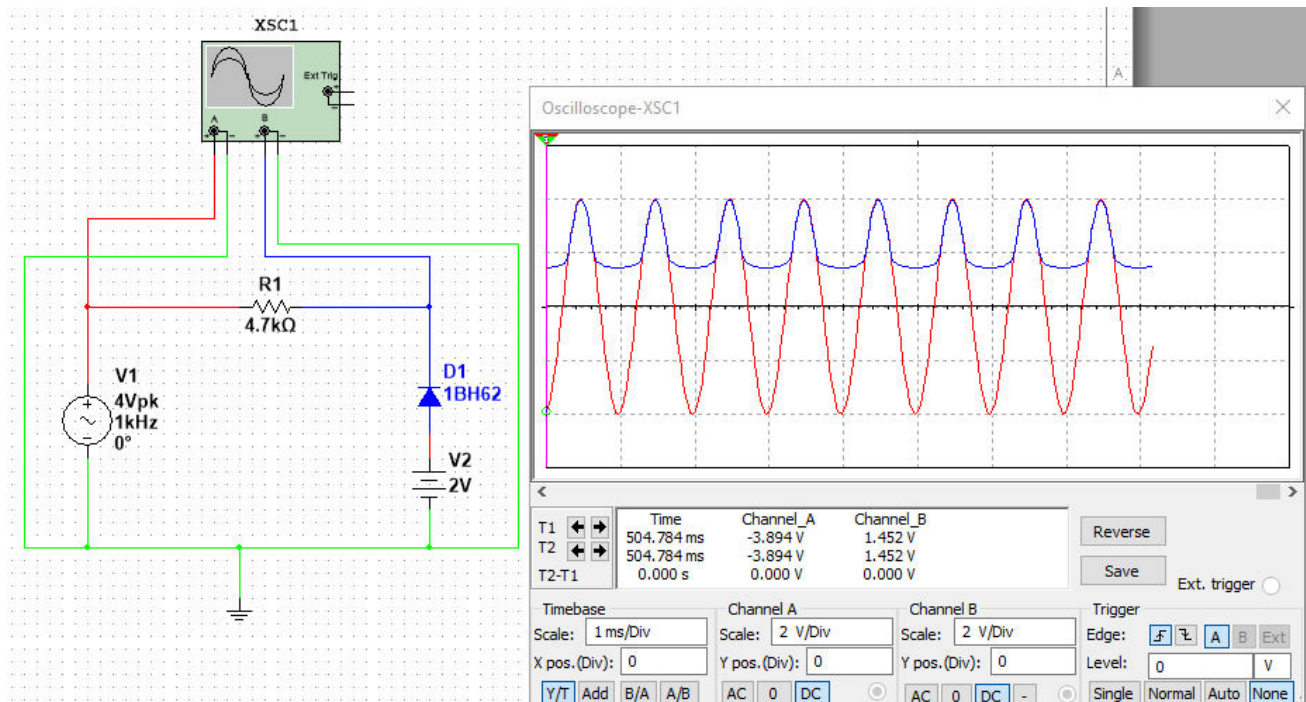


Figure 2.3.4 Waveforms of the scissor circuit with the diode inverted, in "Multisim"

2.4 DIFFERENT SCISSOR CIRCUITS

General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

Theoretical Solving

Figure 2.4.1 shows the schematic of a clipper circuit with 8 V_{p-p} power supply, 1 kHz frequency, 4.7 kOhm resistor and 2 volts DC voltage. The difference is that the diode and resistor have changed position and the diode is facing the AC source.

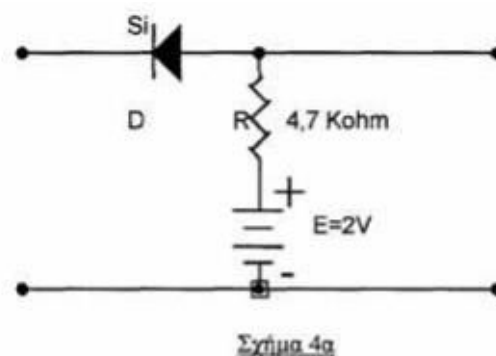


Figure 2.4.1 Schematic of the 1st scissor circuit

The voltage at the anode of the diode is < 2 Volt, due to the voltage drop of the resistor, while the voltage at the cathode is the input voltage from the AC source. Therefore, the diode turns off (right polarity) for voltages < 2 Volts, so that the voltage at the cathode is less than the voltage at the anode. The output signal

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during the right polarization of the diode reaches up to -3.4 Volts due to the potential difference between the anode and cathode (0.6 Volts) during right polarization. Figure 2.4.2 shows the **input** and **output** waveforms.

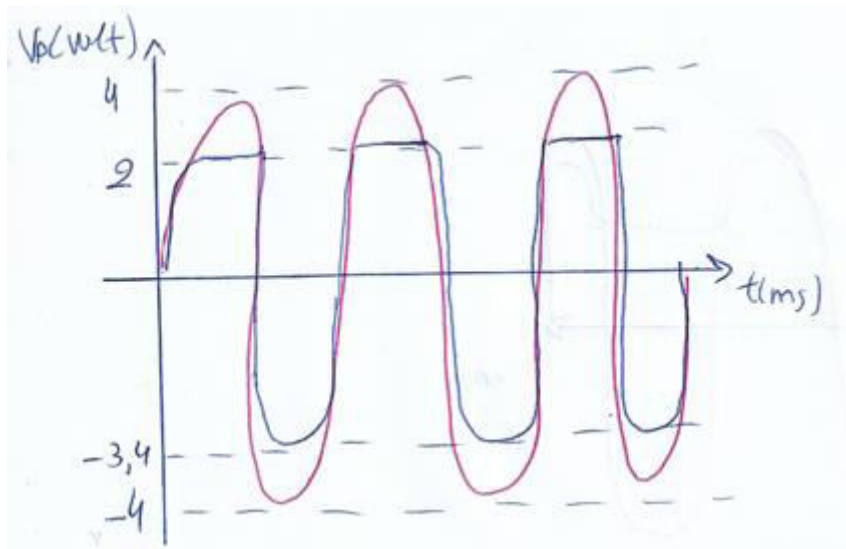


Figure 2.4.2 Waveforms of the 1st clipper circuit

Figure 2.4.3 shows the schematic of a clipper circuit with 8 V_{p-p} supply, 1 kHz frequency, 4.7 kOhm resistor and 2 volts DC voltage. The difference is that the diode and resistor have changed position and the diode is facing the DC source.

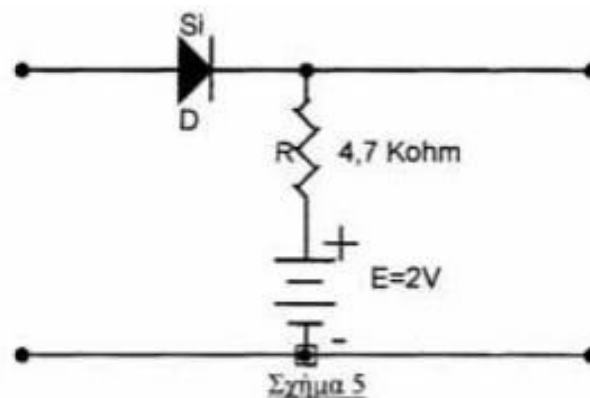


Figure 2.4.3 Schematic of the 2nd scissor circuit

This case is the same as that of Figure 2.4.1 with the difference being in the diode now facing the DC source. The voltage at the diode cathode is < 2 volts due to the voltage drop of the resistor, while the voltage at the anode is the input voltage from the AC source. Therefore, the diode turns off (right polarity) for voltages > 2 Volts, so that the voltage at the cathode is less than the voltage at the anode. The output signal during right polarization of the diode reaches up to 3.4 Volts due to the potential difference between the anode and cathode (0.6 Volts) during right polarization. Figure 2.4.4 shows the **input** and **output** waveforms.

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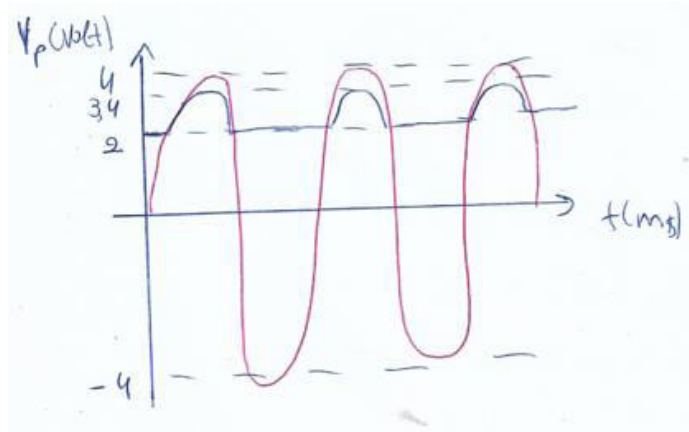


Figure 2.4.4 Waveforms of the 2nd clipper circuit

Figure 2.4.5 shows the schematic of a clipper circuit with 8 V_{p-p} power supply, frequency 1 kHz, 4.7 kOhm resistor, 2 Volt DC voltage, 2 sources of 2 Volt DC voltage and two diode clippers.

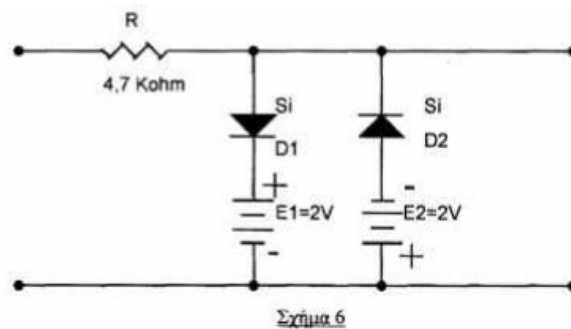


Figure 2.4.5 Schematic illustration of the 3rd scissor circuit

In this circuit the peaks of the positive and negative half-period are clipped. The peaks of the positive half-period are clipped by diode D1, since for a voltage equal to or higher than 2.4 Volts, the diode turns off (right polarity). The peaks of the negative half-period are clipped by diode D2, since for values less than -2.4 Volts, D2 is turned off (right polarity). As a result, the output is the same as the input for voltages from -2.4 Volt to 2.4 Volt. Figure 2.4.6 shows the **input** and **output** waveforms.

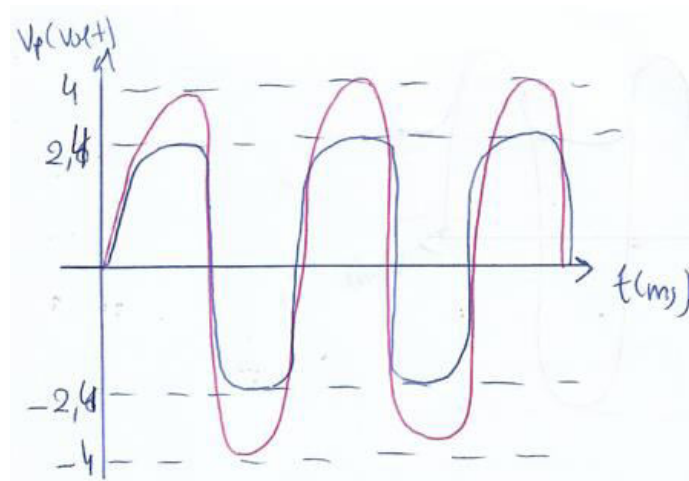


Figure 2.4.6 Waveforms of the 3rd clipper circuit

Simulative Resolution

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Figure 2.4.7 shows the simulation of the 1st scissor circuit (Figure 2.4.1) with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kOhm and 2 Volt DC voltage, in the simulation software "Multisim".

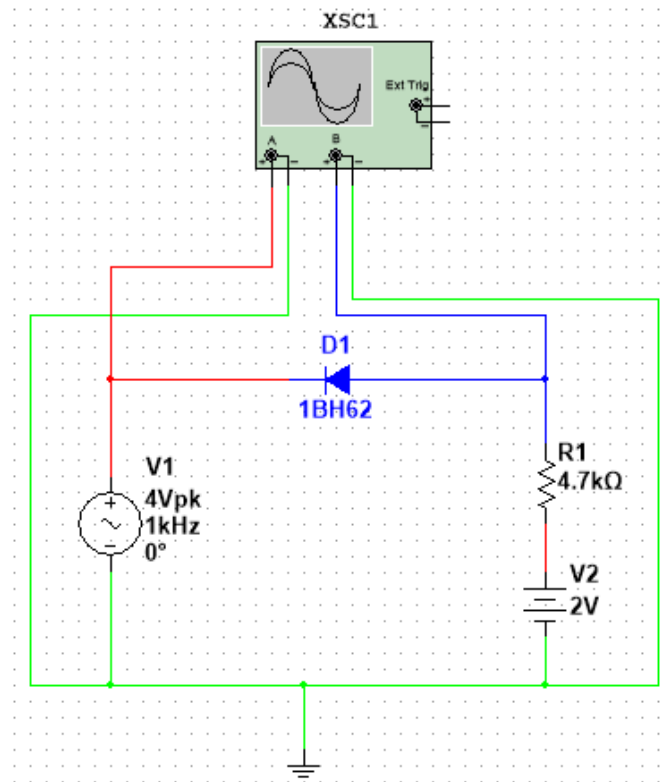


Figure 2.4.7 Simulation of the 1st scissor circuit in "Multisim"

In Figure 2.4.8 and 2.4.9, the observations made in the Theoretical Solution chapter are verified simulatively in the simulation software with the help of the oscilloscope.

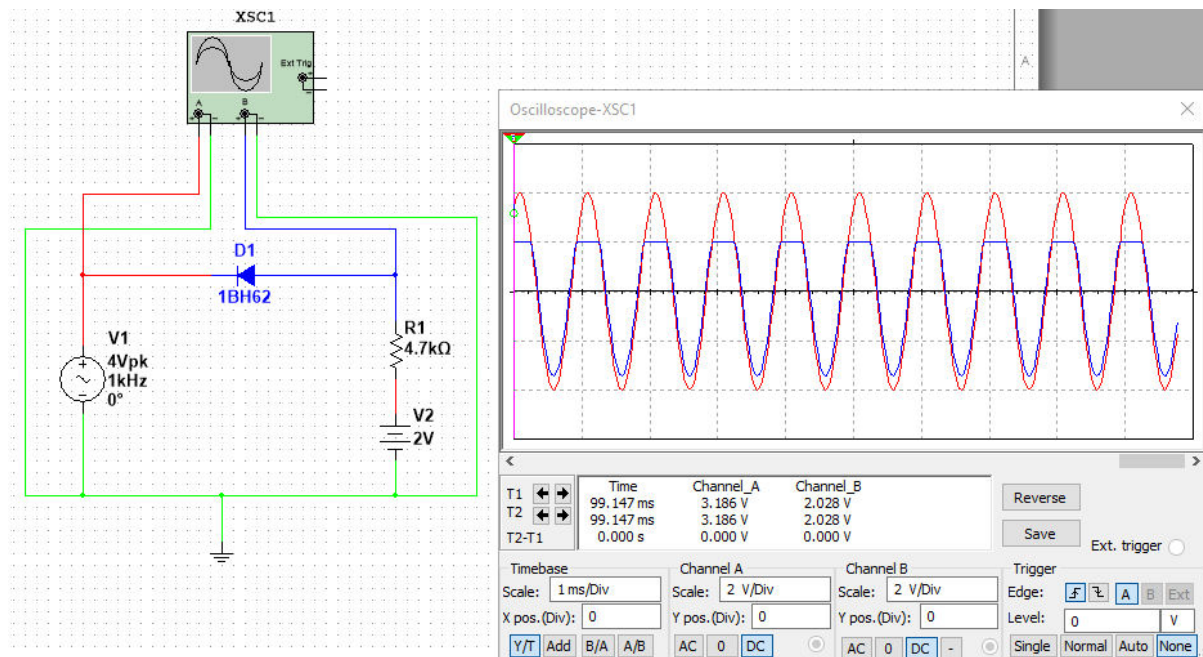


Figure 2.4.8 Waveforms of the 1st clipper circuit, in "Multisim"

ELECTRONICS

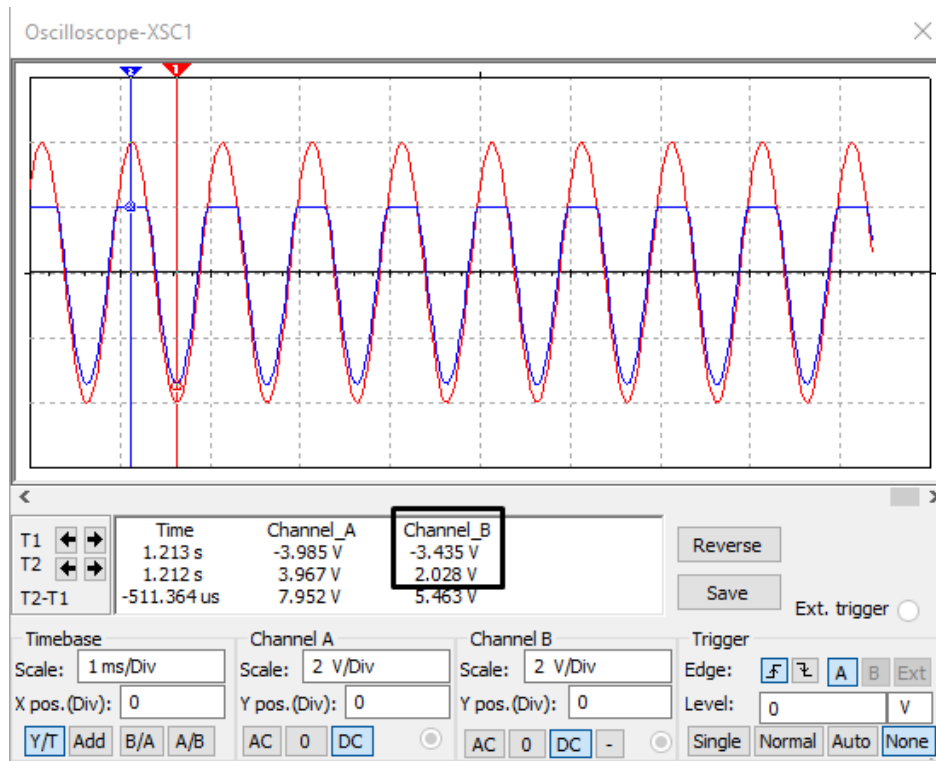


Figure 2.4.9 The peaks of the output signal of the 1st clipper circuit, in "Multisim"

Figure 2.4.10 shows the simulation of the 2nd clipper circuit (Figure 2.4.3) with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kOhm and 2 Volt DC voltage, in the simulation software "Multisim".

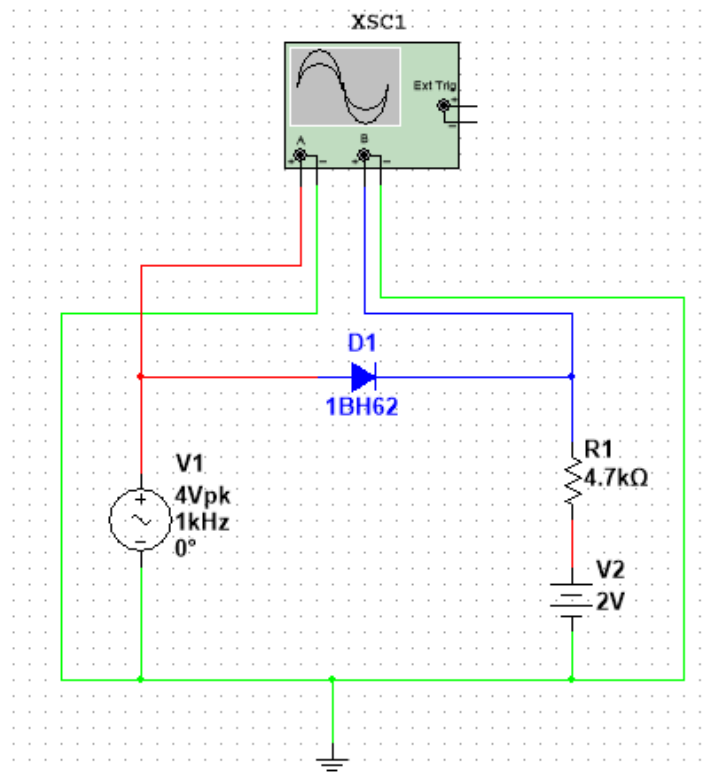


Figure 2.4.10 Simulation of the 2nd scissor circuit in "Multisim"

In Figure 2.4.11 and 2.4.12, the observations made in the Theoretical Solution chapter are simulated in the simulation software with the help of the oscilloscope.

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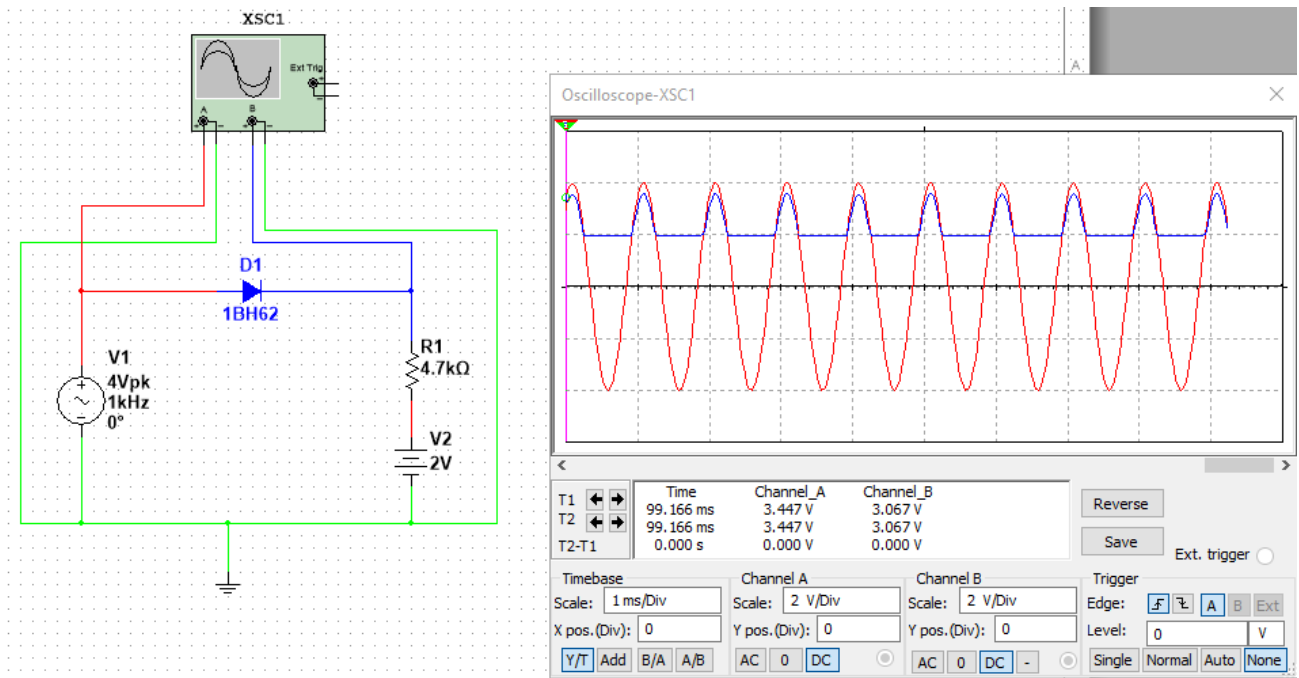


Figure 2.4.11 Waveforms of the 2nd clipper circuit, in "Multisim"

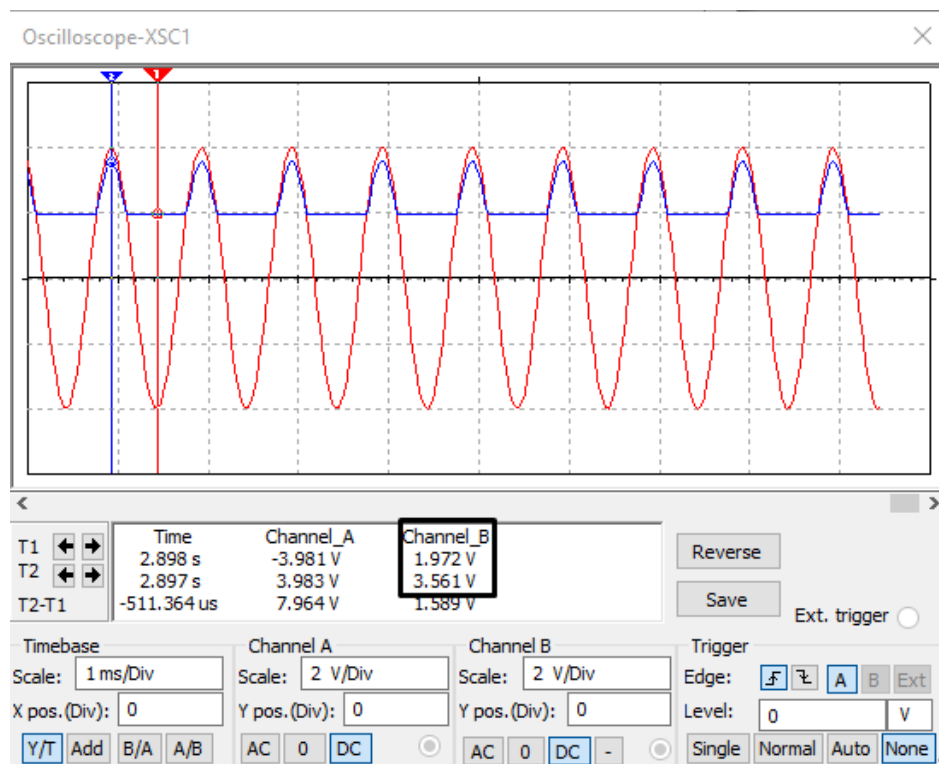


Figure 2.4.12 The peaks of the output signal of the 2nd clipper circuit, in "Multisim"

Figure 2.4.13 shows the simulation of the 3rd scissor circuit (Figure 2.4.5) with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kOhm, 2 DC voltage sources of 2 Volt and two diodes - scissors, in the simulation software "Multisim".

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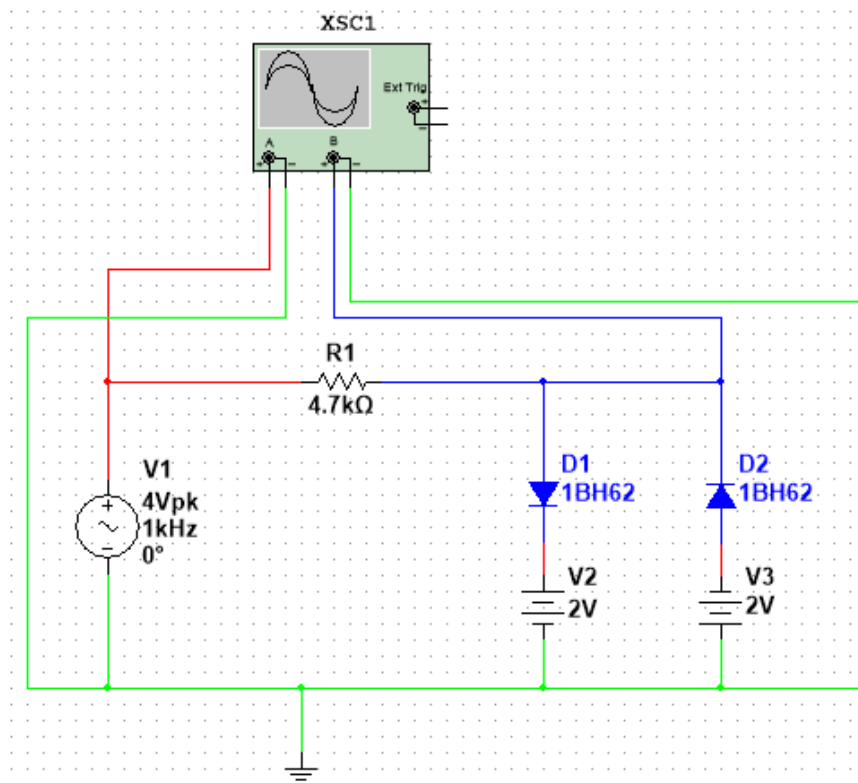


Figure 2.4.13 Simulation of the 3rd scissor circuit in "Multisim"

In Figure 2.4.14 and 2.4.15, the observations made in the Theoretical Solution chapter are simulated in the simulation software with the help of the oscilloscope.

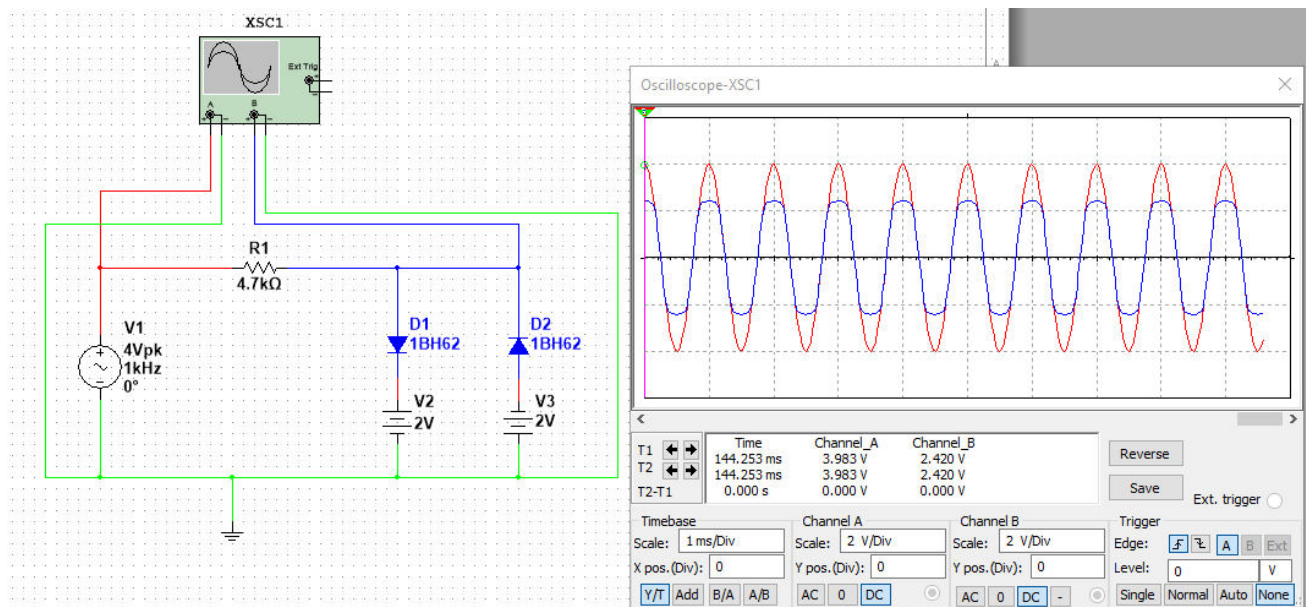


Figure 2.4.14 Waveforms of the 3rd clipper circuit, in "Multisim"

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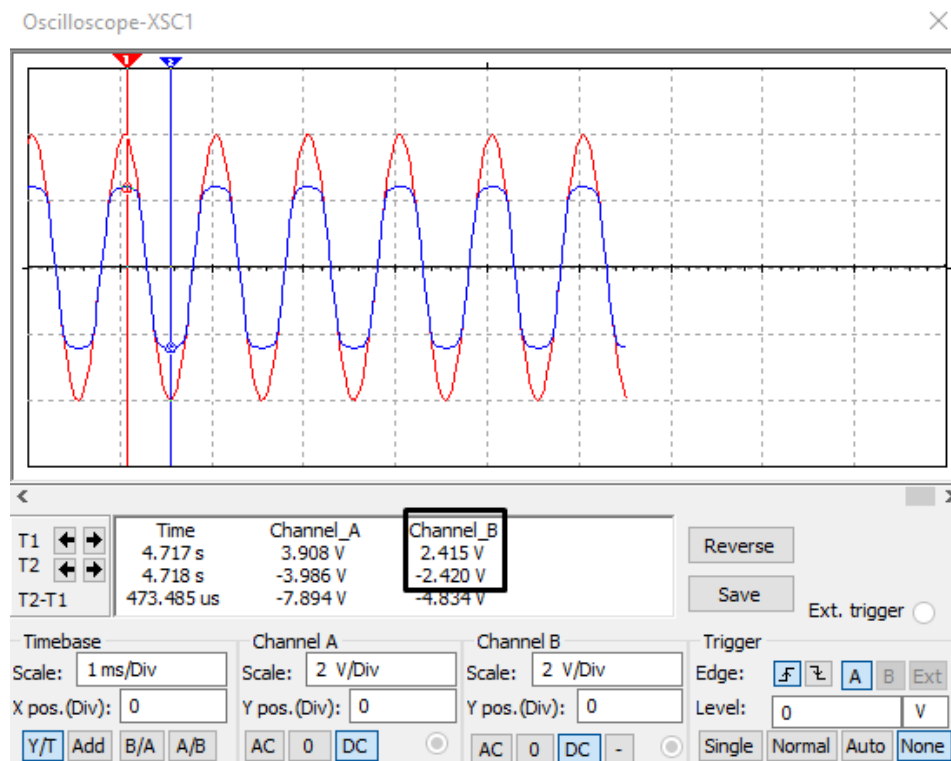


Figure 2.4.15 The peaks of the output signal of the 3rd clipper circuit, in "Multisim"

Experimental Solution

Figure 2.4.16 shows the 1st clipper circuit (Figure 2.4.1) with 8 V_{p-p} power supply, 1 kHz frequency, 4.7 kOhm resistance and 2 Volt DC voltage on the "breadboard", where it was carried out in the laboratory.

The steps in the construction of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 2.4.17).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 2.4.18).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 2.4.18).
4. We take channel A of the oscilloscope and connect the terminals to display the input waveform on the screen (Figure 2.4.18).
5. We set the switch to 2 Volts/Div (Figure 2.4.19).
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 8 V_{p-p} (Figure 2.4.19).
7. Align from the oscilloscope the waveform to be correctly positioned on the axes (Figure 2.4.19).
8. We take the silicon diode (Figure 2.4.20) and connect it in series with the AC voltage source and point it towards it.
9. We take the 4.7 kOhm resistor (Figure 2.4.21) and connect it in series with the diode.
10. We take a cable and connect the DC voltage source 0+15 Volt to the circuit (Figure 2.4.22).
11. We connect to the ends of the DC source a digital bench multimeter configured to calculate DC voltage, so that we can set the DC source to 2 Volts (Figure 2.4.23)
12. We also take channel B of the oscilloscope and connect it in parallel with one end of the resistor and the negative pole of the DC voltage source to get the output signal (Figure 2.4.24).

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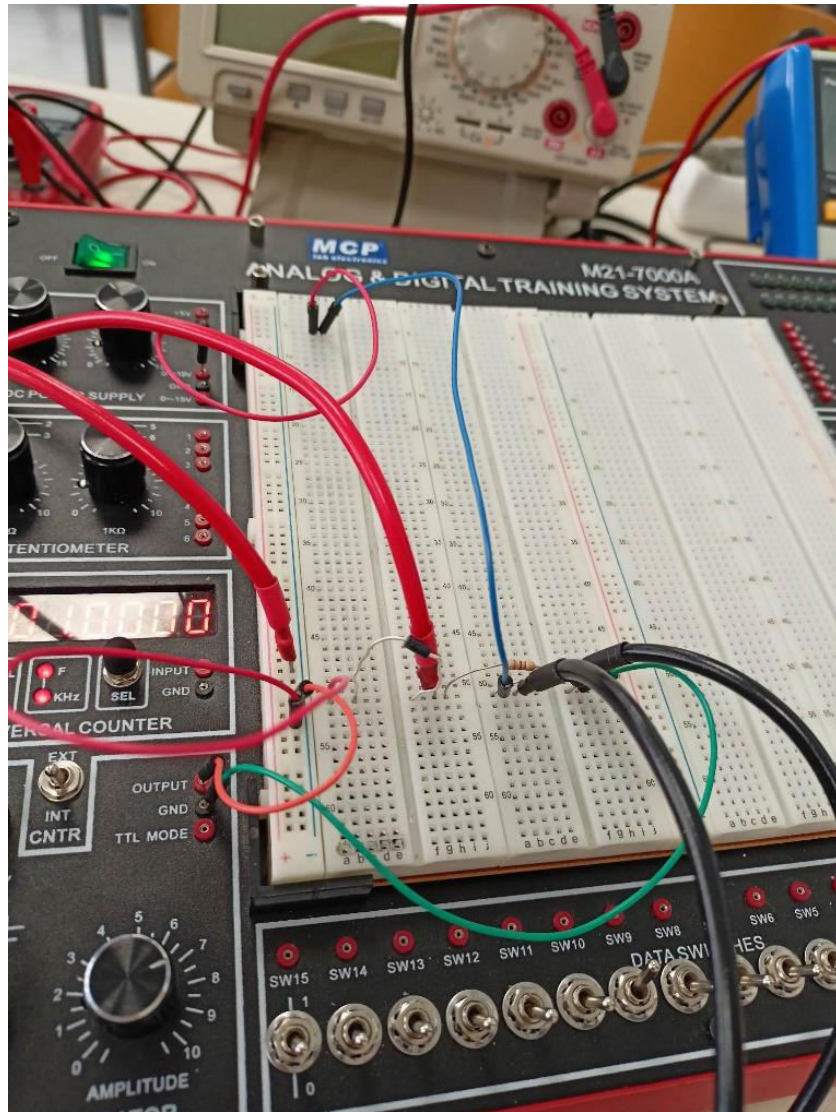


Figure 2.4.16 Experimental illustration of the 3rd clipper circuit in "Multisim"



Figure 2.4.17 Step 1

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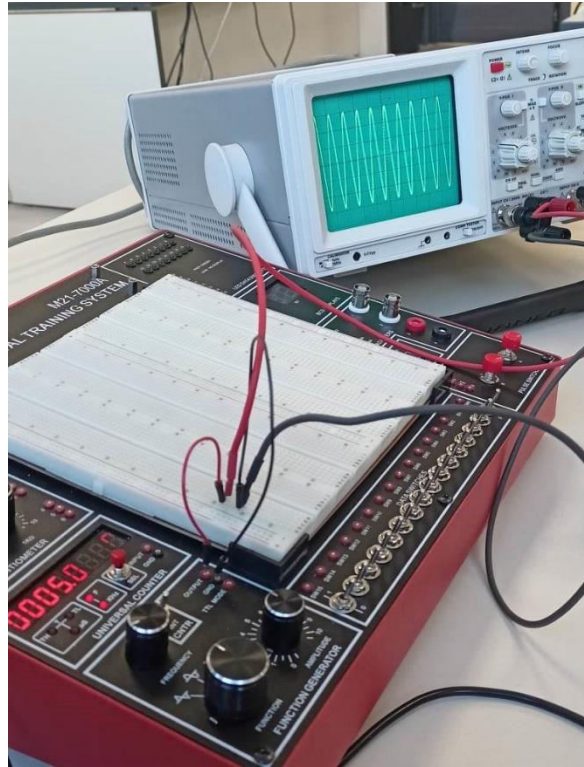


Figure 2.4.18 Step 2, 3, 4

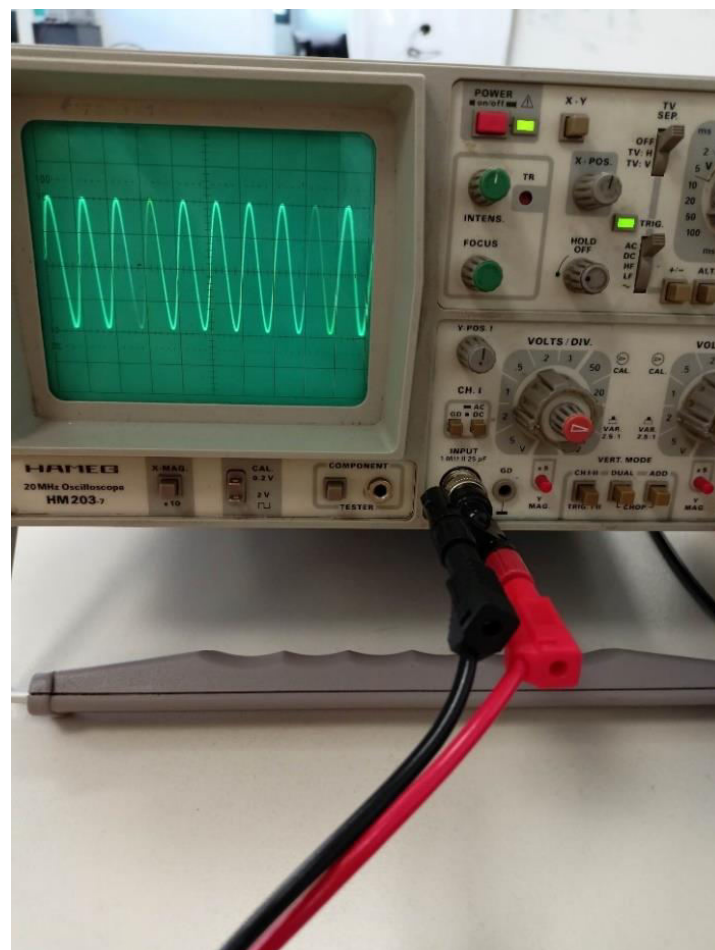


Figure 2.4.19 Step 5, 6, 7

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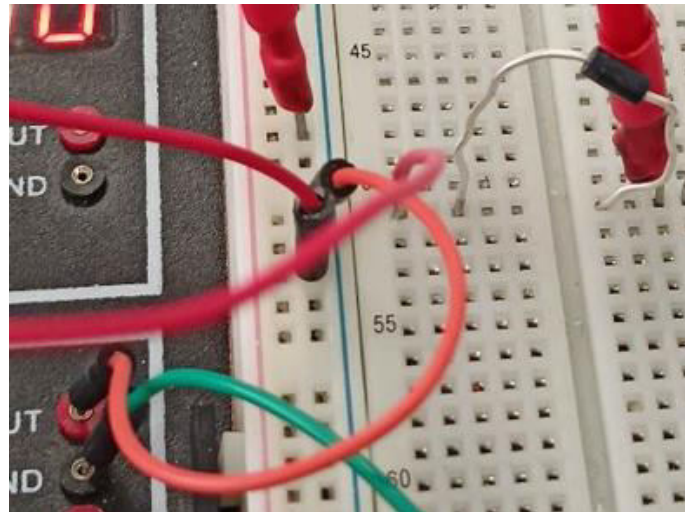


Figure 2.4.20 Step 8

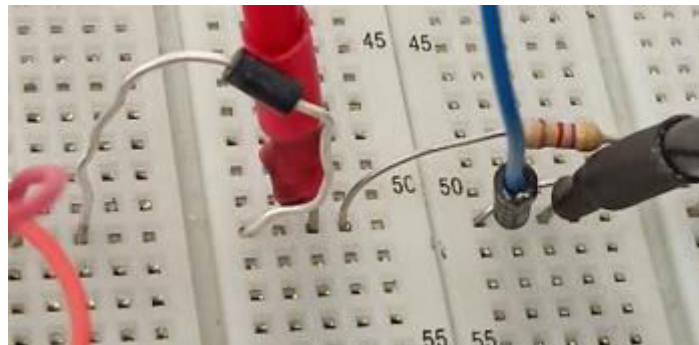


Figure 2.4.21 Step 9

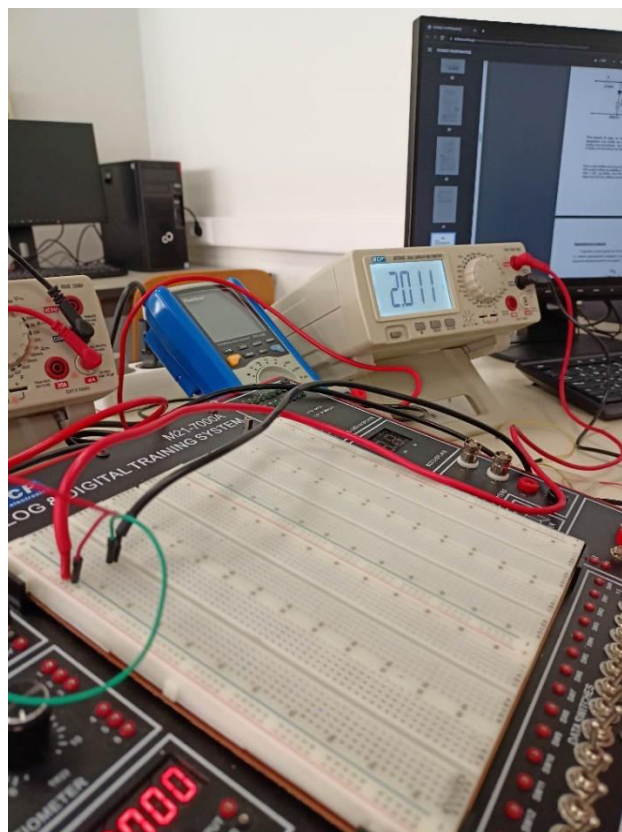


Figure 2.4.22 Step 10, 11

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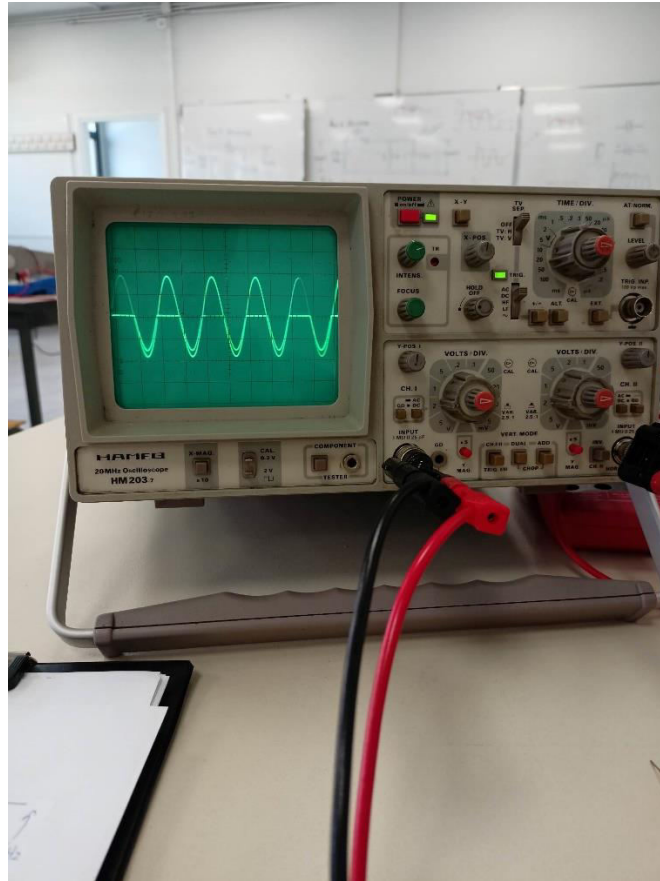


Figure 2.4.23 Step 12

2.5 QUESTIONS

Question 2.5.1

α.) Explain the operation of the shredder circuit (Figure 2.5.1.1.1)

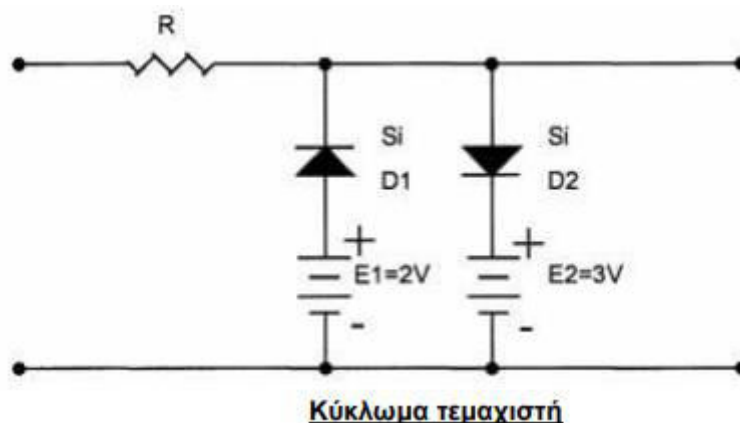


Figure 2.5.1.1 Schematic illustration of the shredder circuit

Diode D1 turns off when the input voltage coincident with the cathode voltage is less than 1.4 Volt, so that it is less than the anode voltage which is the 2 Volt voltage of the DC voltage source. Between the anode and cathode there is a potential difference of 0.6 Volt (silicon diode) and therefore, the output of diode D1 follows the input for voltages greater than 1.4 Volts, since, for these voltages the diode is inversely polarized and therefore, does not turn off.

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Diode D2 turns off when the input voltage coincident with the anode voltage is greater than 3.4 Volts, so that it is greater than the cathode voltage which is the 3 Volt voltage of the DC voltage source. Between the anode and cathode there is a potential difference of 0.6 Volt (silicon diode) and therefore, the output of diode D2 follows the input for voltages less than 3.4 Volts, since, for these voltages the diode is inversely polarized and therefore, does not turn off.

In summary, the input signal is clipped and the output follows the input for voltages from 1.4 Volt to 3.4 Volt. Figure 2.5.1.2 shows the **input** and **output** waveforms.

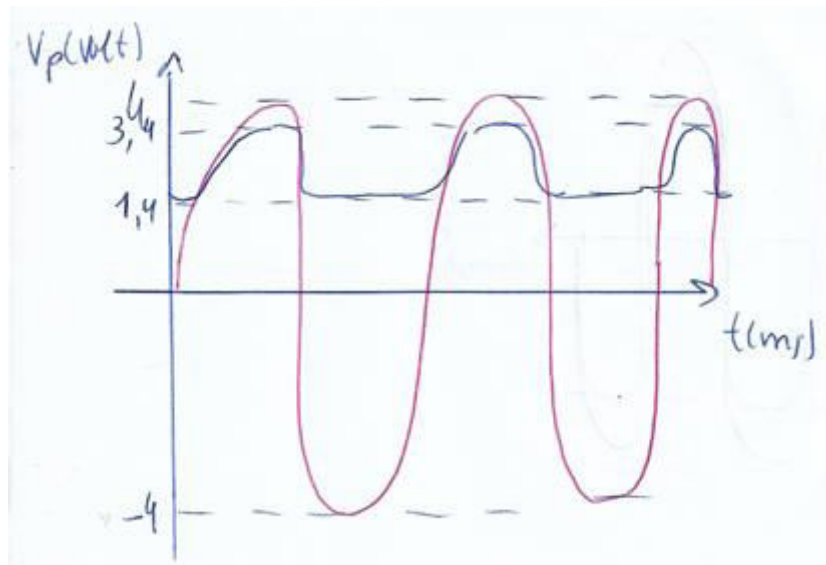


Figure 2.5.1.2 Waveforms of the shredder circuit

Figure 2.5.1.3 shows the simulation of the shredder circuit (Figure 2.5.1.1) with 8 V_{p-p} power supply, frequency 1 kHz, resistance 4.7 kOhm, and two DC voltage sources of 2 Volt and 3 Volt, in the simulation software "Multisim".

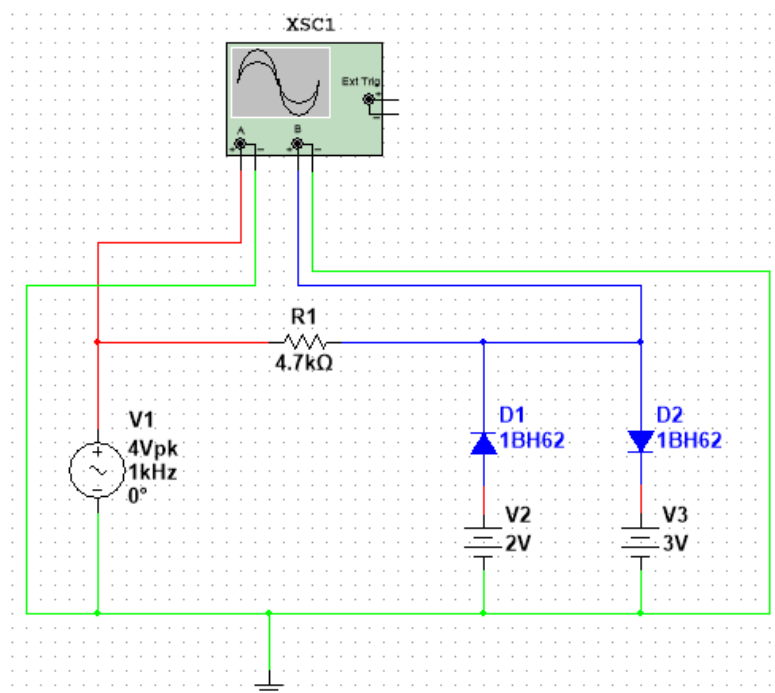


Figure 2.5.1.3 Simulation of the shredder circuit

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In Figure 2.5.1.4 and 2.5.1.5, the observations made above regarding the operation of the shredder circuit are simulated in the simulation software with the help of the oscilloscope.

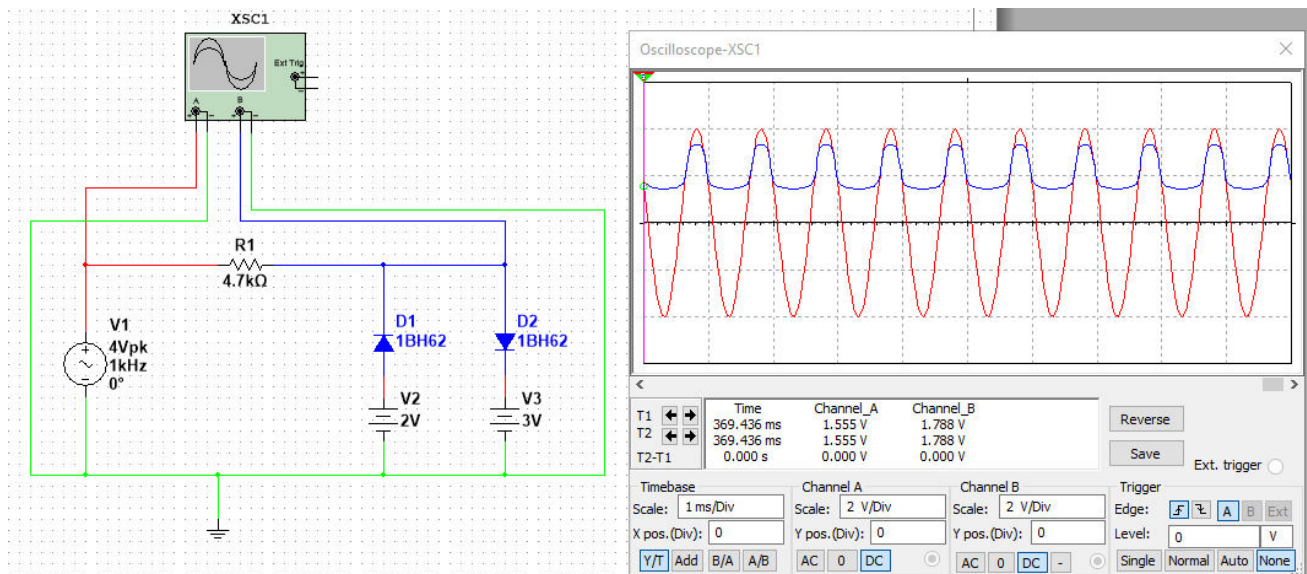


Figure 2.5.1.4 Waveforms of the shredder circuit, in "Multisim"

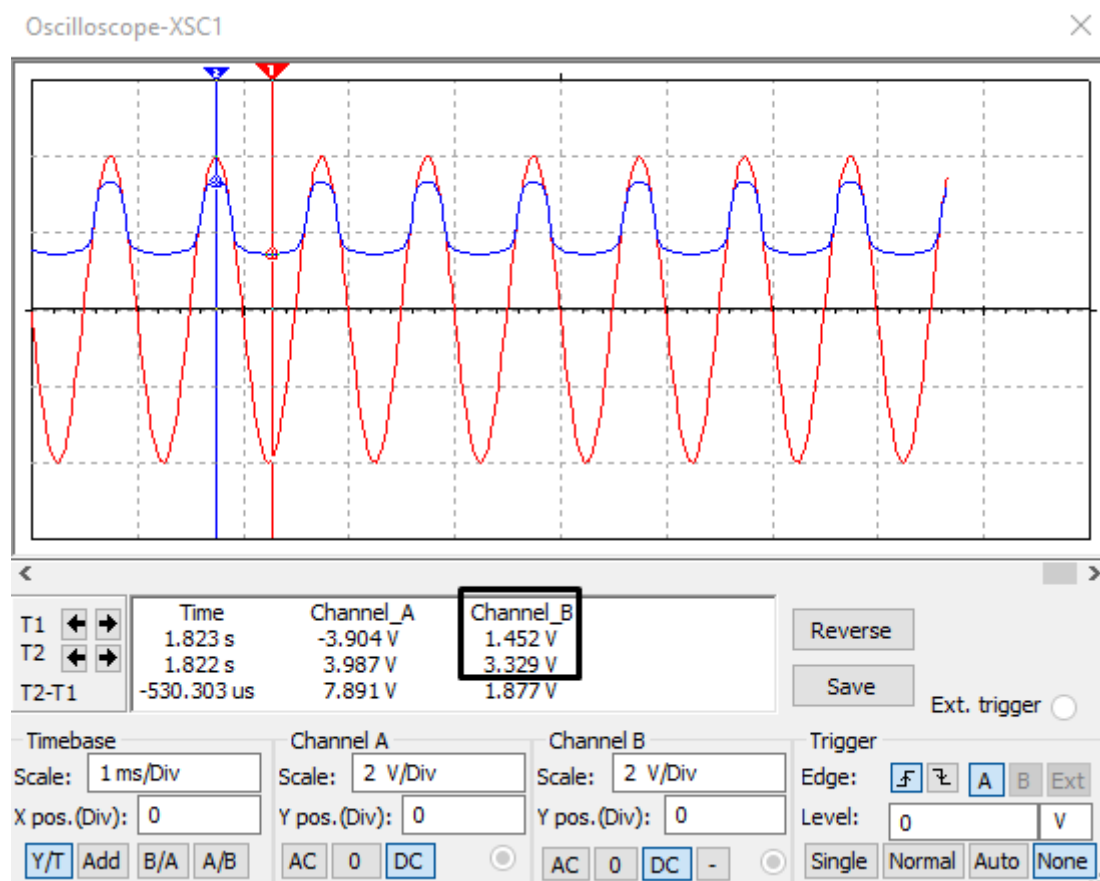


Figure 2.5.1.1.5 The peaks of the output signal of the shredder circuit, in "Multisim"

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Thank you for your attention.

